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Data Enclaves

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For Sheila, Maple, and Saga

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CHAPTER 1

Introduction

Abstract Today, digital personal data has become the defining resource for our societies and economies. Unfortunately, our personal data are increasingly concentrated in the hands of a small number of digital technology businesses often called *Big Tech*. The past decade has been defined by the rise of Big Tech as the dominant social players in our societies, and much of their rise and dominance is down to their control over our personal data. Big Tech has created a series of data enclaves that entrench their power and dominance, limiting the capacity of other businesses to compete in technoscientific capitalism. In building their data enclaves, Big Tech has engaged in a parasitic form of innovation, developing digital technologies designed to limit access to resources, to undermine regulations or social conventions, to undermine or avoid competition, to exploit customers psychology, to lock customers into using a product, to stop customers fixing their own property, or to use information asymmetries to treat customers inequitably. The contention of this book is that we need to rethink data governance in order to address the growing paradoxes and problems engendered by the market and social power of Big Tech.

Keywords Personal data · Big Tech · Parasitic innovation · Data enclaves

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Recently, I spent two weeks without a smartphone. It felt like I was living in a strange, new world, despite being the norm for most of my life—as I only got my first smartphone around 12 years ago. For those two weeks I had to make do with an old-style Nokia brick phone, only able to call and text. I didn't have mobile access to the internet or all the apps I'm now so used to using. In some ways, it was very liberating, and being away from social media like Facebook or Twitter probably did wonders for my mental health. The experience got me thinking, though, especially about the data traces we leave behind through the everyday tapping away we do on our smartphones (and other devices). For that two-week period I was digitally invisible, in that data about my personal life, behaviours, preferences, and so on were not being collected, analysed, and used to grab my attention, to push advertising down my eyeballs, or to sell me stuff that may but, as likely, may not interest me. My data twin now has a blank spot for those two weeks where usually it would have a very detailed social, temporal, and geographical 'graph' of my life; all the places I go, all the things I look up, all the people I interact with, and much more.

Why does this matter? Today, our economies are increasingly driven by the digital personal data collected about almost everything. Smartphones collect data on our physical movement; computers collect data about our online searches; televisions collect data about our viewing habits; websites collect information about our buying habits; and much more beside. All of these data feed into data analytics and algorithms, making inferences and predictions about our everyday decisions, behaviours, and preferences: for example, streaming services like Netflix collect data on our viewing habits in order to recommend series and films to us, or to make decisions about what series and films to make and to renew.[1] Data have become an important economic resource, underpinning the financial decision-making of businesses, of investors, of lenders, of governments, and of individuals. A helpful illustration of this is the way that during Covid both United Airlines and American Airlines used the value of their customer data as collateral to secure billion-dollar loans; in fact, these data were more valuable than the stock market valuation of these corporations themselves.[2]

[1] Birch, K. (2023) Yet another subscription fee: Twitter, Facebook, Netflix are desperate and dying, *The Globe and Mail—Report on Business* (10 Apr).

[2] Laney, D. (2020) Your company's data may be worth more than your company, *Forbes* (22 July).

It's not just businesses that are finding their data are increasingly important to their operations and are, consequently, increasingly understood as valuable assets: governments are trying to open up public data, like weather or traffic information, to improve public services and create collective social benefits[3]; higher education institutions are trying to analyse their student records to improve student performance and learning outcomes[4]; digital activists are trying to promote the idea that we should have ownership rights over our personal data to ensure we capture some of its value[5]; … and so on. Across these diverse individual, organizational, and governmental interests in data there is an assumption that digital data are something different from other information we have always had in our economies: these differences include the massification of digital data collection via new digital platforms; the invention of new digital architectures to enable the mass collection of data; the creation of opportunities for developing new data-driven products and services; the opening up of new data markets, especially for our attention; and the promise of greater efficiencies in the delivery of welfare and social services.

The Rise (and Fall) of Big Tech

All of this sounds well and good, so why are there so many critical books and articles about this burgeoning data-driven economy? At the heart of this critical concern sit a small number of large multinational corporations usually identified as Apple, Alphabet/Google, Amazon, Meta/Facebook, and Microsoft,[6] although other corporations, like Tesla, Netflix, Tencent, and Alibaba, are also mentioned in the same breath. Critics define this group in different ways: popular acronyms include 'GAFAM' and

[3] WEF (2011) *Personal Data: The Emergence of a New Asset Class*, Geneva: World Economic Forum.

[4] https://www.gartner.com/en/industries/education.

[5] Lanier, J. (2014) *Who Owns the Future*? New York: Simon & Schuster.

[6] For ease, I will use "Alphabet/Google" and "Meta/Facebook" throughout this book, although readers should note that Alphabet Inc. was established in 2015, subsuming Google Inc., and Meta Platforms Inc. was established in 2021, replacing Facebook Inc.

'FAANG' but my personal preference is the phrase 'Big Tech'.[7] Rather than having to adapt an acronym to name changes—like Facebook's change to Meta—or the vacillating fortunes of smaller technology companies—like Netflix's recent wobbles—a term like Big Tech can be deployed more consistently to frame and analyse the uniqueness of these digital leviathans, especially when it comes to their scale and scalability.[8] Even with the use of the term Big Tech, though, it's important to recognize that these corporations are not a singular monolith, representing one set of interests, strategies, or logics; instead, we have to remember that they come into conflict with one another and their interests can diverge quite significantly. For example, in 2021 Apple and Meta/Facebook engaged in a very public spat over Apple's decision to introduce more stringent privacy features into its operating system, which undermined Meta/Facebook's advertising revenues.

Nevertheless, Big Tech is a helpful term to understand the dominance of digital technology businesses in our economies today. Like previous generations of 'big' industry—whether that's 'Big Auto', 'Big Pharma', or 'Big Oil'—it's useful to understand what underpins Big Tech's market power. As a starting point, we can see that Apple, Alphabet/Google, Amazon, Meta/Facebook, and Microsoft control access to the digital technologies, infrastructures, and platforms we rely upon in our everyday lives, whether that is searching online for information, staying in touch with friends and family via social media, using our smartphones to find our way around, or buying things from online marketplaces. In particular, though, Nick Srnicek, a leading academic critic, argues that Big Tech represents a new 'platform' model of capitalism "capable of extracting and controlling immense amounts of data" leading to the "rise of large monopolistic firms".[9] The platform, according to Srnicek, is a new type of firm, usually acting as intermediaries between different users, such as people who use Google Search to find information and businesses that use

[7] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58; and Hendrikse, R., Adriaans, I., Klinge, T. and Fernandez, R. (2022) The Big Techification of everything, *Science as Culture* 31(1): 59–71.

[8] Prainsack, B. (2019) Data donation: How to resist the iLeviathan, in J. Krutzinna and L. Floridi (eds.), *The Ethics of Medical Data Donation*, Dordrecht: Springer, pp. 9–22; and Birch, K. and Bronson, K. (2022) Big Tech, *Science as Culture* 31(1): 1–14.

[9] Srnicek, N. (2016) *Platform Capitalism*, Cambridge: Polity Press, p. 6.

Google Ad Manager to buy those people's attention. While in agreement that platforms are an important aspect of Big Tech, my analytical preference is to think of Big Tech as 'ecosystems', reflecting a "heterogenous assemblage of technical devices, platforms, users, developers, payment systems, etc. as well as legal contracts, rights, claims, standards, etc.".[10] I'll come back to why this distinction is important later in the book.

However we define Big Tech, it's evident that they have become dominant economic players—and increasingly influential social and political players too. In 2020, these five Big Tech corporations comprised about 25 percent of the S&P500, which itself represents the market capitalization of the largest 500 US corporations. They are Big! As shown in Fig. 1.1, Big Tech's rise in scale took off after the 2007–2008 global financial crisis (GFC), especially from around 2012. Microsoft has been a major corporation since the mid-1990s, but the antitrust court case brought against them by the US government in 1998, and which was eventually settled in 2001, opened up space for other digital technology businesses, like Alphabet/Google, to prosper. In the fallout from the GFC, especially the implementation of a low interest rate regime by many governments in the Global North, Big Tech was able to tap into cheap capital to expand significantly. Apple started the trend, growing at an annualized 18 percent per year from 2010 through to 2019, followed by Meta/Facebook, Alphabet/Google, and Microsoft in 2013, and finally Amazon from 2015 with an annual growth rate of 45 percent.[11] Despite growing political and public condemnation—more of which below—all of these Big Tech corporations topped US$1 trillion valuations at some point by 2022, before stock markets started to soften: some, like Apple, passed US$2 trillion valuations. In 2022, *The Economist* noted that the combined market capitalization of Big Tech was US$9 trillion, tripling between 2015 and 2020.[12]

[10] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1), p. 45; also see Hein, A., Schreieck, M., Riasanow, T., Setzke, D., Wiesche, M., Bohm, M. and Krcmar, H. (2020) Digital platform ecosystems, *Electronic Markets* 30: 87–98.

[11] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58.

[12] https://www.economist.com/briefing/2022/01/22/what-americas-largest-technology-firms-are-investing-in.

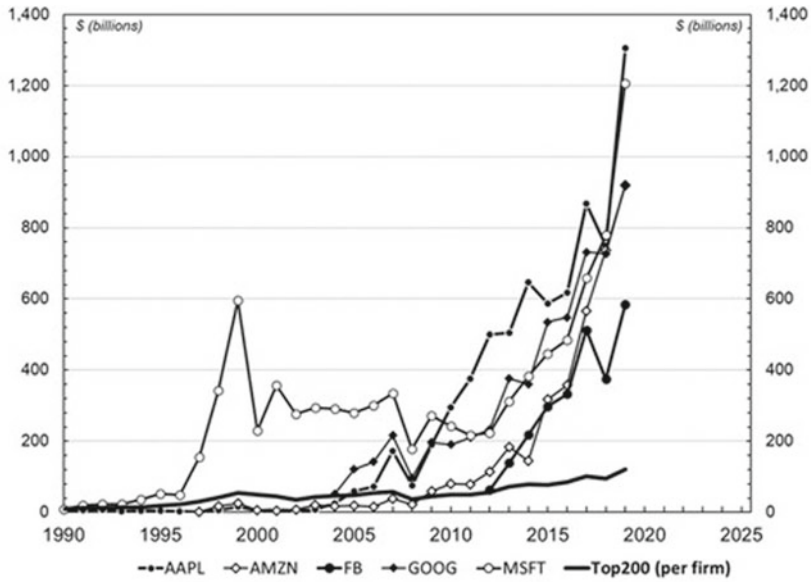

Fig. 1.1 Big Tech market capitalization, 1990–2019 (*Note* Reproduced from Birch and Cochrane [2022], note 7, p. 47. *Note* Compiled by D.T. Cochrane with data from Compustat via Wharton Research Data Service; AAPL = Apple, AMZN = Amazon, FB = Facebook, GOOG = Alphabe, MSFT = Microsoft)

Academics and others have generally pointed to the negative consequences of this scale and power of Big Tech, emphasizing the importance of, for example, platform or ecosystem network effects, winner-takes-all (or -most) competition dynamics, and the financial leverage it gives them vis-à-vis their competitors. By network effects, people mean the benefits that a platform or market user gets from participating in that platform or market as it grows larger with an increasing number of other users. Here, scale and scaling imperatives underpin the growth of Big Tech, enabling them to attract a growing user base from which they can collect

data.[13] By winner-takes-all (or -most) dynamics, people mean the competitive benefits that firms get from network effects, which enable fast scale up of a platform or market and better access to data and better data analytics; this creates a significant barrier to entry for competitors, especially new startups, which don't have similar access to data.[14] By financial leverage, people mean the expectations of investors who, looking at the network effects and winner-takes-all dynamics, decide on who to invest in, choosing those firms with the scale to monopolize entire markets. Subsequently, these 'expected' monopolists benefit from lower financing costs than their competitors as investors lend them capital at lower cost, enabling those firms to dominate, performatively, their markets.[15]

It's exactly this situation that propelled Big Tech to the commanding heights of our economies in the 2010s and into the 2020s. Data and finance created a heady mix which Big Tech rode to dominance, while companies in other sectors hunkered down in the aftermath of the global financial crisis, trying to cut costs or recover from the housing bubble. At the start of this run in the early 2010s, these Big Tech firms seemed to be able to do no wrong, pumping out technological marvels and financial returns seemingly at will. Such success, as with every other instance, did not—could not—last as a growing number of scandals hit the digital technology sector generally and Big Tech specifically. By the 2020s, Big Tech had turned from the darling of many politicians, policymakers, and publics into their *bête noire*.

Emerging Data Enclaves

Sometimes hindsight is easy, other times looking back on ourselves is more difficult to do. I still recall the fuss around the launch of Apple's first iPhone in 2007. I was living in the UK at the time, and I remember people were genuinely excited about the launch of a new 'smart' phone; being somewhat curmudgeonly myself, I didn't personally get what all

[13] Kenney, M. and Zysman, J. (2019) Unicorns, cheshire cats, and the new dilemmas of entrepreneurial finance, *Venture Capital*, 21(1), pp. 35–50; and Pfotenhauer, S., Laurent, B., Papageorgiou, K. and Stilgoe, J. (2022) The politics of scaling, *Social Studies of Science* 52(1): 3–34.

[14] Sadowski, J. (2020) The internet of landlords: Digital platforms and new mechanisms of rentier capitalism, *Antipode* 52(2): 562–580.

[15] Galloway, S. (2018) *The Four*, New York: Portfolio/Penguin.

the fuss was about. To be fair, the iPhone was a cool-looking piece of 'tech', although it did have a hefty price-tag to go with it. My first smartphone was an iPhone, but that didn't last long and I've sworn off them ever since. Around about the same time as the iPhone's launch, I also remember Google launching its Gmail service more widely and also signing up to Facebook as my go-to social network. For some reason I'd avoided MySpace and went straight to Facebook. Obviously, my hazy memories reveal something about my age, but they're also supposed to highlight the difference a decade or more makes to our appreciation of things. Thinking back to those heady days of the late 2000s highlights the way that these digital technologies seemed to offer something new, exciting, and cool—even if it was still largely about making money somehow. Fast forward to today, more than a decade later, and that technology honeymoon is well and truly over.

Interestingly, *The Economist* predicted as early as late 2013 that there would be a backlash against digital technology, something they called the 'techlash'.[16] This backlash only grew throughout the 2010s, especially with fears about the spread of misinformation, interference in political elections and referenda (e.g. 2016s Brexit referendum and Trump's presidential election), fears about entrenching digital and data monopolies, and the growing linkages between digital technologies and their negative impacts on our mental health.[17] Today, Big Tech is the target of numerous takedowns, including political attacks from left and right, legal suits by governments or individual class actions, journalistic investigations, and academic criticism. My bookshelves now groan under the weight of books taking aim at Big Tech with names like *Don't Be Evil*, *The Four*, *The New Goliaths*, *Cloud Empires*, and, of course, *The Age of Surveillance Capitalism*. And much of this criticism is warranted, of course; we haven't ended up with the bright, gleaming future we were promised, or that we imagined back in the early 2010s. In fact, it's probably accurate to say that we've ended up in an all-too-familiar grubby capitalist reality a few people warned us about.[18]

[16] https://www.economist.com/news/2013/11/18/the-coming-tech-lash.

[17] https://www.humanetech.com/key-issues.

[18] Morozov, E. (2013) *To Save Everything, Click Here: The Folly of Technological Solutionism*, New York: PublicAffairs.

I'm not going to provide a list of all the problematic things blamed on Big Tech here, although I'll come back to many of them throughout the book. Instead, I'm just going to discuss one government investigation and its findings about Big Tech. From June 2019 until October 2020, the Subcommittee on Antitrust, Commercial, and Administrative Law—part of the Committee on the Judiciary of the US Congress—undertook an investigation into competition in digital markets; it focused specifically on the actions and strategies of Apple, Amazon, Alphabet/Google, and Meta/Facebook. Its 450-page report came out in October 2020 and made fascinating reading.[19] Although the House of Representatives was dominated by the Democratic Party at that time, giving them control of many committees, the investigation was touted as bipartisan, which became evident in the cross-examination of Big Tech executives by politicians from both Democratic and Republican parties. Noting that Big Tech are diverse, the investigation stressed that there are common issues with their business practices. Allegations included:

- That Meta/Facebook sought to buy out any competitors threatening their dominance in the social media market; they identified these competitive threats through their "data advantage" (p. 14).
- That Alphabet/Google sought to monopolize online search and advertising through establishing their products as defaults on digital devices; as a result of their monopoly, they are able to increase the fees they charge advertisers. Alphabet/Google also "exploits information asymmetries" to "provide it with near-perfect market intelligence" (p. 15).
- That Amazon dominates online retail by buying out competitors, self-preferencing its products and services in its marketplace, and using users "customer data" in "shoring up its competitive moats" (p. 16).
- And that Apple has created an ecosystem of devices, operating system, app sales, etc., which it exploits "through misappropriation of competitively sensitive information" (p. 17).

[19] US House of Representatives (2020) *Investigation of Competition in Digital Markets*, Washington, DC: House of Representatives.

Across these allegations, there is a common thread: namely, the mass collection, use, and monetization of digital data, especially personal data, has expanded and then entrenched market power and concentration. As our economies and technological innovation are becoming increasingly data-driven, this market power of a very small number of Big Tech corporations has the potential, according to the Congressional investigation, to 'weaken' innovation and entrepreneurship—as well as undermine our political and information systems.

Data Enclaves and Parasitic Innovation

In this book, I come at these issues from an analytical perspective I describe as *constructivist political economy*, sitting at the interface between science and technology studies (STS) and political economy. Generally, STS is premised on the idea that knowledge is collectively produced, constituted, and legitimated, meaning that hero metaphors and narratives of genius inventors are nonsense and that there is no inherent logic in the evolution of science or technology. Instead, to understand science and technology necessitates focusing on the social, political, economic, and material context in which it emerges, since these all shape what science and technology are developed, how it is diffused, and how it ends up influencing societal choices and actions. Capitalism is important here as it plays a major role shaping the science and technology we end up with. It's useful to understand science and technology together, leading to the STS neologism of 'technoscience' to characterize the ways that science, technology, and society are entangled with one another. As should be evident from this, technoscience, being socially, culturally, and economically configured, is not neutral or free from social bias and prejudice; technoscience is and can be sexist, racist, homophobic, etc. It can also undermine political-economic alternatives to capitalism. And this is important because technoscience is socially and culturally powerful, both institutionally (e.g. universities) and as a source of authority (e.g. credentialed expertise). Nevertheless, there are challenges to this power in the form of citizen science or lay expertise. Consequently, we can think of technoscience and society as being co-constructed, in that our political economies and our technoscience shape each other. Finally, there is a political economy to technoscience resulting from the decisions about what and how to research and innovate, meaning that it's critically important to examine the allocation of financial resources to technoscientific

developments and how this shapes those developments in particular ways (which may not be equitable, just, or socially beneficial).

Fears about technoscientific innovation and entrepreneurship are raised across the spectrum of policy and political perspectives. As *The Economist* highlights, for example, Big Tech's market power translates directly into significant influence over technological developments since they have become the key economic investors in research and development (R&D) *and* the focus of startups and competitors who find it more financially rewarding to be acquired by Big Tech than become their competition.[20] As a result, Big Tech has had an oversized, even overweening, role in shaping the future of our technologies for a good decade.[21] Such influence has led, in my view, to the growth in what I call *parasitic innovation*. Unlike most people who use the term 'innovation', I don't think there's any reason to assume that (technological) innovation per se is inherently good or beneficial: in fact, I have written, with colleagues, about the problem *of* innovation itself, it's dark side.[22] In this book I'm using the term 'parasitic innovation' to refer to technological developments deliberately designed to limit access to resources, to undermine regulations or social conventions, to undermine or avoid competition, to fool or scam customers, to exploit customers' psychology, to lock customers into using a product, to stop customers fixing their own property, or to use information asymmetries to treat customers inequitably. It covers quite a range of business practices, strategies, and models, many of which are directly associated with digital technologies generally and Big Tech specifically.

Rather than conflating innovation with entrepreneurship, both often considered and treated as unalloyed social goods, I think it's increasingly evident that (technoscientific) innovation is defined and driven by *rentiership*. By rentiership, I mean the techno-economic practices that underpin

[20] https://www.economist.com/briefing/2022/01/22/what-americas-largest-technology-firms-are-investing-in; and Hellman J. (2022) Big Tech's 'voracious appetite' or entrepreneurs who dream of acquisition? Regulation and the interpenetration of corporate scales, *Science as Culture* 31(1): 149–161.

[21] One example of this are the billions of dollars that Meta/Facebook has invested in the 'Metaverse', including investments in virtual, augmented, and extended reality; they have recently backpedaled to focus on artificial intelligence, https://www.cnn.com/2023/03/15/tech/meta-ai-investment-priority/index.html.

[22] Birch, K., Chiappetta, M. and Artyushina, A. (2020) The problem *of* innovation in technoscientific capitalism: Data rentiership and the policy implications of turning personal digital data into a private asset, *Policy Studies* 41(5): 468–487.

the extraction or exaction of revenues through different modes of ownership and/or control over assets or resources in the socio-natural world, constituted by different artificial or natural degrees of scarcity, quality, or productivity.[23] Innovators are searching for new ways to extend ownership and/or control rights over social and natural resources, turning them into private assets they can exact or extract an economic rent from. Here, rentiership is not a passive practice, as often portrayed in analyses of economic rents, it's an active practice pursued by innovators, especially in the digital technology sector where startups mimic past innovators who have been able to find new ways to exact new revenues or resources (e.g. collect personal data), to extract revenues by diverting them (e.g. intermediation), or to shift costs onto others (e.g. avoid labour regulations). With a colleague, we argued that Big Tech has specifically led to the emergence of several new forms of economic rent: 'enclave rents' constituted by control of digital ecosystems; 'expected monopoly rents' constituted by the performative fulfilment of future expectations; 'engagement rents' constituted by digital rankings and metrics differentiating users by their engagement with digital products and services; and 'reflexivity rents' constituted by the exploitation of ecosystem rules and conventions. These examples are less important, though, than the overall trend we can see in the economy towards finding ways to impose new costs on users, customers, and citizens, much of which has been turbo-charged by the mass collection of personal data.

Although not relating to Big Tech specifically, I'd like to illustrate what I mean by parasitic innovation and rentiership with a few examples. First, dating websites like Tinder have introduced forms of price discrimination into their payment model; they charge higher fees to older customers. An investigation found that Tinder on average, across six countries, charges

[23] With and without colleagues, I've discussed the notion of rentiership at some length across a range of publications, including: Birch, K. (2017) *A Research Agenda for Neoliberalism*, Cheltenham: Edward Elgar; Birch, K. (2020) Technoscience rent: Toward a theory of rentiership for technoscientific capitalism, *Science, Technology and Human Values* 45(1): 3–33; Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58; Birch, K., Ward, C. and Tretter, E. (2022) Introduction: New frontiers of techno-economic rentiership, *Competition and Change* 26(3–4): 407–414; and Birch, K. and Ward, C. (2023) Introduction: Critical approaches to rentiership, *Environment and Planning A*.

its customers who are 30–49 years-old about 65 percent more than 18–29 year-olds.[24] Tinder is using personal data to do this. Second, printer companies like Hewlett-Packard (HP) have created an entirely new form of economic rent: 'ink rent'. They charge extortionate amounts for ink cartridge refills and design or programme their printers to clog up or stop working if you use cheaper generic ink cartridges.[25] Again, these practices depend upon the manufacturers' ability to collect data, this time about printer usage. A final example of parasitic innovation and rentiership is the current trend in generative AI, including ChatGPT, Bing Chat, and Bard; these large language models are trained on massive datasets of copyrighted material or by scraping websites like Reddit; the AI developers have used all this information without requesting consent. Such generative AI requires a mass of data to train the models, as well as a huge amount of energy and computing capacity; whether this is worth the while will be something only hindsight can tell us in a few years.[26] Turning to Big Tech, parasitic innovation and rentiership is evident in the range of lawsuits that have been brought against them over the last few decades: for example, according to *The Information*, and as of 2023, there are at least 70 major competition investigations against Big Tech across a range of jurisdictions.[27]

When it comes to personal data, Big Tech has pursued a particular form of parasitic innovation based on the creation of the eponymous *data enclaves* of this book's title.[28] According to mainstream economists, and the policymakers who draw on their economic epistemic expertise, markets are the best social mechanism for decision-making, reflecting individual preferences that are revealed through market choices. Here, markets are imagined as competitive and spontaneously emerging from

[24] https://www.choice.com.au/consumers-and-data/data-collection-and-use/how-your-data-is-used/articles/consumers-international-tinder-investigation.

[25] https://www.telegraph.co.uk/money/consumer-affairs/hp-printers-computers-ink-cartridges-rivals/.

[26] https://www.theguardian.com/technology/2023/apr/11/techscape-zirp-tech-boom.

[27] https://www.theinformation.com/articles/apple-amazon-google-and-facebook-face-at-least-70-antitrust-probes-cases.

[28] Birch, K. (2023) There are no markets anymore: From neoliberalism to Big Tech, *State of Power 2023 Report*, The Transnational Institute (3 Feb): https://www.tni.org/en/article/there-are-no-markets-anymore.

transparent and truthful information about individual decisions and choices. Markets should, in theory, provide everyone with the signals (i.e. prices) we need to efficiently decide on what to produce, consume, and so on, even going so far as making moral choices. However, and it's a significant shift in economic understanding, markets are increasingly seen as social constructs or artefacts resulting from social relations, institutional inheritance, and epistemic claims, meaning that markets reflect often deliberate and conscious design choices to achieve specific ends. And this is done by managing the flows of information we all rely on to make decisions. As policymakers and others have adopted this market or mechanism design view, they have turned the mainstream economic idea of a market on its head.

Individuals are no longer the centre of market thinking, but rather information is: it's constructed and packaged by market designers to provide incentives for people to act in desired ways.[29] Market design specifically underpins digital economies, with Big Tech strategically working out how to monetize the very information on which markets supposedly depend to function: I would even argue that Big Tech has worked out how to capitalize transactions costs themselves. In the place of markets, Big Tech has created privately regulated pseudo-markets, comprising digital infrastructures (e.g. platforms), processes (e.g. algorithmic pricing), and inputs (e.g. personal data) designed to benefit Big Tech at everyone else's expense. The result? We now have a series of enclaves fed by our personal data: but rather than this being just an example of the monetization of attention, as many people frame it, it's useful to examine how Big Tech are able to control and manage the very information markets depend upon (e.g. who wants to buy X, what person Y would pay for Z, how many people viewed A, etc.). This information—comprising our personal data and more—is meant to be transparent and truthful to ensure 'fair' competition and broadly beneficial innovation, but it is increasingly hoarded and hidden in *data enclaves* constructed by Big Tech firms to secure their monopolistic positions.

[29] Mirowski, P. and Nik-Khah, E. (2017) The Knowledge We Lost in Information, Oxford: Oxford University Press; and Viljoen, S., Goldenfein, J. and McGuigan, L. (2021) Design choices: Mechanism design and platform capitalism, *Big Data & Society* 8(2): https://doi.org/10.1177/20539517211034312.

My goal in this book is to show that we desperately need to rethink how our personal data are collected, used, and valued (i.e. how it's generated) in our economies in order to find alternative and socially beneficial forms of data governance, or we'll end up locked into these data enclaves for the foreseeable future.

Outline of the Book

Throughout this book I'm going to provide an overview of the most pressing policy and political issues facing us, in my view, when it comes to dealing with the importance of digital data to our economies and societies. I'm specifically focusing on 'personal data', which I return to in the first chapter, but many of the issues I raise are likely relevant across different types of digital data.

In Chapter 2, I start with a discussion of how digital personal data is always a construct or artefact of the digital architectures implicated in its collection: data cannot exist without this architecture. I define this process of construction as 'techcraft', drawing on James C. Scott's work on *statecraft*, to analyse how digital technology designers and developers find ways to make personal data legible, measurable, and valuable. Unfortunately, this process obscures the fact that personal data are relational, meaning that they have effects beyond the relevant individual(s), and entail emergent properties that change data's qualities when it's combined and aggregated. I end the chapter with a discussion of the reflexive nature of data, highlighting how our framing, collection, and use of data actually changes it in unpredictable and often counter-performative ways as individuals alter their behaviours, preferences, and decisions in light of changing understandings of their 'data twins'.

In Chapters 3 and 4, I analyse the transformation of personal data into a political-economic resource, drawing on the burgeoning interest in the techno-economic configuration and valuation of data. I argue that data are not a commodity: in particular, data are not fungible since they are the artefact of often very specific and particular collection architectures (i.e. different businesses collect data in their own way). As such, data are best understood as an asset, meaning capitalizable property that can be owned/controlled and from which future benefits accrue without a necessary sale. As an asset, data underpin an increasing number of digital and algorithmic technologies and markets (e.g. online advertising, cloud

computing, ecommerce, social media, etc.). Consequently, data are valuable. Everyone agrees on this, but there is no agreement on exactly how valuable data are, or how even to go about valuing it. Current valuation methods—subjective versus objective—create different outcomes, none of which are currently being captured on balance sheets. For me, then, although data are an asset, its value is ambiguous; more importantly, because data are not accounted as an asset, there is currently no accountability for the data enclaves that control our data.

This brings me to the pivot of the book, Chapter 5. Here, I analyse the business practices, models, and strategies of Big Tech to understand how they have dominated their markets. Rather than owning personal data, these corporations have worked out ways to take control of our personal data through the mass collection and hoarding of data in enclaves, which has stymied both competition and innovation by new startups because the latter cannot afford the capital costs required to build up their own data assets. Big Tech's head start is almost unassailable. Data enclaves provide Big Tech with the means to entrench their market dominance through the creation of ecosystems that spread across diverse markets and enrol other businesses, users, consumers, developers, etc., in the success of that ecosystem: everyone becomes tied into buttressing the benefits that access to the data enclave provide (e.g. access to consumers or users). In controlling their ecosystems, Big Tech sets the rules of the (economic) game for others, becoming, in effect, a market to themselves (or what I'm going to call pseudo-market).

In the final part of the book, Chapter 6, I present the paradoxes that are becoming very evident in our data-driven economies and motivate the search for new ways to govern data—both as a political-economic object and as the focus of privacy and data protection regulation. Emerging paradoxes include: the fact that the most socially beneficial use of data entails freely or openly sharing and combining it with other data, while its economic value is defined by its enclaving, so without new regulations to force them to open up their enclaves, Big Tech will simply exploit moves towards open data; the fact that data is reflexive, meaning that people and organizations start to work out how to game the system and exploit unforeseen outcomes; and the fact that the enclaving of our personal data means that Big Tech effectively control the information which underpins markets, including our preferences, choices, decisions, actions, and so on. These paradoxes, alongside the arguments in the previous chapters, illustrate the need to rethink data governance. Markets are not working

anymore, or maybe they never did; consequently, private property rights to data are not a solution to the problems of a data-driven economy. Public or collective governance of data can lead to beneficial outcomes if we can work out how to ensure the data are not enclaved. A range of options confront us, from data co-ops and trusts to open data mandates, but we need to think carefully about how these structures fit with growing concerns about privacy.

CHAPTER 2

Data

Abstract Digital personal data are always a construct or artefact of the digital architectures implicated in its collection: data cannot exist without this architecture. Drawing on James C. Scott's work on statecraft, I define this practice of data construction as *techcraft* to help analyse how digital technology designers and developers find ways to make personal data legible, measurable, and valuable. Unfortunately, this techcraft practice obscures the fact that personal data are relational, meaning that they have impacts beyond the relevant individual(s), and entail emergent properties that change data's qualities when it's combined and aggregated. Perhaps the most important dimension of personal data is their reflexive nature, in that the framing, collection, and use of data—through techcraft practices—actually ends up changing it in unpredictable and often counter-performative ways as individuals alter their behaviours, preferences, and decisions in light of changing understandings of their 'data twins'.

Keywords Personal data · Data collection architecture · Techcraft · Reflexive data · Big Tech

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K. Birch, *Data Enclaves*,
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Introduction

With a wonderful turn of phrase, Lisa Gitelman and Virginia Jackson argue that the idea of 'raw data is an oxymoron'.[1] Here, they contrast sharply with the once ascendant notion that 'big data' will do away with the need for theory altogether because we'll simply be able to analyse 'everything, all at once', to appropriate another phrase.[2] As Gitelman and Jackson point out, however, 'data' are never "raw", simply reflecting information collected about and from the world; that is, a 'natural resource'—like oil, a metaphor to which data has been repeatedly associated—sitting out there, ready for extraction and processing. Rather, they emphasize that data are cultural artefacts of their collection, curation, storage, and analytical infrastructures. A simple way of conceptualizing this is to say that data are made, they are constructions, or effects, of particular techno-economic arrangements designed specifically to generate data (cf. collect it). And, as this would imply, these arrangements can be configured in different ways to generate different types and forms of data depending upon different logics.

A burgeoning literature has emerged in recent decades focused on the particularities and idiosyncrasies of these techno-economic arrangements, variously concerned with the digitalization, datafication, or platformization of society.[3] Much of this literature provides an important analytical contribution to understanding the digital architectures that have been rolled out in the making of digital data. Analysing how these architectures

[1] Gitelman, L. and Jackson, V. (2013) Introduction, in L. Gitelman (ed.), *"Raw Data" Is an Oxymoron*, Cambridge, MA: MIT Press; also see Hoeyer, K. (2023) *Data Paradoxes*, Cambridge, MA: MIT Press.

[2] Anderson, C. (2008) The end of theory: The data deluge makes the scientific method obsolete, *WIRED* (23 June), available at: https://www.wired.com/2008/06/pb-theory/.

[3] Examples include: van Dijck, J. (2014) Datafication, dataism and dataveillance: Big Data between scientific paradigm and ideology, *Surveillance & Society* 12(2): 197–208; Helmond, A. (2015) The platformization of the web: making web data platform ready, *Social Media + Society* 1(2): 1–11; Nieborg, D. and Poell, T. (2018) The platformization of cultural production: Theorizing the contingent cultural commodity, *New Media & Society* 20(11): 4275–4292; West, S.M. (2019) Data capitalism: Redefining the logics of surveillance and privacy, *Business & Society* 58(1): 20–41; Blanke, T. and Pybus, J. (2020) The material conditions of platforms: Monopolization through decentralization, *Social Media + Society* 6(4): https://doi.org/10.1177/2056305120971632; and Wu, A.X., Taneja, H. and Webster, J. (2021) Going with the flow: Nudging attention online, *New Media & Society* 23(10): 2979–2998.

are implicated in the construction of data is critical for any understanding of data enclaves, which I undertake in this chapter. To do this, I deploy the concept of 'techcraft' to analyse how digital technology designers, developers, and businesses find ways to construct digital data that are legible, measurable, and valuable as a political-economic object in techno-scientific capitalism. In developing these ideas, I draw on James C. Scott's work on 'seeing like a state' which helps us to examine the ways that administrative ordering, high-modernist ideology, and authoritarian states come to shape how the world is measured and made legible as a site of policy action. From Scott's perspective, it is the social facts about populations that are artefacts of measurement practices, but with the notion of techcraft it is personal data that is an artefact of particular standards, measurements, and logics build into the monetization of users and their user data—which I'll come back to more fully in the subsequent chapters. My point, like Scott, is that techcraft does not, in the end, create predictable or knowable persons; rather, techcraft turns people into 'users' who may bear no resemblance to their measurement or non-digital selves.

My goal in this chapter is to outline how 'personal data' can be thought of as a distinct political-economic object that is an artefact of its collection and use architecture. This makes it different from other related categories that might be used in data protection and privacy regulations and legal definitions: for example, 'personal information' or 'personal identifiable information'. It's also different from other characterizations of related political-economic objects like 'information' or 'knowledge' that have long been implicated in the functioning of markets (see previous chapter); however, there are overlaps between these categories that require analytical dissection. To get to this point, I start with an outline of some influential definitions of personal information and personal data, before explaining why I think it's important to frame and understand personal data in a particular way. Much of my argument revolves around the idea that 'data' are artefacts of their collection and use architectures; that is, personal data don't exist outwith the infrastructures we use to collect and use them. And this has important policy and political implications for understanding personal data as political-economic objects (e.g. assets) and for working out their value—both of which I come back to in subsequent chapters.

Defining Digital Data …

It's helpful to start by defining digital 'data', before turning to personal data.

Digital data is often used to refer to public information produced by governments (e.g. demographics), health information produced by medical institutions (e.g. health records), business or industrial information produced by businesses (e.g. marketing, industrial performance), alternative information (e.g. sentiment analysis, satellite imagery), synthetic information (e.g. algorithmically created), and personal information produced by and about identifiable individuals.[4] In this book, I focus on the last of these (i.e. personal data) and on the collection and use of personal data by businesses because I'm most interested in understanding how personal data have ended up concentrated in private, commercial enclaves and increasingly treated as a political-economic object of value.

International standards setters are currently discussing many of the difficulties in defining what we mean by 'data'; for example, this is a major point of debate amongst experts and policymakers who are trying to create a common definition in order to update national accounting standards.[5] According to these national accounting experts, digital data can be defined as "information content that is produced by collecting, recording, organising and storing observable phenomena in a digital format".[6] It's immediately obvious that this conceptualization of data entails a number of important characteristics: it has "information content" (meaning that 'information' is something different), it is "produced"

[4] For example, see Edwards, L. (2018) Data protection: Enter the general data protection regulation, in L. Edwards (ed.), *Law, Policy and the Internet*, Oxford: Hart, pp. 77–117; Prainsack, B. (2020) The value of healthcare data: To nudge, or not?, *Policy Studies* 41(5): 547–562; Hansen, K.B. and Borch, C. (2022) Alternative data and sentiment analysis: Prospecting non-standard data in machine learning-driven finance, *Big Data & Society* 9(1): https://doi.org/10.1177/20539517211070701; and Jacobsen, B. (2023) Machine learning and the politics of synthetic data, *Big Data & Society* 10(1): https://doi.org/10.1177/20539517221145372.

[5] See Birch, K. (forthcoming) Assetization as a mode of techno-economic governance: Knowledge, education, and personal data in the UN's System of National Accounts, *Economy & Society*.

[6] Quoted in OECD (2022) *Going Digital Toolkit Note: Measuring the Economic Value of Data*, Paris: Organisation for Economic Co-operation and Development; somewhat tautologically, the OECD defines "observable phenomena" as a fact or situation that can be recorded.

(meaning that it does not exist in a 'raw' form), and it is specifically "produced" through the "collecting, recording, organising and storing" of information (meaning it is an 'artefact' of a broader set of digital infrastructures). I think this policy discussion provides a useful starting point for understanding the different ways people use the term 'personal information' and how I'm trying to understand and analyse 'personal data' in this book.

When it comes to specifically 'personal' data, it's helpful to think of *personal data* as distinct from personal information, or personal identifiable information (PII). The latter has a more limited framing, referring to the 'information content' about identifiable persons: for example, name, address, social welfare records, financial records, and so on. As such, it does not include the 'user information' that is *generated* (i.e. produced) by an individual's use of digital products (e.g. smartphones, devices), services (e.g. apps, search), platforms (e.g. social media), and infrastructures (e.g. government systems). I'll come back to the importance of this difference below when discussing the 'techcraft' practices of Big Tech that configure personal data in very specific ways.

… and Defining Personal Data

For my purposes in this book, then, 'personal data' covers both *personal* information and *user* information. As such, personal data can be categorized in the following ways: (1) by type; (2) by production method; and (3) by its characteristics. First, the OECD and others provide a useful breakdown of different types of personal data, including:

- "User generated content" (e.g. images, videos, comments, likes, etc.);
- "Activity or behavioural data" (e.g. searches, purchases, ad clicks, etc.);
- "Social data" (e.g. contacts, social graph, interactions, etc.);
- "Locational data" (e.g. addresses, cellular geo-locations, IP addresses, etc.);
- "Demographic data" (e.g. age, gender, race, etc.); and,

- "Identifying data of an official nature" (e.g. name, passport or ID number, etc.).[7]

As can be seen from this list, there is significant and broad variety in personal data, not all of which is recognized by different jurisdictions (more on that below). In an interesting article about the EU's 2018 *General Data Protection Regulation* (GDPR), legal scholar Nadya Purtova makes the point that as definitions of personal data have broadened, as they have with GDPR, then data protection and privacy regulations will end up covering almost everything in our lives.[8] This is because datafication has spread throughout our everyday activities, from the behavioural data generated by our smart home appliances through the locational data generated by cars, trucks, and tractors to social data generated from social media.

Second, the OECD again provides a useful breakdown of personal data by production method. They differentiate personal data into:

- "Volunteered" by individuals willingly, although usually as a result of signing up for a product, service, online purchase, social network, etc.;
- "Observed" about individuals through recording their actions, including online browsing histories, online search requests, geolocation, cellular usage, smart device usage, etc.; and,
- "Inferred" from aggregated analysis of individuals, combining user profiles, financial decisions, personal preferences, etc.[9]

From this list, it seems evident that some personal data are more problematic than others in privacy terms, depending upon the digital

[7] OECD (2013) *Exploring the Economics of Personal Data: A Survey of Methodologies for Measuring Monetary Value*, Paris: Organisation for Economic Co-operation and Development; and Eben, M. (2018) Market definition and free online services: The prospect of personal data as price, *I/A: A Journal of Law and Policy for the Information Society* 14(2): 227–281.

[8] Purtova, N. (2018) The law of everything. Broad concept of personal data and future of EU data protection law, *Law, Innovation and Technology* 10(1): 40–81.

[9] OECD (2022) *Going Digital Toolkit Note: Measuring the Economic Value of Data*, Paris: Organisation for Economic Co-operation and Development; and WEF (2011) *Personal Data: The Emergence of a New Asset Class*. Geneva: World Economic Forum.

architectures used to generate them: for example, some are given willingly, if often without much consideration, when we sign up to services or products, but others are generated through various forms of surveillance technologies we are mostly not even aware of, like cookies and website analytics. This has led scholars like Shoshana Zuboff to posit that contemporary economies are defined by a form of 'surveillance' capitalism, in which digital technology businesses collect personal data with few limits.[10] More problematic than this, even, is the personal data that can be inferred through data analytics about individuals who have never volunteered their information and have not used surveillance technologies. As Salomé Viljoen points out, personal data are relational in that the generation of data about a group enables a business to make inferences about people who they've never actually collected data about.[11] There is, in this sense, nowhere to hide even if we're rejecting cookies left, right, and centre.

Finally, personal data can be broken down by characteristics, covering its relative identifiability with an individual, including:

- "Identifiable", relating to identified persons (e.g. national ID);
- "Anonymous" (or 'de-identified'), excluding individually identifying details from a dataset (e.g. IP addresses rather than name); and,
- "Pseudonymous", entailing the use of separate information to identify an individual (e.g. clinical health data).[12]

It's become increasingly easy, as a result of technological developments (e.g. increasing data collection, increasing sharing of datasets, etc.), to re-identify individuals.[13] Consequently, anonymous and pseudonymous data are effectively no different than identifiable data. As Lilian Edwards notes, though, there is a need conceptually and normatively to consider whether personal data can be associated with someone, which is relevant

[10] Zuboff, S. (2019) *The Age of Surveillance Capitalism*, New York: Public Affairs.

[11] Viljoen, S. (2020) Democratic data: A relational theory for data governance, *Yale Law Review*, available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3727562.

[12] Edwards, L. (2018) Data protection: Enter the general data protection regulation, in L. Edwards (ed.), *Law, Policy and the Internet*, Oxford: Hart, pp. 77–117.

[13] Rocher, L., Hendrickx, J. and de Montjoye, Y. (2019) Estimating the success of re-identifications in incomplete datasets using generative models, *Nature Communications* 10(1): 3069.

for privacy concerns, and whether 'de-identified' personal data can still be used to target someone with, for example, online advertising.[14] If personal data are a 'function' (as opposed to 'content'), which I'll come back to below, then de-identified personal data need to be treated the same as identifiable personal data.

The Economics of Personal Data

As it's become increasingly important for our economies, people have sought to understand the economics of personal data. Researchers have been writing about this for at least a couple of decades,[15] with the earlier discussions mostly focusing on the idea that specifically 'digital' personal data represents a market in privacy. Here, the value of personal data was meant to reflect how much individuals value their privacy; that is, those who value privacy more will pay more to use digital products and services that are more privacy-enhancing (and vice versa). Unfortunately, the dominance of Big Tech means there is no real market for privacy anymore since there are so few suitable competitors offering privacy enhancing products or services (e.g. ProtonMail cf. Gmail, or DuckDuckGo cf. Google Search). Hence, debates about the economics of data have shifted. Consequently, it's more useful today to analyse personal data as a political-economic object, specifically as an asset, rather than as a type of market.[16] As such, we need to think carefully about the dimensions of personal data in order to identify how it might be different from other political-economic objects.

Several researchers and others have identified these dimensions of personal data as an economic good.[17] These dimensions usually reflect

[14] Edwards, L. (2018) Data protection: Enter the general data protection regulation, in L. Edwards (ed.), *Law, Policy and the Internet*, Oxford: Hart, pp. 77–117.

[15] Spiekermann, S. and Korunovska, J. (2017) Towards a value theory for personal data, *Journal of Information Technology* 32: 62–84.

[16] Obviously, I'm not the only one to think this: there is a growing debate about how to value data as an economic object, including discussions by statistical agencies in the USA, UK, Canada, Netherlands, and elsewhere, as well as statistical offices of international organizations like the UN and OECD. See Coyle, D. and Manley, A. (2022) *What Is the Value of Data? A Review of Empirical Methods*, Bennett Institute for Public Policy, University of Cambridge.

[17] See, for example: OECD (2022) *Going Digital Toolkit Note: Measuring the Economic Value of Data*, Paris: Organisation for Economic Co-operation and Development; and

pretty mainstream economic thinking on the attributes of other goods and services that are traded in markets, including:

- Personal data are **non-rivalrous**, in that using a particular dataset does not stop others using it.
- Nevertheless, personal data can be **excludable**: collecting, holding, and using data can provide enormous economic benefits.
- There are **no property rights** to personal data per se: companies can benefit from data by selling it, selling access to it, or restricting access to it, but control rests, largely, on de facto control rights rather than de jure property rights.[18]
- Personal data has **positive externalities** when data are combined with one another; the commercial uses and benefits of personal information therefore increase as the amount collected increases.

The reason I emphasize that these attributes reflect the dominant, even mainstream, thinking about economic goods and services is that there are potential differences at play when thinking about personal data as an asset (see subsequent chapters), which complicates this framing above. For example:

- Personal data are only notionally **non-rivalrous** because data can be configured in a particular techno-economic way to make it rivalrous, in that only one person or organization can use it at once (e.g. data access licence agreements).
- As noted, personal data are **excludable** because it requires significant capital to *generate* it in aggregate to be useful through investment in data centres and other physical infrastructure.
- Although there are **no property rights** to personal data and it doesn't appear on balance sheets, the current techno-economic configuration can provide other important benefits to businesses,

Purtova, N. and van Maanen, G. (forthcoming) Data as an economic good, data as a commons, and data governance, *Law, Innovation, and Technology*.

[18] Cohen, J. (2019) *Between Truth and Power*, Oxford: Oxford University Press; although 'data' are not ownable per se, some countries and jurisdictions do allow property rights for databases since they represent a particular arrangement and structuring of data, equivalent to copyright.

especially Big Tech, like the avoidance of tax liabilities (e.g. capital gains, mergers, and acquisitions).[19]

- Personal data are **relational** in that the commercial uses of data reflect the fact that combining data generates new and unforeseen commercial opportunities and capacities; for example, predictions about behaviour are enabled by inferential analytics of large datasets of individual actions and preferences.[20]
- Personal data has **emergent properties**, not simply positive externalities: when data are combined with one another, this can generate emergent properties that are greater than the sum of their parts and can entail unexpected or unpredicted effects.[21]

Whatever its dimensions or characteristics, personal data are important for a range of businesses and business models, being an important resource (or asset) underpinning consumer products and services like online search (e.g. Google Search, Bing); social networking platforms (e.g. Facebook, YouTube, TikTok); online advertising (e.g. Google Ad Manager), especially with programmatic advertising (which I'll come back to in subsequent chapters); analytical services for businesses (e.g. Microsoft); and artificial intelligence or algorithmic products and services (e.g. ChatGPT).[22] Moreover, there is a market for personal data in which data brokers collect, curate, and sell personal data to other businesses. There have also been attempts to create alternative data markets through the establishment of consumer markets in which individuals either sell their personal data or aggregate it with other people to then sell; for

[19] See Parsons, A. (forthcoming) The shifting economic allegiance of capital gains, *Florida Tax Review* 26; U of Colorado Law Legal Studies Research Paper No. 22-19, Available at SSRN: https://ssrn.com/abstract=4152114 or http://dx.doi.org/10.2139/ssrn.4152114.

[20] Viljoen, S. (2020) Democratic data: A relational theory for data governance, *Yale Law Review*, available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3727562.

[21] Esayas, S. (2017) The idea of 'emergent properties' in data privacy: towards a holistic approach, *International Journal of Law and Information Technology* 25(2): 139–178.

[22] Laney, D. (2018) *Infonomics*, New York: Bibliomotion Inc.

example, Datacoup, Personal Black Box, and Handshake are or were businesses set to sell personal information, although this consumer market has not been successful.[23]

Contextualizing Personal Data

Terminological differences and their real-world implications reflect the policy evolution of data governance regimes; these regimes entail the constitutional underpinnings, legislation, and regulations relating to privacy and data protection established by different countries and jurisdictions. Notably, the remit of data governance means that the commercial collection, use, and value of personal data ends up overlapping with privacy and data protection concerns, which is evident when it comes to the literatures that discuss personal data (e.g. legal scholars have often led the way in debating these issues).

In my previous research with Robert Guay, we outline how the different 'constitutional settlements'—by which we mean the generally accepted political and legislative framework—of the EU and USA configure the understanding and treatment of personal data differently in these two jurisdictions.[24] From the early 2000s onwards, for example, significant differences emerged between how the EU and USA frame and regulate personal data as a result of these distinct settlements. On the one hand, the EU has gradually strengthened its data governance regime in response to their fundamental principle of a right to privacy; while, on the other hand, the USA largely abandoned attempts to regulate data collection federally and left data regulation to the courts in line with the USA's 'standing' doctrine (i.e. harm has to be caused before redress

[23] Beauvisage, T. and Mellet, K. (2020) Datassets: Assetizing and marketizing personal data, in K. Birch and F. Muniesa (eds.), *Assetization*, Cambridge, MA: MIT Press, pp. 75–95; and Helmond, A. and van der Vlist, F. (2023) Situating the marketization of data, in K. van Es and N. Verhoeff (eds.), *Situating Data*, Amsterdam: Amsterdam University Press, pp. 279–286.

[24] Guay, R. and Birch, K. (2022) A comparative analysis of data governance: Sociotechnical imaginaries of digital personal data in the USA and EU (2008–2016), *Big Data & Society* 9(2): 1–13.

can be sought).[25] Nevertheless, and despite these differences, both jurisdictions are concerned with enabling the commercial use of personal data, premised on the assumption that this will generate significant social benefits.

The data governance regimes of these two important jurisdictions differ quite significantly. In the USA, because data privacy is not treated as a human right, personal data is better understood more narrowly (and somewhat ambiguously) as 'personal information', or 'personal identifiable information' (PII), which businesses can collect if they inform users and consumers about collection. Such 'choice-and-consent' regimes have relatively few limits on data collection and use; for example, most of the limits in the USA relate to health, financial, and marketing information. Data collection and use is regulated by the Federal Trade Commission (FTC), rather than a dedicated privacy agency, reflecting fears about safeguarding consumer's personal information.[26] Businesses understand that data privacy in the USA entails ensuring data security and limiting data breaches, rather than limiting the collection and use of personal data: simply put, privacy = security.[27]

In sharp contrast, the EU has enacted a series of data protection and privacy policies culminating in the 2018 *General Data Protection Regulation* (GDPR), which built on a long history of defining personal data more broadly than the USA.[28] Based on the idea of a fundamental human right to privacy, the GDPR was designed to curtail the wholesale collection and use of personal data and has increasingly influenced other countries around the world through its impacts on multinational businesses, which have to abide by the regulation if they process EU citizen's

[25] Nissenbaum, H. (2017) Deregulating collection: Must privacy give way to use regulation? Available at SSRN: https://ssrn.com/abstract=3092282 or https://doi.org/10.2139/ssrn.3092282.

[26] Esteve, A. (2017) The business of personal data: Google, Facebook, and privacy issues in the EU and the USA, *International Data Privacy Law* 7(1): 36–47.

[27] Waldman, A.E. (2022) *Industry Unbound*, Cambridge: Cambridge University Press.

[28] Schwartz, P. and Solove, D. (2014) Reconciling personal information in the United States and European Union, *California Law Review*, available at SSRN: https://ssrn.com/abstract=2271442 or http://dx.doi.org/10.2139/ssrn.2271442.

personal data.[29] According to Article 4 of the GDPR, personal data is defined as:

> …any information relating to an identified or identifiable natural person ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person.[30]

Unlike the USA, then, personal data in the GDPR goes beyond the idea of 'personal information' (e.g. name, age, ID number, etc.) to cover what could be termed 'user data' which is only generated through the data subject's use of new digital platforms and systems (e.g. geolocation data, online identifiers, online search histories, etc.). As a legal definition, GDPR provides a broad framing of personal data that reflects the expansion of the specifically digital and mass collection, use, and exploitation of personal data, which is distinct from earlier eras. Importantly, this definition addresses the fact that contemporary digital and algorithmic technologies have enabled the massification of personal data collection, use, and commercial exploitation, combining both new techno-economic objectives (e.g. inferential analytics enabled by 'big' data) and new techno-economic structures of collection (e.g. collection of data on our online and cellular activities).

Techcraft and the Construction of Personal Data

It's important to note the longstanding, in-depth, and ongoing debate about the definition of information, knowledge, and data across several jurisdictions, as well as across various academic disciplines. I don't have the space in this book to delve into these debates here but acknowledge that how I'm going to frame 'information' and 'data' in this section—and the differences between them—contrasts with other interpretations. Generally speaking, though, I think it's useful to distinguish between

[29] Edwards, L. (2018) Data protection: Enter the general data protection regulation, in L. Edwards (ed.), *Law, Policy and the Internet*, Oxford: Hart, pp. 77–117.

[30] https://gdpr-info.eu/.

content and *function* in these definitional discussions, which can represent information and data respectively. Why I'm doing this will become more pertinent as the book progresses, but it also relates to my theoretical starting point in science and technology studies (STS), which is a field that emphasizes the need to examine the social construction of knowledge claims and technological systems. Because of this STS background, I conceptualize information as knowledge claims and data as part of a technological system (i.e. the production infrastructure and produced output): that is, without a technological system of collection, storage, curation, etc., data could not be generated and would simply not exist.

The question then becomes, how are personal data constructed? I've argued elsewhere with colleagues—and which I mentioned in the introduction to this chapter—that personal data are generated and made legible, measurable, and valuable through practices of 'techcraft'.[31] Corresponding to James C. Scott's concept of 'statecraft', colleagues and I sought to build on his idea of 'seeing like a state' to understand how to 'see like Big Tech'. To simplify his approach considerably, Scott argues that modern states deploy a range of administrative tools and techniques to make their populations measurable and then legible as sites for intervention. Consequently, social facts about national populations are artefacts of measurements processes which are themselves attempts to simplify the diversity of localized practices and make them legible as *homogeneous* practices to a centralized bureaucracy. Scott uses examples of land tenure in Soviet Russia, collectivization in Tanzania, and modern city-building in Brasilia, amongst others. He specifically emphasizes that measurement tools and techniques are not simply understood as "mere tools of observation" by state actors, but rather that they are understood as having the "power to transform the facts they take note of".[32]

In a similar fashion, we argue that techcraft entails the construction of personal data through particular standards, measurement tools and techniques, and commercial logics that simultaneously generate users (and their data) *and* the monetization of those users. Consequently, and like Scott, our aim is to highlight that techcraft does not necessarily 'observe' and record 'information' (i.e. content) about people;

[31] Birch, K., Cochrane, D.T. and Ward, C. (2021) Data as asset? The measurement, governance, and valuation of digital personal data by big tech, *Big Data & Society* 8(1): 1–15.

[32] Scott, J. (1998) *Seeing Like a State*, New Haven: Yale University Press.

rather, it generates personal data of users, reflecting their function within the techno-economic configuration rolled out by Big Tech (and others) to create a monetizable user base. Again, like Scott, this entails the transformation of the 'facts' they collect into something that is legible, measurable, and valuable to these and other digital businesses. This is evident in three ways:

- Personal data are often framed as an intangible good or asset but it's difficult to measure as such (see subsequent two chapters); Alphabet/Google's Chief Economist, Hal Varian, notes that the value of a good or asset relies upon some form of market sale, but personal data are not usually sold or licensed.[33] That means that businesses have to use other measurement tools to identify personal data as a political-economic object and as something of value: in this case, that means identifying and measuring 'users'. In his book, *Subprime Attention Crisis*, Tim Hwang argues users can be standardized and measured as "attention assets" through the deployment of techno-economic infrastructures.[34] These infrastructures include digital technologies for collecting data as well as the standards created by organizations like the Interactive Advertising Bureau: for example, the 'viewable impression' is a standard defined as 50 percent of an online ad occupying a browser's viewable space for more than one second. Here, techcraft entails the creation of such user metrics and standards to generate personal data as a political-economic object.
- Personal data need to be understood (i.e. made legible) in terms of property rights, which is a problem because information about people (i.e. personal information) are facts (e.g. names, addresses, relationships, etc.) and therefore not ownable.[35] Hence, personal data have to be *made* legible as a techno-economic object, which is not an organic or spontaneous process. This is done through the definition of 'users' and the control over access to them, which

[33] Varian, H. (2018) Artificial intelligence, economics, and industrial organization. Report, NBER Working Paper 24839, National Bureau of Economic Research.

[34] Hwang, T. (2020) *Subprime Attention Crisis*, New York: FSG Originals x Logic.

[35] Laney, D. (2018) *Infonomics*, New York: Bibliomotion.

requires the monetization of 'attention' (e.g. impressions) articulated as metrics like 'daily active users' or 'monthly active users'. As users become legible as such, their use becomes the means to understand them, and this can be augmented via the deployment of digital technology architecture that encourages particular kinds of measurable use, such as auto-play, constant scrolling, notifications, etc.[36]

- Personal data are valuable for particular reasons, once it is made measurable and legible; it is user engagement, in particular, that has value. User engagement is a specific measurement of the time, regularity, and activeness of the use of a digital product or ecosystem. Engagement transforms people and their subjectivities into a techno-economic object (an asset), being measurable and legible as something of value; for example, engagement represents the 'attention' of users (e.g. time and effort spent online), which can be monetized. However, personal data are *only* measurable and legible as user engagement (e.g. searches, views, scrolling, impressions, etc.); it does not, then, represent some form of truthful representation of an individual, their preferences, or their behaviours.

Ultimately, techcraft entails a reflexive and performative transformation of users into measurable and legible techno-economic objects, which need not change individuals nor make them any more predictable than the populations James C. Scott studies in *Seeing Like a State*. In fact, and which I'll come back to in Chapter 6, techcraft, like statecraft, can lead to a range of dysfunctional effects *as a result of* its underpinning architectures and logics.

At this point, it's probably helpful to illustrate techcraft with some examples of the digital infrastructures and architectures that underpin the generation of personal data as a measurable, legible, and valuable political-economic object. To do this, I'll go over some of the key metrics used in online advertising, drawing from TikTok's, Meta/Facebook's, and Alphabet/Google's websites, and other sources.

[36] Wu, A.X., Taneja, H. and Webster, J. (2021) Going with the flow: Nudging attention online, *New Media & Society* 23(10): 2979–2998.

- Button Click: TikTok metric that measures the number of app downloads, or phone consultations.[37]
- Cost-per-X:
  - Mille (CPM): used by most online advertising platforms like Meta/Facebook and Alphabet/Google to represent the cost per 1000 impressions.
  - Click (CPC): as above but representing cost per click on a link.
  - Unique User Accounts Reached: another TikTok metric that measures the cost to reach 1000 "unique" accounts, which is "estimated".
- Engaged-view Conversions: Alphabet/Google metric that combines measurement of user watching an ad and clicking on the ad (i.e. 'converting').[38]
- Follower Demographics: Meta/Facebook enables businesses to track audience metrics so that they can target specific demographics.[39]
- Impressions: as noted above, there are standards for what counts as an 'impression' (e.g. 50 percent visible for 1-second).
- Views: TikTok has several metrics about "views", including 2-second, 6-second, and 6-second (Focused) views; the last of these includes engagement like shares or clicks, which are split between hashtags or other add-ons.

Notably, these examples of user metrics represent only a few of the total developed by these businesses: for example, TikTok alone tracks close to 300 metrics for advertisers.[40] My point, unlike some critics (e.g. Shoshana Zuboff), is not that these user metrics are somehow scary or intrusive, invading our privacy and shaping our behaviours in underhanded or nefarious ways: rather, my point is that personal data are an artefact—that is, a *construct*—of these user metrics and the techno-economic configuration in which they operate (e.g. devices, platforms, ecosystems, standards,

[37] https://ads.tiktok.com/help/article/all-metrics?redirected=2#.

[38] https://support.google.com/google-ads/answer/2375431?hl=en&ref_topic=3119144&sjid=2053450622374324604-NA.

[39] https://blog.contentstudio.io/facebook-metrics/.

[40] See this website for other examples of metrics: https://www.adriel.com/blog/advertising-metrics-benchmarks.

etc.). And this has certain implications for how we should understand data enclaves and their implications.

So, What Is Personal Data?

Like others, I don't think that personal data are best understood as a natural resource, but nor do I think that personal data are a set of datapoints about us just lying around in our worlds waiting for businesses to collect, aggregate, and curate with the ambition of monetizing them. Something else is going on that is worth unpacking. Unfortunately, I think that a number of critical thinkers fall into the analytical trap of treating personal data this way, as something existing in the world, mostly because, in my view, a lot of critical takes of personal data build on legal and economic thinking as their starting point. Although this is a broad generalization that misses a lot of nuance in legal and economic thinking, I often find that legal and economic concepts tend to start from the assumption that 'data' are something that require intervention to ensure privacy (as a human right or not), as well as protection (against misuse), security (against breaches), and some form of property right (to ensure fairness). I've also adopted this analytical angle myself in prior writing about personal data, which reflects a desire for policy and practical relevance when it comes to data governance. One side effect of this thinking, however, is that it can lead us to think of personal data as something that exists before the point of its 'collection, storage, and curation', for want of better terminology. However, we have to unpack and complicate personal data as an analytical or empirical issue since personal data are now considered to be so foundational to our economies and societies.

Here, the legal privacy expert Elizabeth Renieris offers one option, namely that since 'data' is such an ambiguous term then it's probably better to go beyond data in our criticism, focusing instead on people and their situations.[41] While I don't necessarily disagree with this point and think that context is indeed crucial, I also think that it has the potential to miss the very real impacts and implications of personal data as a political-economic object, especially implications that result from its emergent and reflexive properties. Another legal expert, Sean Martin McDonald, provides another, intriguing take on these issues, going beyond some

[41] Renieris, E. (2023) *Beyond Data*, Cambridge, MA: MIT Press.

of the usual legal and economic assumptions about personal data.[42] He argues that personal data has two sides to it which often get conflated. One side consists of the idea that personal data are 'truthful' statements of fact (e.g. my friend's name is Dave), while on the other side is the idea that data are 'fallible' representations (e.g. I like Dave). More interesting, though, is McDonald's argument that personal data generally are 'constructed' representations defined by their context: for example, a statement or knowledge claim—from an STS perspective—can be highly consequential in one context (e.g. in court), but not in another context (e.g. in our homes). I can make a statement in my home that I cannot in a court without facing significant potential consequences (e.g. perjury). The reason this is an issue is because our digital economies are defined by the mass reuse of personal data within quite different and contrasting contexts, thereby mixing up 'truthful' and 'fallible' representations but treating them the same.

In my view, these analytical and practical difficulties with our understandings of personal data can be illuminated by taking an STS perspective: that is, examining how personal data are artefacts of digital architectures of collection, storage, and curation. Rather than understanding personal data as a series of facts or truthful statements about us (i.e. information), it's more useful to think of data as constructions that result from *and* reinforce a particular techno-economic configuration. In taking this approach, I find myself coming back to the work of the French social theorist Jean-François Lyotard, especially his book *The Postmodern Condition*.[43] Originally written as a 'report' on knowledge, Lyotard argued in this book that "Knowledge is and will be produced in order to be sold" (p. 4), meaning that it becomes a productive force (i.e. asset) and that it will become increasingly 'performative'. Here, Lyotard used a slightly different sense of performative than the one we often use theoretically today—especially in STS with the work of Michel Callon and others[44]—to mean (self-)reinforcing the system in which it originates. Performativity

[42] McDonald, S.M. (2021) *Data governance's new clothes*, Waterloo: Centre for International Governance Innovation, available at: https://www.cigionline.org/articles/data-governances-new-clothes/.

[43] Lyotard, J.-F. (1984) *The Postmodern Condition*, Manchester: University of Manchester Press.

[44] Callon, M. (1998) (ed.) *The Laws of the Markets*, London: Blackwell Publishers.

becomes the optimization of a system, meaning that truthful informational content is less important—if we can even say it exists at all—than its data function (i.e. self-reinforcing the system that produces it).

This theoretical perspective is helpful when considering three aspects of personal data that are less well-defined or discussed than other aspects. First, there is limited discussion of the emergent properties of personal data, although see the work of Samson Esayas.[45] In one paper, Esayas points out that individualistic approaches to data (e.g. personal privacy) are often inadequate because they frame data in individual terms (e.g. *this* person's data), rather than analysing the emergent properties that result from its aggregation and combination, which can be unpredictable and unexpected—as well as lucrative (e.g. opening up new and unforeseen markets like programmatic advertising). Here, data reuse—using datasets collected for one purpose for another purpose—entails a higher chance of generating these emergent properties, fuelling the further generation of personal data as businesses seek to collect as much data as possible, even though they might have limited foreknowledge about its value.[46] This, obviously, encourages the expansion of things like data collection architectures and permissive terms and conditions that enable data collectors to use personal data however they want in their development of products, services, and markets—whether or not individuals from whom the data is generated agree with the premises of said developments (e.g. facial recognition technologies developed by scraping images from social networks).

Second, the reflexivity of personal data refers back to the arguments of Lyotard on performativity about the idea that certain data are system-reinforcing (while others might not be). Thinking of personal data as 'generated' helps to explain how it reinforces the prevailing political-economic systems that generate it—as well as social and political ones.[47] Reflexivity can be understood as the way that our descriptions and explanations about the world end up changing the world, our behaviours,

[45] Esayas, S. (2017) The idea of 'emergent properties' in data privacy: Towards a holistic approach, *International Journal of Law and Information Technology* 25: 139–178.

[46] Sadowski, J. (2019) When data is capital: Datafication, accumulation, and extraction, *Big Data & Society* 9: 1–12.

[47] https://theconversation.com/personal-data-isnt-the-new-oil-its-a-way-to-manipulate-capitalism-126349.

and our attitudes in ways that can be self-reinforcing and/or contradictory. Thinking of personal data as a set of generated descriptions and explanations about individuals—rather than any essential quality of those individuals—requires us to think about how this generated personal data can change our behaviours and attitudes: for example, as data are generated, we become aware that they are generated and how they are generated, meaning that we are able to game the data generation process (see Chapter 6). An important point to emphasize from this perspective is that personal data constantly change as we adjust to other people's understanding of us, rendering already generated data out-of-date as soon as it's generated. Fears about this fluidity of data are evident in the requirements that certain digital platforms, like Facebook, place on users to be truthful about their identity: for example, Facebook's terms of service state that: "The name on your profile should be the name that your friends call you in everyday life" and "Pretending to be anything or anyone isn't allowed".[48]

And this brings me to the third point: there is a problem with the generation of 'impersonal data', especially generated by 'bots' that imitate human behaviours and attitudes. Some estimates put the proportion of online traffic generated by bots at 50 percent, meaning that half of online activity entails *impersonal* data generated to mimic or parrot humans.[49] Why people programme bots to generate these data might seem like a puzzle, until you realize (or remember) that our digital economies have been structured around the pursuit of engagement—and, in many ways, it doesn't actually matter what sort of engagement that is. Businesses selling ad inventory—the space on their website dedicated to ads—benefit from metrics showing high website traffic, or high views on ads displayed on their websites; similarly, individual influencers and content creators benefit from metrics showing that they attract large and active audiences, enabling them to attract higher ad fees.[50] Over the last few years, a whole industry has emerged to generate *impersonal* data through the creation of content or click farms. These businesses hire people to click, or comment, or like, or whatever on numerous devices they stack on shelves, all in

[48] https://www.facebook.com/help/112146705538576.

[49] https://nymag.com/intelligencer/2018/12/how-much-of-the-internet-is-fake.html.

[50] Tepper, J. and Hearn, D. (2019) *The Myth of Capitalism*, New Jersey: Wiley.

order to imitate the online behaviour of 'real' individuals.[51] Despite some people raising alarm about the consequences of this bot-driven internet,[52] there is actually no incentive to do anything about it because it could undermine the whole system.

[51] Birch, K. (2020) Automated neoliberalism? The digital organisation of markets in technoscientific capitalism, *New Formations* 100–101: 10–27.

[52] One example is Augustine Fou, who writes regularly about online ad scams: https://www.forbes.com/sites/augustinefou/?sh=42b0b2dbdb68.

CHAPTER 3

Data Assets

Abstract Digital personal data are a political-economic object: techno-scientific capitalism increasingly depends upon data as its underlying resource base. However, I argue that data are not a commodity, in that they are not fungible since they are an artefact or construction of digital collection architectures that usually have very specific and particular purposes. Different businesses construct data to serve their own business models and innovation strategies, including online advertising, cloud computing, ecommerce, platform delivery, and social media. Data are better understood as an asset, meaning capitalizable property that can be owned and/or controlled by a person or organization and from which future economic benefits accrue without a necessary sale. Assets are techno-economic configurations of legal rights, knowledge claims, management practices, and especially contractual arrangements, all of which configure data in specific ways entailing several problematic policy implications.

Keywords Personal data · Data assets · Assetization · Innovation · Platform capitalism · Contract law

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K. Birch, *Data Enclaves*,
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Introduction

I became interested in 'digital' personal data sometime in the early 2010s after I'd read a report by the World Economic Forum (WEF) titled *Personal Data: The Emergence of a New Asset Class*.[1] Funnily enough, and on the back of reading this policy report, I later bought a painting called 'Emergence of a New Asset Class' from a Toronto-based artist called Tony Taylor—see Fig. 3.1. I especially like the grinning zebra. At that time, I was already theoretically intrigued by assets and how the asset form was coming to dominate our economies and societies; something that has only become more pronounced in the twenty-first century and to which I'll return in this chapter when discussing the assetization of data. Coming back to the WEF report, it stated: "As some put it, personal data will be the new 'oil' – a valuable resource of the twenty-first century. It will emerge as a new asset class touching all aspects of society" (p. 5). Obviously, this contrasts somewhat with the main arguments I outlined in the previous chapter about personal data being 'made', rather than simply existing.

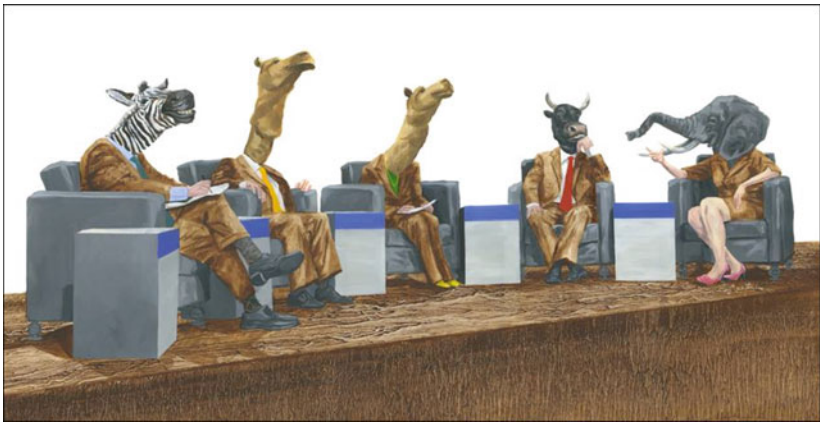

Fig. 3.1 'The Emergence of a New Asset Class' (*Credit* Tony Taylor)

[1] WEF (2011) *Personal Data: The Emergence of a New Asset Class*, Geneva: World Economic Forum.

For the World Economic Forum, their personal data report was starting point for the establishment a broader programme to pursue research into the policy importance of personal data to our economies. Notably, the WEF report and subsequent research programme seemed to have been primarily instigated by businesspeople and stakeholders in the telecommunications sector, rather than in what we'd call the digital technology sector today. Nevertheless, it was prescient in noting, in 2011, that:

> From a private sector perspective, some of the largest Internet companies such as Google, Facebook and Twitter clearly show the importance of collecting, aggregating, analysing and monetising personal data. These rapidly growing enterprises are built on the economics of personal data. (p. 7)

In 2011, the WEF stood out from many other stakeholders and policymakers with their concern with personal data. Fast forward to today, and we can see this concern has stretched far beyond one global think tank to include an array of national and supranational governments, international and intergovernmental institutions, international standards setters, multinational businesses, and advocacy or public interest groups. Recently, for example, international institutions like the International Monetary Fund (IMF) and World Bank have produced their own reports on the importance of personal data in and for our economies and societies: examples include the IMF's report *Towards a Global Approach to Data in the Digital Age* and the World Bank's report on *Data For Better Lives*.[2] There are also growing calls to develop international data governance frameworks from a range of stakeholders and policymakers.

Amidst all these clarion calls for action, there is a tension. Instead of assuming that personal data will 'emerge' organically as a political-economic object (e.g. asset), the question driving this chapter is: if personal data are an important resource or asset, then how is it *made* so? This requires a diversion into broader debates about the direction of our economies and societies, and the growing importance of the asset form

[2] IMF (2021) *Towards a Global Approach to Data in the Digital Age*, Washington DC: International Monetary Fund; and World Bank (2021) *World Development Report: Data For Better Lives*, Washington DC: World Bank Group.

in both of these,[3] before returning to the wider implications of understanding and treating personal data as an asset. While it might sound as though treating data as an asset is a problematic approach to dealing with the complexities of the digital economy, implying the further commercialization of our lives, there are actually good reasons to reach a consensus on exactly this outcome. Currently, Big Tech corporations—and other digital technology businesses—don't have to account for their personal data holdings on their balance sheets; this is a key part of what makes them data enclaves. They have little and limited accountability for what they do with personal data. Making them account for personal data, by recording it on their balance sheets, introduces an important means of societal and policy oversight: we can see what they collect, what they do with it, what its value is, how we might govern it, and how we might tax it. And this is increasingly important as our personal data are exploited to scam us or trick us.[4]

A World of Assets

The world is increasingly dominated by the asset form. Now, the world was not always this way, it's important to point out. A growing literature in science and technology studies (STS) and other academic disciplines is looking at this increasing importance of assets in our societies and how an increasing number of things are being turned into assets.[5] Generally defined as a process of 'assetization' by myself and others, this literature tries to unpack this transformation of things into assets, including personal

[3] See Birch, K. and Muniesa, F. (eds) (2020) *Assetization: Turning Things into Assets in Technoscientific Capitalism*, Cambridge MA: MIT Press.

[4] https://www.theguardian.com/technology/2023/jun/16/victims-speak-out-over-fraud-on-instagram-facebook-and-whatsapp.

[5] The literature on assetization is growing all the time, so it's difficult to provide a good selection of materials here; instead, I'll point the reader to one of my earlier books dealing with it, Birch, K. (2015) *We Have Never Been Neoliberal: A Manifesto for a Doomed Youth*, Winchester: Zer0 Books, and to a more recent review of the assetization literature, Birch, K. and Ward, C. (2022) Assetization and the 'new asset geographies', *Dialogues in Human Geography*, https://doi.org/10.1177/20438206221130807.

data.[6] As researchers, we're interested in the specifics of this transformation, covering what kinds of knowledge claims (e.g. accounting), devices (e.g. term sheets), rights (e.g. intellectual property), relations (e.g. external analysts), and organizational practices and structures (e.g. discounting) constitute something as an asset. By asset, Fabian Muniesa and myself "mean something that can be owned or controlled, traded, and capitalized as a revenue stream, often involving the valuation of discounted future earnings in the present"[7]; and we go on to argue that "the point is to get a durable economic rent from them [assets], not to sell them in the market today" (p. 2). Obviously, our conceptualization of the asset form doesn't come from nowhere; we're drawing on a range of ideas and concepts from academic literatures, policy discourses, accounting standards, media commentary, and business informants, all of which frame assets in similar fashions. However, our take on contemporary technoscientific capitalism does contrast, somewhat, with other critical takes on capitalism.

Most criticisms of capitalism, including of contemporary economies dominated by digital technologies, tend to start with the 'commodity' (or 'commodity form'), harking back (at least) to Karl Marx's writing in *Capital* in which he stated:

> The wealth of those societies in which the capitalist mode of production prevails, presents itself as "an immense accumulation of commodities," its unit being a single commodity. Our investigation must therefore begin with the analysis of a commodity.[8]

Generally, products and services are defined as commodities, in that both are seen as a form of 'economic good'. Often resources—whether produced (e.g. structures, equipment, etc.) or non-produced (e.g. land, oil, etc.)—are defined and treated as if they are commodities as well,

[6] An example of this is Beauvisage, T. and Mellet, K. (2020) Datassets: Assetizing and marketizing personal data, in K. Birch and F. Muniesa (eds), *Assetization*, Cambridge MA: MIT Press, pp. 75–95.

[7] Birch, K. and Muniesa, F. (eds) (2020) *Assetization: Turning Things into Assets in Technoscientific Capitalism*, Cambridge MA: MIT Press, p. 2.

[8] https://www.marxists.org/archive/marx/works/download/pdf/Capital-Volume-I.pdf.

although this is less clearcut and sometimes requires different terminology: for example, Karl Polanyi famously described land, labour, and money as *fictious commodities* because they were not produced specifically for capitalist exchange.[9] A key aspect to this analytical understanding of everything as a commodity is the idea that these economic goods are fungible, in that it does not matter who produces them or puts them to work: the economic good will have the same characteristics and function either way.

An immediate question emerges here: are personal data a commodity? It's often treated as such, but my argument in the last chapter—along with other people's arguments elsewhere—emphasizes that data are the consequence of the techno-economic arrangement that generate them: that is, their collection, aggregation, and curation architectures. Data enclaves themselves are constituted by distinctive digital architectures that enable them both to collect data from different people and to limit access to it. Here, then, data are defined by their non-fungibility on two levels: first, data enclaves are deliberately creating siloes of data which require dedicated and distinct architectures to use and benefit from (and which they control)[10]; and second, data are defined by the distinctions between people that mean every data point or dataset are distinct from one another. Consequently, the idea that personal data are like any other economic good becomes problematic. As highlighted in the last chapter, we might be able to conceptualize the rivalrous and excludable characteristics of data as an economic good, but it's more difficult to theorize their property relations and their emergent and reflexive aspects (e.g. how to value the fact that combining datasets leads to unknown and unpredictable outcomes, like new markets in programmatic advertising). Instead, I would argue that data are unique and increasingly so by design, and consequently that data don't really fit into the orthodox notion of the commodity form or economic good; instead, I and others think data are better understood as an 'asset' and that as assets they have

[9] Polanyi, K. (2001 [1944]) *The Great Transformation*, Boston: Beacon Press.

[10] Helmond, A. (2015) The platformization of the web: Making web data platform ready, *Social Media + Society* 1(2): 1–11.

specific characteristics that make them very interesting political-economic objects.[11]

Funnily enough, Jesus Suaste Cherizola points out that, in contrast to their critics, capitalists (i.e. businesspeople) are more interested in assets than in commodities; this is especially the case when it comes both to understanding *and* to structuring the world through the lens of a particular political-economic object, with the asset form and the investment thinking that underpins it being dominant today.[12] Cherizola argues that whole areas of economics research, like finance, have emerged to study assets and to understand the world as a portfolio of assets; that a range of news media have emerged dedicated to reporting on assets, whether financial (e.g. stocks, bonds) or non-financial (e.g. oil, intellectual property); and that a plethora of analysts, commentators, and promoters extol the virtues, economic and even moral, of thinking about the world through an asset lens (e.g. natural capital accounting solving climate change).[13]

Cherizola provides a helpful conceptual approach for differentiating assets from commodities. He notes, first, that assets are a set of rights: these include ownership rights, but also entitlements rights to the revenues generated by an asset. Entitlements rights can be separated from ownership and sold, which makes assets distinct from commodities. Second, the entitlements derived from an asset must be recognized by an authority (e.g. state), and third, these entitlements have value only as a result of their performative enforcement: that is, (1) someone claims these entitlements and (2) someone else enforces that claim. Last, assets are vendible claims, meaning that they can be traded with their prices established on the basis of an evaluation of the entitlements that an asset confers. All of these characteristics are important for any understanding of personal data as an asset, and for any examination of the data enclaves that dominate our data-driven economies.

[11] See my previous work on this: Birch, K., Cochrane, D. and Ward, C. (2021) Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech, *Big Data & Society* 8(1), https://doi.org/10.1177/20539517211017308.

[12] Cherizola, J.S. (2021) From commodities to assets: Capital as power and the ontology of finance, *Review of Capital as Power* 2(1): 1–29.

[13] An example of this is the 2021 'Dasgupta Report': HM Treasury (2021) *The Economics of Biodiversity: The Dasgupta Review*, London: HM Treasury.

Others, like myself and colleagues, have also written about what makes assets distinctive. For my part, I think there are at least seven aspects that differentiate assets from commodities *and* from the usual notion of economic goods.[14] I'll briefly sketch them out here before returning to some of them below with specific reference to personal data. First, assets are legal constructs, requiring the state enforcement of ownership or control rights and entitlements; these rights and entitlements can be separated from the 'thing' in question. Second, assets entail specific forms of ownership and/or control, defined by contractual arrangements to determine how an asset or its derivatives/copies can be used. Third, assets are often implicated in the search for economic rents, especially rents derived from their natural or constructed uniqueness. Fourth, their unique qualities mean that assets have distinct supply and demand logics; price rises may not lead to the entrance of substitutes (since there may be none) and, therefore, rising demand leads to rising prices. Fifth, an asset's value often reflects the discounting of future expected revenues and expected capital gains, which are constituted by a broader set of political-economic trends and decisions (e.g. long-term interest rates). Sixth, asset prices and capital gains are subject to the actions of owners/controllers who may seek to increase or reduce an asset's value, or to transfer it or turn it into something else; there is, in this sense, no fundamental quality or value to an asset. Last, asset values and valuation are dynamic, reflecting an ongoing set of organizational and governance practices, which include internal and external social actors.

My point here is that assets are peculiar things *and* they are reflexively understood as such by businesses, governments, individuals, and others. Before I get to the peculiarities of the assetization of personal data, however, it's worth opening the Pandora's box of debates about specifically intangible assets, since data would seem to fall into this bracket.

[14] These aspects are spelled out in: Birch, K. and Tyfield, D. (2013) Theorizing the bioeconomy: Biovalue, biocapital, bioeconomics or …what? *Science, Technology and Human Values* 38(3): 299–327; and Birch, K. (2017) Rethinking *value* in the bioeconomy: Finance, assetization and the management of value, *Science, Technology and Human Values* 42(3): 460–490.

The Intangibles Puzzle

International accounting standards are especially interesting when it comes to understanding assets and assetization. The International Accountings Standards (IAS) Board, which develops global business accounting standards for most countries, defines an asset as "a resource controlled by the entity as a result of past events and from which future economic benefits are expected to flow to the entity".[15] Assets can be both financial and non-financial, as well as tangible and intangible; both of which are relevant when considering personal data as an asset.

On the one hand, a financial asset requires that there is a counterparty with a corresponding liability (i.e. obligation from past events which entails an outflow of economic resources), while a non-financial asset does not necessitate a counterparty but might simply be recorded in terms of its nature and use on a balance sheet by a reporting entity (e.g. business).[16] On the other hand, tangible assets include 'property, plant, and equipment', representing the physical material owned or controlled by a business; in contrast, intangible assets are defined by the IAS as "identifiable non-monetary asset without physical substance".[17] Intangible assets include a range of things, but they are still characterized by identifiability/separability, control, and the generation of future benefits: examples include intellectual property (e.g. patents, copyright, trademarks), software, databases, customer lists, and licensing/royalty agreements. In thinking about personal data, it would seem to make sense to think of them as another form of intangible (and non-financial) asset, since data do not have physical form and since data do not (seem to) create a liability for a counterparty. However, it's not necessarily that simple.

Unpacking accounting concepts and definitions is important because it helps us to get at how businesses, like Big Tech, understand, frame, and treat different assets: do they understand, for example, personal data the same way as they understand intellectual property? In business accounting, certain digital resources, like databases and software, are defined as intangible assets, as are certain intellectual resources, like

[15] https://www.iasplus.com/en/standards/other/framework

[16] Chiapello, E. (2023) So what is assetization? Filling some theoretical gaps, *Dialogues in Human Geography*, https://doi.org/10.1177/20438206231157913.

[17] https://www.iasplus.com/en/standards/ias/ias38.

patents, copyright, and trade secrets.[18] Today, most people—from businesses through policymakers to academics—think and treat intangible assets as the key drivers of our economies or the main drivers of capital accumulation. Accounting academics like Baruch Lev stress the former, while critical political economists like Cedric Durand, Cecilia Rikap, and Herman M. Schwartz stress the latter.[19] It would, then, make sense to frame personal data as an intangible asset; however, if you examine the financial reports and earnings calls of Big Tech (and other digital technology businesses), then you will not find personal data defined or accounted for this way. In fact, you will find relatively little discussion of personal data as a resource, and you won't find it defined as such on any business balance sheets.[20]

Interestingly, the ambiguities with personal data reflect some of the longstanding and ongoing debates in more mainstream accounting literature about intangible assets: people like Baruch Lev, for example, have been arguing for years about the failure of accounting (and accounting standards boards) to adequately define and conceptualize intangible assets. According to Baruch, the failure to properly conceptualize intangibles has a negative impact on both business performance, because business decision-making is impacted, and broader economic performance, because macro-economic policymaking ends up unable to address the particularities of intangibles.[21] Now, with personal data, if it's not (and cannot be) treated as a distinct intangible asset, then it might be better understood as part of 'goodwill'. Goodwill is defined as an intangible asset representing everything that cannot be clearly identified and categorized during a merger or acquisition, especially assets that cannot

[18] See, for example, Haskel, J. and Westlake, S. (2018) *Capitalism without Capital*, Princeton: Princeton University Press.

[19] See: Lev, B. (2019) Ending the accounting-for-intangibles status quo, *European Accounting Review* 28(4): 713–736; and see Durand, C. and Milberg, W. (2020) Intellectual monopoly in global value chains, *Review of International Political Economy* 27(2): 404–429; Rikap, C. (2021) *Capitalism, Power and Innovation*, London: Routledge; and Schwartz, H.M. (2022) Intellectual property, technorents and the labour share of production, *Competition & Change* 26(3–4): 415–435.

[20] Birch, K., Cochrane, D., & Ward, C. (2021) Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech, *Big Data & Society* 8(1), https://doi.org/10.1177/20539517211017308.

[21] Lev, B. (2001) *Intangibles: Management, Measurement, and Reporting*, Washington DC: The Brookings Institution.

be clearly separated or distinguished from a firm. Again, however, financial reports and earnings calls don't indicate that data are considered to be goodwill, at least when it comes to Big Tech. We'll come back to some of these issues in the next chapter, as they complicate how we understand the value of personal data.

Personal Data Assetization

So, personal data appear to be neither an intangible asset nor part of goodwill from this mainstream accounting perspective. To understand personal data as an asset and the assetization of personal data, we have to look at how revenues streams are constituted and configured by the techno-economic configuration of the epistemic and political-economic boundaries that distinguish data as a political-economic object. This means focusing on the transformation mechanisms and processes involved in connecting revenue streams with the claims on those revenue streams. To illustrate this, I'm going to discuss a project I undertook with colleagues to look at the ways Big Tech transforms personal data into an asset.[22] We started with the assumption, which we thought was both methodologically and epistemically sensible, that personal data would be understood and treated as an 'intangible' asset: that is, personal data would be understood, framed, and accounted for by Big Tech and their investors as something that can sit easily on a balance sheet even if it doesn't have a physical form. As we got further in our project, however, it became evident that our starting assumption missed the point somewhat.

In examining personal data, we looked at the differences between the asset bases of Big Tech corporations (i.e. Alphabet/Google, Amazon, Apple, Meta/Facebook, and Microsoft) and other large US corporations. Generally, these corporate asset bases can be split between financial assets (e.g. cash, investments), other assets (e.g. receivables, inventory), and fixed assets (e.g. intangibles, tangibles). As evident in Fig. 3.2, there is a general tendency amongst the largest US corporations that mirrors wider claims about the increasing importance of intangible assets to businesses: tangible assets have declined from 60 percent (early 1980s) to around 30 percent (2019) of total assets, while intangible assets have

[22] Birch, K., Cochrane, D. and Ward, C. (2021) Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech, *Big Data & Society* 8(1), https://doi.org/10.1177/20539517211017308.

risen from 1 percent (early 1980s) to surpass tangibles in 2016. What is most interesting about these general trends is that they contrast quite sharply with changes in the asset bases of Big Tech, even when taking into account the differences between Big Tech corporations themselves. For example, Amazon, Alphabet/Google, and Meta/Facebook all have a growing tangible asset base, doubling their share between their IPOs and 2019. These trends are even more stark when reading through their financial reports, which show that Big Tech investment in tangible assets (e.g. land, buildings, machinery, equipment) has risen dramatically over the last decade, reaching over US$100 billion in most cases (e.g. Apple, Amazon, Alphabet/Google, and Microsoft), and certainly far ahead of spending on intangibles.[23]

So, our findings showed that Big Tech had a *lower* proportion of intangibles on their asset bases than other large US corporations, suggesting that personal data not showing up here as an asset. Rather, personal data are made legible and measurable by Big Tech as 'users' and 'user base' and defined by 'user engagement',[24] which has problematic economic,

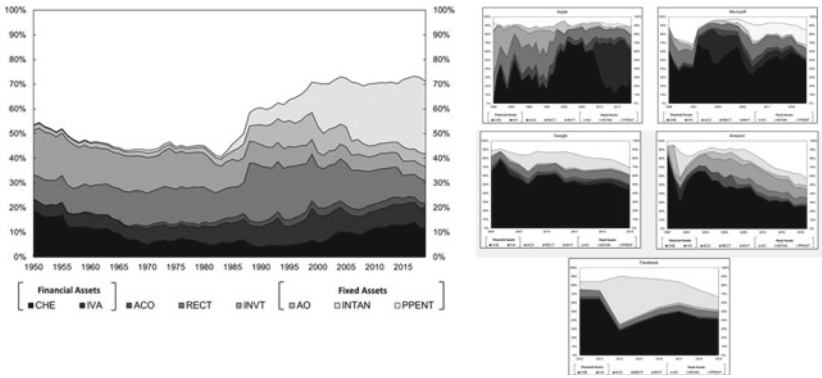

Fig. 3.2 Breakdown of total assets—Top 200 US Corporations vs. Apple, Microsoft, Google, Amazon, Facebook (*Note* Reproduced from Birch et al. [2021], note 22)

[23] Thanks to Jacob Hellman for collecting and collating this empirical material, which we haven't published yet.

[24] See also, Parsons, A. (forthcoming) The Shifting Economic Allegiance of Capital Gains, *Florida Tax Review* 26; U of Colorado Law Legal Studies Research Paper No.

political, and social implications (see below). Consequently, it's helpful to unpack the steps entailed in the assetization of personal data:

- **Measurement:** data assetization starts with the deployment of collection, aggregation, and storage metrics, standards, and technological architectures to define, measure, and delineate users and their usage.
- **Engagement:** it then proceeds through the configuration of users and their engagement within an ecosystem comprising a heterogenous assemblage of devices, platforms, developers, payment systems, etc., ('technical') and terms of service, privacy policies, standards, etc. ('socio-legal').
- **Enclave:** these technical and socio-legal dimensions of the ecosystem are defined by interoperability and contractual restrictions respectively that generate user data from users, establishing an enclave and thereby creating a resource (i.e. users and user data) that can be used for different purposes (e.g. training AI, accessing customers, data analytics).
- **Monetization:** within this enclave, revenues can be generated through different monetization mechanisms, including locking in users to products and services (e.g. Apple), offering subscription services (e.g. Microsoft), selling access to users (e.g. Alphabet/Google and Meta/Facebook), or collecting fees for use of a platform (e.g. Amazon).

Within this data assetization process, techcraft turns users and user data into political-economic objects that have value whether or not they reflect 'real-world' people and their preferences and behaviours. As Jake Goldenfein and Lee McGuigan argue, these users are resources because they are proxies for 'attention': here, attention (and use, or user engagement) can be measured and understood by digital technology businesses as something valuable.[25] Users must engage with ecosystems in particular

22–19, available at SSRN: https://ssrn.com/abstract=4152114 or http://dx.doi.org/10.2139/ssrn.4152114.

[25] Goldenfein, J. and McGuigan, L. (forthcoming) Managed Sovereigns: How Inconsistent Accounts of the Human Rationalize Platform Advertising (March 18, 2023), *Journal of Law and Political Economy*, available at SSRN: https://ssrn.com/abstract=4392875 or http://dx.doi.org/10.2139/ssrn.4392875.

ways to be counted and valued: consequently, *use* becomes a performative construction, as with Lyotard's arguments regarding system optimization, of legible and measurable users, understood by digital technology businesses in terms of metrics like 'daily active user' or 'viewable impression' or 'click-through rate' or any number of similar actions that reinforce control over access to users and user data, reinforced and augmented by digital technologies that drive or encourage users to engage more (and in particular ways) with(in) digital ecosystems.[26]

Implications of Turning Personal Data into an Asset

Data assetization leads to a number of problematic implications; many of these relate specifically to the distinctive characteristics of the asset form. Coming back to the seven aspects of assets I mentioned earlier, it's possible to analyse their theoretical implications for understanding personal data as an asset and the broader policy implications of this. Rather than run through all seven aspects, I'll focus on what I see as the most pertinent.

To start, if assets are legal constructs, then personal data are constructed through legal mechanism: as legal scholars like Lothar Determann, Julie Cohen, Katharina Pistor, and Josef Drexl point out,[27] data are not the result of property or ownership rights, but rather generated through control rights over *access* to data. Such access rights are usually enshrined in contractual agreements, both the ones that users (often unknowingly) agree to when signing up to a digital service or product and the ones that other businesses agree to when trying to access the

[26] It's important to emphasize that 'data' may not always have been or been considered an asset; Sabina Leonelli provides a short but interesting outline of the history of scientific data as it moved from being institutionalized as a commodity in the nineteenth century to becoming a reusable asset in the twenty-first century: Leonelli, S. (2019) Data—from objects to assets, *Nature* 574, 317–320.

[27] Determann, L. (2018) No one owns data, *Hastings Law Journal* 70(1): 1–43; Cohen, J. (2019) *Between Truth and Power*, Oxford: Oxford University Press; Pistor, K. (2020) Rule by data: The end of markets? *Law and Contemporary Problems* 83: 101–124; and Drexl, J. (2021) Data access as a means to promote consumer interests and public welfare—An introduction, in German Federal Ministry of Justice & Consumer Protection and MPI for Innovation & Competition (eds), *Data Access, Consumer Interests and Public Welfare*, Baden: NOMOS.

data for their purposes (e.g. to access customers). Contracts are a form of private ordering,[28] sitting outside public law, and it is through contractual arrangements that data enclaves are able to establish the (private) rules of the game. Contracts mimic property rights by conveying exclusion rights, but only between the contractual parties.[29] Two things are important here: first, and up to now, the collection, aggregation, and analysis of personal data has often been treated as a private matter between users and businesses, leaving redress for harm to the courts; and second, following on from the last point, there has been a proliferation of digital technologies implicated in the collection, aggregation, and analysis of personal data as businesses have assumed that 'notice-and-consent' is enough to do as they please. Both of these situations are changing, but only in certain jurisdictions like the EU: for example, a court case that came before the Court of Justice of the EU (CJEU) in 2022 has resulted in an opinion that digital technology businesses, Meta/Facebook in this particular case, cannot simply collect as much data as they like beyond the original consent they've received (e.g. through the collection of personal data from third-parties or apps integrated into their ecosystem).[30]

Second, if assets are defined by their contractual nature, as some of us argue,[31] then it's worth unpacking what this means when it comes to personal data. In legal terms, a contract is an agreement between individuals for *consideration*, meaning a promise of value made by both parties to the contract: notably, these promises of value need not be equitable. Access to data is only valuable through the establishment of contractual terms, specifically to do with things like how you can use something, for what time period, entailing what obligations, and whether any output is tradeable. Some things, for example, can be restricted, like the resale of a contractual term (e.g. access to data), while other things cannot (e.g. resale of data received).[32] Data enclaves control personal data through contractual terms on accessing the data they hold, which is in demand

[28] Birch, K. (2017) *A Research Agenda for Neoliberalism*, Cheltenham: Edward Elgar.

[29] Determann, L. (2018) No one owns data, *Hastings Law Journal* 70(1): 1–43.

[30] https://curia.europa.eu/juris/document/document.jsf?docid=265901&doclang=EN.

[31] Birch, K. and Muniesa, F. (eds) (2020) *Assetization: Turning Things into Assets in Technoscientific Capitalism*, Cambridge MA: MIT Press.

[32] Zech, H. (2017) Data as a tradeable commodity—Implications for contract law (September 2017), in J. Drexl (ed.), *Proceedings of the 18th EIPIN Congress: The*

because they have the capital resources to build the collection, aggregation, and curation architecture needed to collect a mass of personal data. Even though personal data has reflexive and emergent properties, the benefits that a data licensor might acquire from accessing data can be controlled via the contractual terms. The same would apply to individuals who agree to the 'collection' of their personal data in exchange for access to a product or service; access rights are determined by the data enclave. In contrast, this exchange of data for a service or product might be better thought of as an example of counter-performance because consent for data collection is not independent of the delivery of service.[33] Consequently, data can be thought of as the price users (have to) pay for the service.[34]

Last, if assets are defined by forward-looking expectations, then it's helpful to examine the particular logics that underpin these expectations. Here, of particular relevance is the idea of the investors' gaze and how investors understand assets, since assets are conceptually an investment (cf. market exchange). As Fabian Muniesa and I argue, businesspeople are increasingly expected to 'think like an investor' rather than manager: that is, to orientate themselves to a particular future and a particular point of view (i.e. that of an investor). As an asset, personal data can be considered as an investment, which entails a need to consider the implications of international investment law. In their work, legal scholars Rochelle Dreyfuss and Susy Frankel argue that intellectual property (IP) gradually shifted from being treated as an incentive to a commodity to an asset in international trade and investment law.[35] As a consequence, IP has ended up covered by international investment law regimes which entails a change in legal protection: first, it means enforcement shifts to investor-state arbitration, which can be brought by a business and is undertaken in opaque

New Data Economy between Data Ownership, Privacy and Safeguarding Competition, Cheltenham: Edward Elgar, available at SSRN: https://ssrn.com/abstract=3063153.

[33] Metzger, A. (2017) Data as Counter-Performance: What Rights and Duties do Parties Have? *JIPITEC* 8(1): 2–8.

[34] Eben, M. (2018) Market definition and free online services: The prospect of personal data as price, *I/A: A Journal of Law and Policy for the Information Society* 14(2): 227–281.

[35] Dreyfuss, R. and Frankel, S. (2015) From incentive to commodity to asset: How international law is reconceptualizing intellectual property, *Michigan Journal of International Law* 36 (4): 557–602.

investor-state dispute settlement panels; and second, it means greater protection against both direct and indirect expropriation by states. As an investment, assets have considerably more protection than commodities or tradeable goods, even to the extent that investors can bring a suit against a state when they perceive an *expected* expropriation resulting from policy changes. Assets, therefore, protect the expected future returns of investors. If personal data are considered an asset, then it would benefit from the same protections. Currently, data are in an ambiguous position, being considered as both falling within[36] and outwith[37] the remit of investment law.

To end this chapter, I'm going to outline the policy implications of data assetization, focusing on four major issues.[38] First, as assets, personal data generates revenues for their owner/controller, whoever that might be. Potential threats to those revenues face significant pushback: so, for example, government policies, like privacy and data protection regulation, are inevitably attacked for a variety of reasons and through a variety of means. We see this already with the lobbying efforts of Big Tech in jurisdictions that are trying to strengthen these policies, but we're also likely to see these businesses (and others) turn to international law as their business models are challenged. In the last paragraph, I mentioned international investment law, which I think will end up being used to limit the regulatory and policy actions of governments and undermine the political will of citizens. Second, there is no clarity, nor is any likely, about who owns personal data: does it belong to the individuals it's derived from? Or the businesses that collect and collate it? From my perspective, it makes some sense to think of personal data as belonging to individuals whose actions produce it with their online searches, viewing habits, likes and dislikes, comments, and so on. But there doesn't seem to be a way, currently, to translate this into a limit on the collection of our

[36] Bian, C. (2022) Data as asset in foreign direct investment: Is China's national data governance compatible with its international investment agreements? *Asian Journal of International Law*, https://doi.org/10.1017/S2044251322000595

[37] Horvath, E. and Klinkmuller, S. (2019) The concept of 'investment' in the digital economy: The case of social media companies, *Journal of World Investment and Trade* 20: 577–617.

[38] Birch, K., Chiappetta, M. and Artyushina, A. (2020) The problem of innovation in technoscientific capitalism: Data rentiership and the policy implications of turning personal digital data into a private asset, *Policy Studies* 41(5): 468–487.

data by digital technology businesses, especially when it comes to Big Tech. We increasingly rely upon the digital technologies they produce to live our lives, so it's difficult to abstain or boycott them, while regulatory and policy action is often stymied by lobbying and other political conflicts. Some policy changes are happening, like the introduction of the EU's suite of regulations covering digital platforms and their use of personal data, so we'll have to see how this plays out (see Conclusion). In other jurisdictions, though, there is limited political will to do anything as direct.

Third, as personal data are treated as a private asset, we end up having little control over how businesses use it, outside of the data governance regime particular to each jurisdiction. Some states, like the EU, are doing something about personal data, but others, like Canada, are dragging their feet on updating their laws and regulations, whether that's privacy or competition policy.[39] Economic control rests with businesses, especially Big Tech, which thereby derive private benefit from it, even though those economic benefits are a direct result of data's collective nature (i.e. from our unremunerated actions and behaviours). Last, then, our lack of economic control opens up space for the ascendance of *parasitic innovation* driven by new forms of *rentiership* that digital technology businesses pursue.[40] Here, rentiership includes the hoarding of personal data to create monopolies; the use of personal data to underpin the introduction of micro-transactions or predatory pricing; the exploitation of personal data to undermine competitors; and so on. Everyday examples of parasitic innovation include the extension of new control and use rights to mundane consumer products like automobiles, smartphones, printers, tractors, electric toothbrushes, and much more: these products are being turned into subscription services through techno-economic controls over functionality and usability, entailing the collection of personal data.[41]

[39] Birch, K. and Adediji, D. (2023) *Rethinking Canada's Competition Policy in the Digital Economy*, ITS Policy Report #01–2023, available at: https://www.yorku.ca/research/its/wp-content/uploads/sites/728/2023/02/ITS-Policy-Report-01-2023-Rethinking-Competition-d2-2.pdf.

[40] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58.

[41] Perzanowski, A. and Schultz, J. (2016) *The End of Ownership*, Cambridge MA: MIT Press.

Unfortunately, to put it lightly, across the full gamut of parasitic innovation, treating personal data as an asset leads to the sidelining of other policy objectives: as investments with future returns expected by investors, the performance of data assets trumps other social, political, and economic considerations, even where they may be democratically decided.

CHAPTER 4

Data Value

Abstract Digital personal data are valuable; most people agree on this. However, we do not agree on either how valuable data are *or* how to value data. I run through the various valuation methods experts, policymakers, and businesses have developed and use to assess data value, contrasting subjective and objective approaches and examining the different implications of each. Notably, data are not currently captured on balance sheets, meaning that they are missing from our accounting of business practices and valuation, which has significant political-economic and policy consequences. Without accounting for data and its value, data remains an ambiguous asset that lacks accountability for its construction, enabling certain businesses to collect and use it as they see fit with little social oversight.

Keywords Personal data · Data assets · Data value · Data valuation · Accounting · Taxation

Introduction

I read a great article earlier this year, 2023, about the online and multiplayer computer game Planet Zoo. Author Nate Crowley wrote an entertaining piece about how the design of the game economy—and there

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K. Birch, *Data Enclaves*,
https://doi.org/10.1007/978-3-031-46402-7_4

is now a growing field of actual work called 'economy design', which looks fascinating—meant that the game had descended into a hellish grind of breeding warthogs (and ostriches and Indian peafowl).[1] The game itself involves creating a zoo and trading with other players to stock that zoo with animals. But new players ended up having to spend hours and hours breeding warthogs in order to trade with other players for the chance—and it was really only a small chance—to buy any other kind of animal; and when they were able to find any other animals for sale, which was rare, those animals would be extortionately priced and often suffering one disease or another. As Crowley explained, the reason why this happened concerns the way the game's economy is designed and what this meant for the valuation of different animals: "economics has happened to Planet Zoo". The Planet Zoo economy runs on cash and 'conservation credits' (CC), the latter of which is meant to incentivize players to breed and then release endangered animals into the virtual wilds: rarer animals required significant CC to buy for your zoos, as well a cash, but new players start with limited CC. Consequently, the only way to build up CC was to grind 'low prestige' and non-endangered animals like warthogs, thereby flooding the game's economy with them and cratering their monetary *and* CC values.

As the Planet Zoo example illustrates, the 'value' of something isn't easy to ascertain and often has a rather odd life of its own. Much debated in mainstream economic and critical political-economic fields, the concept of value itself can currently be understood in at least two different ways: first, as *value* resulting from an intrinsic or fundamental characteristic, such as the work or labour put into producing something (which can be called 'objective' value); and second, as *valuation* resulting from inter-subjective transactions, such as market prices resulting from the negotiation of personal preferences and resources (which can be called 'subjective' value).[2] My aim in this chapter is not to engage in these debates, which are ongoing and won't be settled by myself, even if I tried, or by anyone else while there are such different epistemic traditions and perspectives vying for our attention. A recent book by the academic Frederick Harry Pitts, called *Value*, goes some way to outlining the range of

[1] https://www.rockpapershotgun.com/planet-zoo-is-temporarily-a-game-about-mass-producing-knackered-warthogs.

[2] For example, Mazzucato, M. (2018) *The Value of Everything: Makers and Takers in the Global Economy*, London: Allen Lane-Penguin.

these debates in a way that may be helpful to readers should they want to head into this theoretical quagmire.[3] For now, a useful starting point is to think about how value *and* valuation play out together in the digital economy and how this concerns the peculiarities of the asset form.

My intention in this chapter is to think through the value *and* valuation of personal data as an asset. This entails a different approach than an assumption that data value is constituted by the flow and flux of personal data in a naturalized market; rather, and as STS scholars and others have been pointing out for some time, value is an achievement entailing an array of social actors, technological devices, knowledge claims, metaphors, and more. Consequently, data value is highly contested: there a numerous analytical and methodological approaches to valuing personal data, including approaches that focus on organizational activities, markets, user preferences, labour or work, social benefits, and the data's attributes. In discussing this variety of approaches, my intention is to set up the final section in which I outline the role of accounting in the techno-economic configuration of personal data's value and valuation. Importantly, this analysis helps to illuminate some of the broader policy issues with data value that are now coming to a head.

Valuation: What Makes Personal Data Valuable?

Everyone agrees that personal data has value or is valuable, but there is little agreement on how we know its value or how we know how valuable it is. In part, that relates back to the fact that there are differing analytical ways of understanding value/valuation, which are often incompatible with one another. To start, though, it's probably worthwhile outlining how personal data are used in the economy. The political economist Jathan Sadowski provides a useful outline of some of the key ways that personal data are deployed today:

- Profiling and targeting people, especially through programmatic advertising (see Chapter 5): personal data are used to segment indi-

[3] Pitt, F.H. (2020) *Value*, Cambridge: Polity.

viduals into market groups, which was recently discussed in-depth in an article in *The Markup*.[4]

- Optimizing systems: personal data are used to generate efficiencies in production, logistic, retail, etc., operations.
- Managing and controlling things: personal data can be used as an input into other activities, whether that's policing (e.g. bodycams) or exercise (e.g. FitBit).
- Modelling probabilities: personal data are analysed to make inferences and predictions about behaviours, choices, and decisions, often using some rather shady and shoddy assumptions about how one factor or other shapes individual lives.
- Building new products and services: personal data are an input into the development of new products and services, like 'smart' technologies underpinning the Internet of Things.
- Growing the value of assets: personal data are useful for extending the life of existing infrastructures (e.g. buildings) or machines (e.g. automobiles) or finding new uses for those things.[5]

As these examples illustrate, personal data are being deployed in a range of political-economic activities and through a range of business and innovation strategies. It's for these reasons that digital data generally and personal data specifically have been lionized by a range of people as the foundation for our economies.

First, there are businesspeople like Douglas Laney, who works for the consultancy firm Gartner Inc. and who has been a big proponent of thinking about, and treating, data as an asset.[6] He developed the concept of 'infonomics' to get at how data are different from other assets, especially because it's not recorded on balance sheets and is difficult to value. He's interested in how new standards could be used to identify data value, especially because current accounting standards, principles, and practices don't agree that data are an asset. Second, there are technology thinkers like Jaron Lanier who see personal data as a conflict zone for winning

[4] https://themarkup.org/privacy/2023/06/08/from-heavy-purchasers-of-pregnancy-tests-to-the-depression-prone-we-found-650000-ways-advertisers-label-you.

[5] Sadowski, J. (2019) When data is capital: Datafication, accumulation, and extraction, *Big Data & Society* 9: 1–12.

[6] Laney, D. (2018) *Infonomics*, New York: Bibliomotion.

back our rights through the establishment of data ownership: according to Lanier, for example, if we own our data we can decide not only whether to let businesses use it, for a suitable price, but also which businesses we definitely don't want to use it (for whatever individual reasons we have).[7] Lanier places data value alongside data privacy, where having the capacity to own and price our personal data leads to an assertion of a wider set of rights we're currently lacking. And third, there are data experts like Viktor Mayer-Schönberger and collaborators who argue that digital data will replace money as the key transactional unit in our economies.[8] Defining price as an information proxy, they argue that digital data will replace money because it provides transactors with more information about their activities, thereby reducing costs. Although it's not well-defined how this will happen, there are already business examples of this way of thinking, including the Data Economics Company whose goal is to develop technologies that enable individuals and organizations to collect and trade their data.[9]

Now, although these ideas reflect a range of understandings of personal data and its value in our economies, they also reflect a shared positive attitude regarding the political-economic opportunities that personal data can offer. I think, though, that these understandings of what data can be and what value they can have are in some sense already too late to the game. Data assetization is already happening, and that means that personal data and its value are already understood, framed, and treated in a particular way, and this will become increasingly difficult to derail. In my view, thinking about data as an asset and about data's value in these terms helps us to get a better sense empirically of where our economies and societies are already heading and what that might mean for us.

When it comes to data assets and their value, a good starting point is to think of value/valuation in terms of the 'circuits of capital' with a dash of constructivist political economy.[10] Here, it makes sense to frame *value* as the (objective) effect of the primary circuit of capital (i.e. production) and *valuation* as an (subjective) effect of the secondary circuit (i.e. finance).

[7] Lanier, J. (2014) *Who Owns the Future*? New York: Simon & Schuster.

[8] Mayer-Schönberger, V. and Ramge, T. (2018) *Reinventing Capitalism*, New York: Basic Books.

[9] https://deco.lydion.com/home.

[10] Harvey, D. (2010) *The Enigma of Capital and the Crises of Capitalism*, Oxford: Oxford University Press.

Simplistically, value represents the objective 'limit' imposed by the wages earned by workers, which cycle back into the economy in the form of consumption, generating revenues constituted by people's consumption from their wages—and people can't spend beyond wages without turning to the secondary circuit. Valuation represents a subjective calculation of those primary circuit revenues within the secondary circuit where financial instruments are used to capitalize those revenues using different discount rates; these subjective calculations can lead to widely different multipliers of revenues as expected returns on investment depend upon an investor's risk appetite.[11] As such, an asset's valuation ends up reflecting a set of inter-subjective relations between investors, their expectations, and levels of risk usually framed by macro-economic dynamics like long-term interest rates. Between these two circuits of capital sit assets: both part of production (generating revenues) and part of finance (generating risk). As such, we can understand the value of an asset, like personal data, as a consequence of both the revenues it generates and the valuation of those expected revenues in the future, suitably discounted by their 'riskiness'. Future expectations are critical because they drive investment in the technologies that collect, aggregate, and analyse personal data, engendering different innovation strategies.

Earlier thinkers made similar analytical connections. People like Thorstein Veblen, writing at the turn of the twentieth century, argued that intangible assets specifically reflect the capitalization of the 'habits of life', including things like loyalty (e.g. customer lists), reputation (e.g. brands), social conventions (e.g. fashion), and so on.[12] Assets are, in this sense, capitalizable property because their value reflects the subjective evaluation—within a techno-economic arrangement of knowledges, devices, and practices—of the expectations and risks underlying the entitlement to future income streams (i.e. yields). Others like Jonathan Nitzan and Shimshon Bichler, building on Veblen, define this as the ownership of 'earning power'.[13] Here, Veblen specifically conceptualized an asset as denoting both ownership, entailing an entitlement to future

[11] An investor's attitude or appetite for risk is constituted by a range of macro-factors; for example, interest rates can significantly alter the valuation of an asset on the basis of the subjective valuation of the returns (which may not change in amount).

[12] Veblen, T. (1908) On the nature of capital: Investment, intangible assets, and the pecuniary magnate, *Journal of Economics* 22(4): 517–542.

[13] Nitzan, J. and Bichler, S. (2009) *Capital as Power*, London: Routledge.

income streams, and an evaluation of those entitlements, where the latter reflects a broader set of techno-economic relations. On this latter point, for example, the historian Jonathan Levy notes that there was a major rethinking of asset valuation at the end of the nineteenth century: as people debated how to value assets, they concluded that "to capitalize the expected pecuniary income streams of the newly consolidated asset" required it be "discounted against a uniform market interest rate", which was "something historically novel, being made possible by the recent geographical integration of capital markets".[14]

In this sense, there is a long history to understanding asset values as relationally constituted, reflecting a broader set of political-economic indicators rather than some notional (and suspect) tendency towards equilibrium between supply and demand. Consequently, and as I and others are increasingly highlighting with our work on assets, we have to understand assets and their value/valuation as governed and managed within a complex set of techno-economic relations, devices, and practices defined by inter-subjective and reflexive judgements. And that includes personal data.

Methods for Valuing Personal Data

Personal data have value: we can agree on that.

How we should value personal data: we probably don't agree on that.

The literature on how to value digital data, not just personal data, has grown over the last few decades on the back of my first statement but has, so far, failed to solve my second statement. Consequently, we know that personal data are valuable, but we don't know either *how* valuable it is, or *how* to value it. Lots of people have made suggestions about both of these issues. My aim in this section is to go over some of these suggestions to illustrate the different assumptions and analytical starting points underpinning the main methodological approaches people have proposed as ways to value personal data. If you are coming to this topic cold, then I'd recommend the following literature as a good starting point:

[14] Levy, J. (2017) Capital as process and the history of capitalism, *Business History Review* 91(3): 483–510.

- For some history, check out the work of Sarah Spiekermann and colleagues on personal data markets, especially the conceptualization of them as markets for privacy[15]; other useful literature on data as privacy markets include Roosendaal et al. and Magali Eben.[16]
- Also for some history but beyond the economics field this time, then it's worth reading Beauvisage and Mellet's chapter in the book Fabian Muniesa and I co-edited on assetization. In it, they provide an overview of real-world/practical attempts to value personal data, most often ending in failure.[17]
- For a business perspective, then the main person to read is probably the already-mentioned Douglas Laney, whose book *Infonomics* is all about trying to understand 'information' as an asset.[18]
- For a professional perspective, then check out Canada's Chartered Professional Accountants report on data value.[19]
- For a government view, see the 2018 report by the UK's Treasury on the 'economic value of data'.[20]
- For an international institution perspective, then the OECD has done a lot of work on the topic, most of which is worth reading not only because it's interesting but also because their work feeds into broader global governance frameworks.[21]

[15] Spiekermann, S., Acquisti, A., Bohme, R. and Hui, K.-L. (2015) The challenges of personal data markets and privacy, *Electronic Markets* 25: 161–167; and Spiekermann, S. and Korunovska, J. (2017) Towards a value theory for personal data, *Journal of Information Technology* 32: 62–84.

[16] Roosendaal, A., van Lieshout, M. and van Veenstra, A.F. (2014) *Personal Data Markets*, TNO Report (R11390): TNO Innovation for Life; and Eben, M. (2018) Market definition and free online services: The prospect of personal data as price, *I/A: A Journal of Law and Policy for the Information Society* 14(2): 227–281.

[17] Beauvisage, T. and Mellet, K. (2020) Datassets: Assetizing and marketizing personal data, in K. Birch and F. Muniesa (eds), *Assetization*, Cambridge MA: MIT Press, pp.75–95.

[18] Laney, D. (2018) *Infonomics*, New York: Bibliomotion.

[19] Girard, M., Lionais, M. and McLean, R. (2021) *What Is Your Data Worth? Insights for CPAs*, Toronto: Chartered Professional Accountants Canada.

[20] HM Treasury. (2018) *The Economic Value of Data: Discussion Paper*, London: HM Treasury.

[21] OECD. (2013) *Exploring the Economics of Personal Data: A Survey of Methodologies for Measuring Monetary Value*, Paris: Organisation for Economic Co-operation and Development; and OECD (2022) *Going Digital Toolkit Note: Measuring the Economic Value of Data*, Paris: Organisation for Economic Co-operation and Development.

- For a data science perspective, then Fleckenstein et al. provide a useful overview and presentation of their own 'dimensional' model.[22]
- And finally, Diane Coyle and colleagues work at Cambridge University's Bennett Institute for Public Policy is an amazing resource, summarizing a lot of other research that's been done on the topic.[23]

Obviously, these suggestions are a small snapshot of a complicated field, which can be overwhelming in terms of the analytical and methodological diversity on show. Across the literature, though, there is a general agreement that assessing the value of personal data—or information more generally—is difficult, and that there is no agreed upon or standardized method for valuing personal data. There are ongoing attempts to establish a standard, especially in national accounting circles, but these are unlikely to be completed for several years and, moreover, will likely only start to impact policymaking and business accounting towards the end of the 2020s.[24]

Drawing on this literature, it's possible to split data valuation approaches into six key categories. It's important to stress that most of these valuation approaches are underpinned by an assumption that personal data have an intrinsic or fundamental value we can measure and calculate if we develop a good methodological approach to do so. These key six valuation approaches include:

- Business/organizational approaches.
- Market approaches.
- User/data subject approaches.

[22] Fleckenstein, M., Obaidi, A. and Tryfona, N. (2023) A Review of data valuation approaches and building and scoring a data valuation model, *Harvard Data Science Review* 5(1): https://doi.org/10.1162/99608f92.c18db966.

[23] See, for example: Coyle, D. and Diepeveen, S. (2021) *Creating and Governing Social Value from Data*, available at SSRN: https://ssrn.com/abstract=3973034 or http://dx.doi.org/10.2139/ssrn.3973034; and Coyle, D. and Manley, A. (2022) *What is the Value of Data? A Review of Empirical Methods*, Bennett Institute for Public Policy, University of Cambridge. Further resources are available at: https://www.bennettinstitute.cam.ac.uk/blog/value-data/.

[24] Birch, K. (forthcoming) Assetization as a mode of techno-economic governance: Knowledge, education, and personal data in the UN's System of National Accounts, *Economy & Society*.

- Labour approaches.
- Social benefits approaches.
- 'Dimensional' approaches

There are other ways to differentiate between data valuation approaches, such as distinguishing between 'stated preferences' (e.g. surveys) and 'revealed preferences' (e.g. prices, auctions),[25] but these draw more specifically on assumptions derived from mainstream economics, which miss some of the analytical and empirical peculiarities that make personal data so interesting.[26]

Starting with the business/organizational approaches, these are premised on calculating data value by analysing (1) *business performance*—which could refer to a business' asset base, market capitalization, and revenues—or (2) *business costs*—which could mean production/replacement costs, damages, relief of royalty, and cost of revenues. As discussed in the previous chapter, we could try to work out the value of personal data by examining the *recorded assets* of a business, like the intangible assets on its balance sheet. But, as also mentioned in the last chapter, this might not be possible because personal data are not currently recorded on balance sheets.[27] The reason this is the case is because personal data are not recognized as an asset by either the International Financial Reporting Standards (IFRS), established by the International Accounting Standards Board (IASB), nor the US Generally Accepted Accounting Principles (GAAP), established by the Financial Accounting Standards Board (FASB)—while most countries follow the IFRS, US corporations follow GAAP. One way to get around this accounting omission is to use *market capitalization* (or private valuation) as a means to calculate data value, reflecting market sentiment about data value or what investors are willing to pay for it. As colleagues and I note in previous research, market sentiment (e.g. net present value) reflects the financial

[25] Coyle, D. and Manley, A. (2022) *What is the Value of Data? A Review of Empirical Methods*, Bennett Institute for Public Policy, University of Cambridge.

[26] Drexl, J. (2021) Data access as a means to promote consumer interests and public welfare—An introduction, in German Federal Ministry of Justice & Consumer Protection and MPI for Innovation & Competition (eds), *Data Access, Consumer Interests and Public Welfare*, Baden: NOMOS.

[27] Xiong, F., Xie, M., Zhao, L., Li, C. and Fan, X. (2022) Recognition and evaluation of data as intangible assets, *SAGE Open*: https://doi.org/10.1177/21582440221094600.

expectations of future benefits (e.g. revenues) derived from a business' assets.[28] However, the problem with market sentiment is that it's subjective and can change rapidly and often because of external factors (e.g. interest rates) that have no relation to personal data per se; as such, market capitalization really reflects the expectations and perceptions of a limited number of investors more than anything else.

The final business performance indicator is *revenues*: this is quite a popular method to measure data value, being used frequently to calculate 'average revenues per user', or ARPU. This measure makes some sense as it reflects the importance of 'users' to businesses and can be applied relatively easily to single product or service businesses, especially social media platforms (e.g. Instagram, TikTok, LinkedIn, etc.). It's less useful, however, with businesses with combined product lines, covering a range of different market segments; for example, Alphabet/Google operates across quite diverse markets, like social media with YouTube, online search, and online advertising. Consequently, if you don't have precise and disaggregated information on user numbers and product revenues, it's difficult to calculate ARPU. Despite the limitations across these methods, researchers have still tried to calculate the data value of various firms. For example, Feijoo et al. sought to calculate the data value of businesses like Facebook, LinkedIn, Xing, Experian, and Google by analysing market capitalization, revenues, and net income per 'data record', leading to a very diverse range of values—the authors acknowledge the significant methodological difficulties in their attempt.[29]

Another business/organizational approach entails analysing the costs incurred in the collection or generation of data, and then using these costs as proxies for data value, assuming a more historical accounting approach. One method is to work out the costs of *producing or replacing* personal data held by a business, assuming that these costs reflect its value to the business. Although a relatively coherent method, in that costs represent a useful measure of data value, researchers note that it reflects the 'lower bounds' of valuation; that is, it reflects the historical costs rather than

[28] Birch, K., Cochrane, D., & Ward, C. (2021) Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech, *Big Data & Society* 8(1): https://doi.org/10.1177/20539517211017308.

[29] Feijóo, C., Gómez-Barroso, J.L. and Voigt, P. (2014) Exploring the economic value of personal information from firms' financial statements, *International Journal of Information Management* 34(2): 248–256.

a measure of current market or fair value. In their analysis, Coyle and Manley suggest that historical costs could be updated over time with fair value costing estimates,[30] but identifying the future benefits derived from the emergent properties of personal data would still be hard to calculate. Other cost approaches entail using substitutes as proxies for value: for example, legal *damages* incurred by businesses in light of data breaches can be seen as reflecting the costs experienced by users/customers who've suffered those breaches; and a *relief of royalty* method could be used to calculate the costs a business would incur if it had to license in data, whose value could be derived from data markets (see below).[31] A final cost approach uses the *cost of revenues* to calculate the costs incurred by a business to generate their revenues, including investments in the collection or generation of personal data. An example of this is Alphabet/Google's calculation of 'traffic acquisition costs' (or TAC), which they have highlighted in their financial reports since 2004 and discuss in their earnings calls. TAC represents the costs Alphabet/Google pays to attract users to their ecosystem where they can be monetized through advertising; consequently, TAC is measured as both an absolute cost and as a percentage of advertising revenues. Analytically, these costs of revenues reflect the value of data as a resource/asset in the delivery of a service or product.

Market approaches to data value reflect a different epistemic starting position, namely that data value can be determined via prices revealed in market exchange. Consequently, market approaches entail a valuation through market exchange, whether through a sale in legal or illegal *data markets*. While legal 'data markets' do exist, especially when it comes to trade in personal data by data brokers like Acxiom, Experian, and Equifax, the price set by these data markets may not accurately reflect the value of personal data. For one, if the market price reflects the price for an individual dataset or data point, then it's likely to undervalue data which has value in aggregate and as a result of its emergent properties. There have been several attempts to work out the price of an individual data point, such as someone's name or favourite colour or whatever, but in most of these cases the market price for single data points is quite varied (see Fig. 4.1). The difficulties in identifying a market price for an individual

[30] See Coyle, D. and Manley, A. (2022) *What is the Value of Data? A Review of Empirical Methods*, Bennett Institute for Public Policy, University of Cambridge.

[31] Girard, M., Lionais, M. and McLean, R. (2021) *What Is Your Data Worth? Insights for CPAs*, Toronto: Chartered Professional Accountants Canada.

data point means that the value of a single data point can be quite low. For example, the *Financial Times* newspaper created a website for people to calculate how much their personal data are worth and the *FT* argue it's basically worth a pittance: "General information about a person, such as their age, gender and location is worth a mere $0.0005 per person, or $0.50 per 1000 people".[32] Furthermore, the market price of personal data may be different depending on whose selling and whose buying, disrupting notions of market fungibility (see Chapter 3); for example, the price on data markets generally reflects business-to-business transactions rather than user-to-business transactions, which are seen as particularly important for individuals to assert control over their personal data.[33] Similarly, the value of personal data may be higher in illegal markets where credit card details can cost US$1 top US$30 per record.[34]

Now, the obvious limitation of market approaches is that personal data are rarely sold explicitly in markets and that it's been legally difficult to identify its value. Key reasons for this go back to the notion that personal data are non-rivalrous (see Chapter 3)—that is, selling it to A does not limit its sale to B, C, D, and so on—and lack property rights, relying more on contractual arrangements.[35] Consequently, it's not possible to identify personal data as an alienable commodity; this raises some interesting legal implications, at least in the USA. For example, the judgement

INFORMATION	VALUATION
Age	US$57
Weight	US$74
Location / Geolocation	€17 to €588
Favourite colour or year of birth	€1
Browsing/search history	€2 to €7
Social network interactions	€12

Fig. 4.1 Personal data valuation (*Note* Data from Beauvisage and Mellet [2020], note 17)

[32] https://ig.ft.com/how-much-is-your-personal-data-worth/.

[33] Posner, E. and Weyl, E. (2019) *Radical Markets*, Princeton: Princeton University Press.

[34] Roosendaal, A., van Lieshout, M. and van Veenstra, A.F. (2014) *Personal Data Markets*, TNO Report (R11390): TNO Innovation for Life.

[35] Cohen, J. (2019) *Between Truth and Power*, Oxford: Oxford University Press.

in the 2014 *Operman vs. Path Inc* case stated that the "Court does not read these decisions to be holding that consumers do not have property rights in their electronically stored private information", but rather "that the copying of such information without any meaningful economic injury to consumers is insufficient to establish standing on that basis". As such, American courts have decided there are no property rights to personal data that limit its collection and, moreover, that there is no 'standing' (i.e. injury) to individuals from data collection when someone else collecting your personal data (e.g. Big Tech) does not per se stop you from selling your own data—also evident in 2016's S*venson v. Google Inc.* (WL 8943301—2016). Cases like *Svenson* and *In re Google, Inc.* (Privacy Policy Litig, 2015 WL 4317479—2015) conclude that Big Tech's collection of personal data does not impinge on a plaintiff's ability to monetize their own personal data; hence, it does not cause harm. Interestingly, courts have required plaintiffs to identify a market for their personal data, which plaintiffs often struggle to do: the reason why is that personal data do not neatly fit into a fungible commodity market.

All of this becomes more relevant when considering user/data subject approaches. These are more subjective, reflecting how individuals view the value of their personal data, and concern the stated preferences of individuals, rather than 'revealed' preferences that is supposed to result from market exchange. There are two main approaches here: *willingness-to-pay* (WTP) and *willingness-to-accept* (WTA). With WTP, there is an assumption that individuals will pay to protect their privacy and information against disclosure: methods generally entail surveying people about how much they'd pay for their privacy.[36] For example, survey respondents might be asked how much they'd be willing to pay Facebook or Instagram or TikTok *not* to collect their personal data, in contrast to getting it (seemingly) for free—but really having to hand over their personal data. Similar methods are used with WTA, but this time survey respondents are asked how much money they'd be willing to accept for their personal data (if someone was paying them). Generally, researchers find that these surveys show that people tend to expect a higher payment when asked to sell their data than when asked to pay for their privacy.[37] While both

[36] Brynjolfsson, E. and Collis, A. (2019) How should we measure the digital economy? *Harvard Business Review*, Nov–Dec.

[37] There is a debate about the so-called "privacy paradox" concerning people's desire for privacy but willingness to let businesses collect their data in exchange for access to

are useful approaches for understanding subjective viewpoints on personal data, they are also both based on stated preferences meaning that they don't provide much help in understanding the idea of data value at an intrinsic or fundamental level.

The three other valuation approaches I've mentioned above are all worth discussing, but perhaps in less detail.

- **Social Benefits:** personal data has more than an economic value, it can also have significant social benefits (e.g. improving transport and healthcare or stimulating innovation). It's not easy to work out the social value of data, and having been asked to do so myself, I can say I floundered somewhat; nevertheless, it's an important issue because of the opportunities presented by opening up public data (e.g. weather, transit, medical, taxation, etc.) to wider analysis.[38]
- **Labour:** there is considerable interest in thinking about data value in terms of the labour or effort that's gone into their production. Personal data are produced, especially when it comes to the digital architectures used to generate user, social, locational, etc., data (see Chapter 2). A lot of personal data actually requires considerable effort on our part, as producers of that data: for example, we use online search to make enquiries (which feed into improving the product); we click on links to view websites (which generates attention that can be sold); we view content on social media (which, again, generates attention); we review things, we comments on things, we write blogs, we generate all sorts of material (which reveals details about our lives that can be monetized); we buy things online (which provides useful customer data); and we do lots more. According to Posner and Weyl, we are 'data producers' whose 'data work' is largely taken for granted and unremunerated: they

free goods and services; others point out that this paradox isn't really a paradox after all since it ignores that fact that most people cannot simply not use the products and services that collect our data because our lives are so wrapped up in them.

[38] Prainsack, B. (2019) Logged out: Ownership, exclusion and public value in the digital data and information commons, *Big Data & Society*: https://doi.org/10.1177/2053951719829773; Coyle, D. and Manley, A. (2021) *Potential Social Value from Data: An Application of Discrete Choice Analysis*, Bennett Institute for Public Policy, University of Cambridge; and Coyle, D. and Diepeveen, S. (2021) *Creating and Governing Social Value from Data*, available at SSRN: https://ssrn.com/abstract=3973034 or http://dx.doi.org/10.2139/ssrn.3973034.

specifically argue that "*data* about users were the central assets for technology giants".[39] Their general point is that we need to treat data as labour, and pay people to produce it, because otherwise there is no incentive for users/consumers to actually contribute their much-needed data.

- **Dimensional:** Fleckenstein et al. provide a useful outline of the dimensional approach.[40] They discuss how different people have tried to create a model to value the different dimensions of specific datasets, including their quality, timeliness, relevance, context, and so on. Focusing on the attributes of datasets and the context of their use highlights the fact that datasets are non-fungible. While detailed in its analysis, this rather technical approach suffers from the lack of accepted or standardized accounting methods for assessing the different qualities of datasets, meaning that it would be based on potentially idiosyncratic criteria.[41]

Having outlined these six approaches to valuing personal data, it's probably worthwhile turning to a discussion of why all of this matters and what it means for the governance of personal data. To do that requires delving into the intricacies of accounting.

Accounting and Accountability: The Importance of Working Out Data's Value

Personally, I think that the greatest trick accountants ever pulled was convincing the world accounting is super boring; they're not quite like the devil, but not far behind! I'm being deliberately cheeky, here, because I've very much come to appreciate accounting as a discipline and as an ordering mechanism for the world: for me, it's a really good example of the performativity thesis put forward by Jean-Francois Lyotard and taken up later by others, although in different ways. And as the sociologist Jens Beckert points out, accounting is constituted by diverse knowledges

[39] Posner, E. and Weyl, E. (2019) *Radical Markets*, Princeton: Princeton University Press, p. 213.

[40] Fleckenstein, M., Obaidi, A. and Tryfona, N. (2023) A review of data valuation approaches and building and scoring a data valuation model, *Harvard Data Science Review* 5(1): https://doi.org/10.1162/99608f92.c18db966.

[41] Laney, D. (2018) *Infonomics*, New York: Biblomotion Inc.

(e.g. demand, supply, costs, etc.), laws, regulations, and techno-economic arrangements: it's not simply a set of neutral or objective practices that can be deployed to understand businesses.[42] We can see much of this already in discussions of accounting as a governance technology, going back several decades in the Foucauldian *governmentality* literature (and elsewhere): businesspeople have to be taught to think in a certain way, usually as a result of their business education and training.[43]

Business education has changed quite significantly over time, especially with the ascendance of so-called 'neoliberal' ideas from the 1970s onwards.[44] Several academics, though, argue that the growth of financial economics in the 1960s and afterwards is specifically associated with the rise of an 'investor' mentality—strongly implicated in the rise of the asset form—and the training of businesspeople to think as an investor, rather than manager. The sociologist Richard Whitley, however, argues that the rise of financial economics, and specifically Eugene Fama's amusingly named 'efficient market hypothesis', actually "says remarkably little about market valuation processes".[45] Nevertheless, changes in financial economics feed into changes in accounting, especially the emergence of 'fair value' accounting as an alternative to 'historical cost' accounting: the latter is based on valuing things at the cost you paid for them, while the former is based on valuing things at the cost you can currently get them for in a market. Fair value accounting is supposed to be a better way to value assets because it better reflects the 'actual' value of an asset, which may have gone up since purchase. However, the argument that fair value accounting is more 'market-based' are not as clearcut as their proponents might want us to believe. As Michael Power notes, the supposed market valuation of assets on which fair value accounting is premised do not necessarily entail an actual market valuation; rather, it entails an

[42] Beckert, J. (2016) *Imagined Futures*, Cambridge MA: Harvard University Press, p. 133.

[43] Miller, P. and Rose, N. (1990) Governing economic life, *Economy & Society* 19(1): 1–31.

[44] For this discussion here, I draw on my previous work: Birch, K. (2016) Financial economics and business schools: Legitimating corporate monopoly, reproducing neoliberalism?, in S. Springer, K. Birch and J. MacLeavy (eds), *The Handbook of Neoliberalism*, London: Routledge, pp. 320–330.

[45] Whitley, R. (1986) The Transformation of Business Finance into Financial Economics: The role of Academic Expansion and Changes in US Capital Markets, *Accounting, Organizations and Society* 11: 171–92; quote taken from pp. 175-6.

expert value judgement, thereby "shifting the focus from transactions to economic valuation methods", which comes to "embed further the principle of fair value accounting as the '*mirror*' of the market".[46] Bignon et al. make a similar point about fair value accounting, arguing that it's "based more on the estimates of certified experts than on the current market price".[47] Consequently, others argue, "Values do not passively reflect the 'objectivity' of the market, but are the product of a measurement technology that, by demarcating and measuring resources, assists in the construction of the marketability of assets".[48]

What does all this mean for personal data, especially when it's understood as an asset? Well, the value and valuation of data assets are not derived from market transactions—such as sale of data, which I've already noted is rare for most digital technology businesses[49]—and instead reflect expert evaluations and investor sentiment; consequently, value becomes an assessment (i.e. valuation) of future expectations. There is significant room for public corporations, even as the businesses with the most public oversight, to report their financial accounts in particular ways, at least in the USA where the key Big Tech firms I'm primarily concerned with in this book are based. There are obviously established accounting rules, but when it comes to intangible assets with indefinite lifespans like personal data, which can be reused long after its collection, then business executives and managers have the latitude to assess the impairment of those intangible assets and goodwill each year, relying upon managerial judgement to do so. Apple provides a good example of this relative freedom. In their 2017 summary of their accounting policies, they note:

> The Company does not amortize goodwill and intangible assets [amortization is depreciation for intangible assets] with indefinite useful lives;

[46] Power, M. (2010) Fair value accounting, financial economics and the transformation of reliability, *Accounting and Business Research*, 40: 197–210; my emphasis, using quotes from p. 201 and p. 205.

[47] Bignon, V., Biondi, Y. and Ragot, X. (2009) *An Economic Analysis of Fair Value*, Cournot Centre for Economic Studies, Prisme No. 15.

[48] Napier, C. and Power, M. (1992) Professional research, lobbying and intangibles: A review essay, *Accounting and Business Research* 23: 85–95; quote at p. 87.

[49] For example, Alphabet/Google state quite clearly that they do not sell personal information, including having a "security and privacy principle" on their website to "Never sell our users' personal information to anyone"; this means that market-based valuation is not suitable for understanding their data's value.

> rather, such assets are required to be tested for impairment at least annually or sooner if events or changes in circumstances indicate that the assets may be impaired.[50]

A year later, however, Apple stopped reporting their intangible assets and goodwill on their balance sheet altogether. It's no wonder that accountants are raising concerns about the valuation of intangibles, like data assets[51]: we've ended up in an economy where the 'book value' (i.e. their assets' value) of the dominant businesses, specifically Big Tech, can reflect less than 10 percent of those firms' market valuations (i.e. investor sentiment).[52]

While all of this might seem like an esoteric argument about accounting practices and the urgent need to update them, there are a set of important practical and policy implication here that we need to consider when it comes to data assets and how to account for their value. An obvious policy implication, mentioned by Mazzucato et al. amongst others, is that it becomes very difficult to oversee and regulate competition in markets where there is no information on the key assets that businesses use to make revenues.[53] There are a range of other policy implications too.

First, businesses and their investors do make value/valuation judgements about data assets, it's just that we don't get to see them: they're not necessarily publicly available or accessible. Hence why the measurement of data value often ends up being an artefact of market capitalization, with the gap between recorded assets and capitalization used to represent the (probable) value of data. Unfortunately, this is tautological because market capitalization's difference from recorded assets ends up being used as evidence of the value of personal data while the value of personal data

[50] https://www.sec.gov/Archives/edgar/data/320193/000032019317000070/R9.htm.

[51] Meredith, P. (2022) *Accounting for the Digital Economy: Time for a Fresh Approach*, CIGI Papers No. 262, Waterloo: Centre for International Governance Innovation.

[52] There is the possibility that having to include data as an asset on their balance sheets would damage corporate return on equity (ROE) metrics, negatively affecting Big Tech corporations' ROE: see Feijóo, C., Gómez-Barroso, J.L. and Voigt, P. (2014) Exploring the economic value of personal information from firms' financial statements, *International Journal of Information Management* 34(2): 248–256.

[53] Mazzucato, M., Strauss, I., O'Reilly, T. and Ryan-Collins, J. (2023) Regulating Big Tech: The role of enhanced disclosures, *Oxford Review of Economic Policy* 39(1): 47–69.

is evidenced by the gap between capitalization and recorded asset values. We therefore get no sense of how businesses or investors put a valuation on data assets; that is, *how* personal data are being valued. There are ways to get at this information, but it's difficult, opaque, and might even be hidden. It's possible, for example, to examine different 'moments of valuation' that a typical digital technology business passes through, each of which provides an opportunity to see how they and their (potential or real) investors understand data value, including:

- **Initial public offerings**, when businesses have to establish their value: a startup needs to attract investment by detailing its assets and potential future performance.
- **Mergers and acquisitions**, when businesses have to produce statements to justify their decisions; in the USA, these are called 'purchase price allocations'. These statements distinguish between at least seven asset classes, including intangible assets; unfortunately, in the USA these statements are tax documents submitted to the Internal Revenue Service and are not publicly available.
- **Debt collateral**, when businesses seek lending: when a business wants to borrow money, it needs to put up collateral and data has been used to do this (as discussed in the Introduction).
- **Compensation**, especially for data breaches: legal cases need to establish compensation for harm caused by a data breach. It's unclear whether such cases have happened yet, rather than being settled, when it comes to personal data, so it's worth further study.
- **Bankruptcy proceedings**: the final moment is when a business goes into bankruptcy and their assets are assessed; whether this happened with personal data needs further study.

All of these 'moments' reflect empirical examples of when businesses and their investors have to make valuation judgements about data value that go beyond analytical models or methodological propositions discussed earlier in this chapter.

Second, the lack of accounting principles and regulations for personal data and its value—that is, its absence from financial records and reports—means that it's difficult to make taxation assessments of many digital technology businesses, including Big Tech. Expenses of collecting data can be deducted as costs, reducing taxable income, but the taxation

of 'data-derived income' is an increasingly important and tangled international issue. As the Canadian Chartered Professional Accountants association points out,[54] large digital technology businesses can and do transfer their data portfolios to lower-tax jurisdictions, following practices established previously with intangible assets like intellectual property.[55] Working out the fair value of these data is difficult, but several jurisdictions seem keen on working out how to do so: for example, the US state Maryland introduced a Digital Ad Tax in 2021 as one way to tax data-driven businesses.[56] It appears that other US states are considering similar measures. As a consequence of the peculiarities of data assets (e.g. global collection, reuse, emergent properties, etc.), the tax law scholar Amanda Parsons argues that:

> The work of digital laborers [basically, all of us who generate personal data] has been discussed in tax academia and policy as part of conversations on the broader crisis of multinational companies, particularly digital companies, being able to conduct substantial economic activities in countries without paying taxes there.[57]

In subsequent work, Parsons also highlights the disparities in taxing capital gains when it comes to data-driven businesses; generally, investors are taxed in their own countries rather than where the businesses they invest in operate, meaning that any capital gains accrued from an investment because of its rising value of personal data are not taxed where that data comes from.[58] And since personal data are non-fungible, Parsons

[54] Girard, M., Lionais, M. and McLean, R. (2021) *What Is Your Data Worth? Insights for CPAs*, Toronto: Chartered Professional Accountants Canada.

[55] Bryan, D., Rafferty, M. and Wigan, D. (2017) Capital unchained: Finance, intangible assets and the double life of capital in the offshore world, *Review of International Political Economy* 24(1): 56–86.

[56] https://news.bloombergtax.com/tax-insights-and-commentary/marylands-digital-ad-tax-ruling-leaves-no-time-for-indecision.

[57] Parsons, A. (2021) Tax's digital labor dilemma, *Duke Law Journal* 71, Available at SSRN: https://ssrn.com/abstract=3902852 or http://dx.doi.org/10.2139/ssrn.3902852.

[58] Parsons, A. (2022) The shifting economic allegiance of capital gains, *Florida Tax Review* 26, and U of Colorado Law Legal Studies Research Paper No. 22–19, Available at SSRN: https://ssrn.com/abstract=4152114 or http://dx.doi.org/10.2139/ssrn.4152114.

argues that this creates inequities in the tax treatment of data: your data may be more valuable than my data because of where you live. Again, this might seem like an esoteric accounting issue, but it has direct implications for our economies and societies. For example, Douglas Laney explains that businesses that don't capitalize their intangible assets, like personal data, on balance sheets have an advantage over other businesses with more 'traditional asset' bases: that's because data's value/valuation ends up determined by future expectations and investor sentiment as opposed to some notion of fundamental or intrinsic value.

Last, but perhaps most important, these issues with data value are not really on the radar of major accounting standards bodies like the US Financial Accounting Standards Board (FASB), which currently doesn't seem to have any plans for working out how to account for personal data. Consequently, the framing and treatment of data assets and their value are not accountable to any transparent set of standards or conventions, requiring businesses to not only disclose their data assets but also work out the data value. The latter is important for tax reasons, as mentioned, but the former is as important because it will enable different countries and citizens to determine what personal data they want Big Tech and other digital technology businesses to collect or to hold. For me, understanding data as an asset means understanding that data has a value determined by specific valuation practices; however, it's not formally accounted as an asset at present and, therefore, the data enclaves that hold it cannot be held accountable—socially as well as economically. Furthermore, since data value is currently opaque, it's largely constituted by a set of future expectations that end up driving broader techno-economic trends in potentially problematic ways, such as stimulating investment in digital products and services that are parasitic upon societal institutions (e.g. regulations, labour protections and benefits, pensions, etc.), thereby undermining those societal institutions rather than supporting them through economic growth.

CHAPTER 5

Data Enclaves

Abstract Digital personal data are increasingly hoarded in data enclaves controlled by Big Tech. These large digital technology businesses dominate their markets through the pursuit of a particular form of *parasitic innovation*. They are able to create data enclaves by extending their digital ecosystems comprising an assemblage of technical devices, platforms, users, developers, and payment systems as well as legal contracts, rights, claims, and standards. Data enclaves provide Big Tech with the means to entrench their market dominance through the enrolment of other businesses, users, consumers, developers, and so on in the success of Big Tech's ecosystems: everyone becomes tied into buttressing the benefits that access to the data enclave provide. Here, I use the notion of parasitic innovation to define the strategic attempt to dominate markets, avoid competition, and undermine competitors, leading to the erosion of markets and their replacement by ecosystems whose rules are set by Big Tech.

Keywords Personal data · Data assets · Big Tech · Data enclaves · Digital ecosystem · Parasitic innovation · Markets · Competition · Adtech · Online advertising

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K. Birch, *Data Enclaves*,
https://doi.org/10.1007/978-3-031-46402-7_5

Introduction

We are now at the pivot of the book. In this chapter, I analyse the business practices, models, and strategies of Big Tech in order to understand how they have come to dominate their respective markets and the markets around them: that is, how they have managed to create the data enclaves we're increasingly dependent upon in our daily lives. If we look at the tech-hype *de jeur*, artificial intelligence, we can see how this plays out in potentially new and transformative areas of the economy. Personally, I think that the growing chorus of voices, some more vociferous than others and stretching across the political spectrum, calling for caution when it comes to AI developments and also for the regulation of AI technologies kind of miss the most important point.[1] Absent regulation, the main threat from new AI technologies is that it will further entrench Big Tech's dominance over our economies and the future direction of our technologies. From an STS perspective, what we call AI today is an effect of the socio-political context in which it emerges, which is dominated by Big Tech. Consequently, mainstream fears about AI will not be solved by tinkering with markets and consumer demand because permissive terms and conditions agreements mean that Big Tech can do whatever they want with our personal data: it is their asset. And that means they can use our personal data as the building blocks for whatever AI technologies they want to develop, whether we like that or not. Big Tech has invested massively in computing infrastructures to exploit these data assets, meaning that the intellectual and computing capacities underpinning AI are increasingly concentrated in the hands of Big Tech.[2]

My core argument in this chapter is that rather than owning personal data, Big Tech corporations have worked out the ways to control our personal data through the techno-economic configuration of the mass collection and hoarding of data in siloed enclaves. As these enclaves have emerged, they have stymied the usual imperatives or pressures promoting innovation and competition in the economy, leading to the growth of a specifically 'parasitic' innovation. Generally, startups are simply unable to compete with incumbents because they cannot afford the capital costs

[1] For example, https://futureoflife.org/open-letter/pause-giant-ai-experiments/.

[2] Whittaker, M. (2021) The Steep Cost of Capture, *interactions* 28 (November–December 2021): 50–55, available at https://doi.org/10.1145/3488666 and https://ssrn.com/abstract=4135581.

necessary to build up their own data assets and to create their own data enclaves, without opening themselves up to acquisition. Consequently, Big Tech's head start is almost unassailable without some form of government or collective intervention, such as strengthening competition policy. As data enclaves, Big Tech have entrenched their dominance in their original markets *and* extended this dominance into other parts of the economy through the creation of ecosystems that spread their influence across diverse markets and enrol other businesses, consumers, policymakers, and so in the success of those ecosystems. Everyone else is tied into buttressing these ecosystems in order to benefit from them, primarily through access to the data hoarded by Big Tech. Data enclaves end up becoming an obligatory passage point—to use Michel Callon's phrase—for other businesses seeking access to consumers or users or even citizens: control of these data enclaves means that Big Tech can set the rules of the (economic) game for themselves and others.

All of this implies that future techno-economic developments, especially with AI, are going to follow the previous trends we've seen with the growth of Big Tech and their data enclaves: that means increasing market concentration (e.g. social media), limited product or services competition (e.g. online search), limited user control (e.g. personal data), and so on. We can see how these outcomes have resulted from the hoarding of personal data and its siloing in enclaves that control access to both the 'data' driving innovation and the 'information' on which markets depend to function properly. By controlling both, Big Tech have become more than dominant market players, they have stepped outside the economic game altogether to create their own economic ecosystems. As such, it's worth analysing whether there are markets anymore and what the consequences of this are if there are not. Before we get to that, however, I need to start with an outline of where data enclaves came from, which means a digression into the rise of platform economies.

The Rise of Platform Economies?

A lot of academics, commentators, and other writers use the term 'platform' to frame their discussions of the digital economy, so why am I not using that term too? The notion of a 'platform', as a conceptual framework for understanding the digital economy, has been around for some time: an early example of this is the work of business scholar Annabelle Gawer who writes about the need to understand the "constantly evolving

nature of platforms such as Windows and Google" and their implications for understanding both where market transactions happen and how to characterize those transactions in contemporary capitalism.[3] Gawer's most well-cited publication is an edited volume, so she doesn't spend much time defining what a platform actually is but she frames it as "a building block, providing an essential function to a technological system": it sits between markets; benefits from network effects from its users, in a positive feedback loop; and is highly compatible or interoperable with other technologies or technological systems (p. 2). As Gawer also notes, platform dynamics mean that beyond a certain 'tipping point' they become entrenched as a techno-economic standard creating barriers to entry for new platforms. All of this is well and good, but it doesn't do much to really unpack the specifics of platforms as techno-economic forms or systems.

Others, especially more critical political economists, started to engage with platforms from the mid-2010s. An early and influential thinker on platforms is Nick Srnicek whose book *Platform Capitalism* provides an important introduction to thinking about platforms as more than a technological system. Srnicek positions platforms as central components in a new form of capitalism that is "centred upon extracting and using a particular king of raw material: data", which requires a large techno-economic system to collect, record, store, and analyse; this system is a material system, comprising data centres, cables, electricity, etc.[4] Defining platforms as "digital infrastructures that enable two or more groups to interact", Srnicek likewise emphasizes that platforms are intermediaries that engender network effects, tend towards monopoly, and control governance rules in their systems (pp. 43–47). Moreover, through cross-subsidization, he argues that platforms enable businesses to extend their operations beyond the initial market: for example, Alphabet/Google can start as an online search engine but move into new areas like online maps and GPS. Not all platforms are the same: Srnicek and the geographers Paul Langley and Andrew Leyshon analyse the diversity in platforms,[5]

[3] Gawer, A. (ed.). (2009) *Platforms, Markets and Innovation*, Cheltenham: Edward Elgar, quote at p. 4.

[4] Srnicek, N. (2016) *Platform Capitalism*, Cambridge: Polity Press, p. 39; see also discussion of data as raw material on p. 56.

[5] Langley, P. and Leyshon, A. (2017) Platform capitalism: the intermediation and capitalisation of digital economic circulation, *Finance and Society* 3(1): 11–31.

including, but not limited to, advertising platforms (e.g. Facebook), cloud platforms (e.g. Microsoft Azure), product platforms (e.g. Spotify), and marketplace platforms (e.g. Amazon). Across these discussions, platforms are presented as a new business model, or a shift in how capitalism works. Of particular importance are the ways that platforms enable a shift away from ownership, including of assets, towards control; just think of businesses like Uber, Lyft, Airbnb, and similar that have limited physical assets—which is explicitly evident in their annual financial reports—but are able, as a consequence of their intermediary position—to exert considerable control over the use of all sorts of other people's physical assets (e.g. cars, trucks, houses, etc.). They control the function of those assets without owning them.

This shift from ownership to control highlights the need to go beyond thinking of platforms as a particular state or form of capitalism, and instead to think about platforms as a process. Making a major contribution to such debates, Anne Helmond has proposed the concept of 'platformization' to capture this different emphasis.[6] Helmond's research focuses on the technological architecture that enables a platform or platform business to expand its influence beyond their starting digital infrastructure: this includes things like application programming interfaces (APIs), software development kits (SDKs), and other plugins. In a recent paper, Kelly Bronson and I conceptualized these plugins as 'boundary assets', reflecting their value to the platform owner; I'll come back to this below.[7] Such boundary assets are deeply implicated in the platformization of Big Tech corporations like Facebook, something Helmond and collaborators illustrate with their research on mobile computing.[8] Here, Facebook, as a social media website, represents the platform, while the various APIs, SDKs, etc., that enable other businesses to connect to its website or app represent the boundary assets. These plugins let other businesses access the data, users, and so on that Facebook has collected in its 'social graph', providing a ready-made customer base or user base for those other businesses without the need for them to invest in collecting

[6] Helmond, A. (2015) The platformization of the web: making web data platform ready, *Social Media + Society* 1(2): 1–11.

[7] Birch, K. and Bronson, K. (2022) Big Tech, *Science as Culture* 31(1): 1–14.

[8] Nieborg, D. and Helmond, A. (2019) The political economy of Facebook's platformization in the mobile ecosystem: Facebook messenger as a platform instance, *Media, Culture & Society* 41(2): 196–218.

personal data. In so doing, these other businesses reinforce Facebook's functional usefulness to Facebook's users and its centrality to other businesses. Gradually, platformization leads to growing dependence on a data enclave as other businesses (e.g. app developers, retailers, news media) have to orient their own operations to the techcraft logics of that enclave; for example, it's possible to argue that the growth of 'clickbait' is directly related to the logic of user engagement as a proxy for value (because of the advertising revenues that user engagement can generate).

Data Enclaves

At this point, it's interesting to note that businesspeople (and business scholars) tend to refer to themselves not as 'platforms' (as understood by critical political economists) but as 'ecosystems'; the self-understanding of these businesspeople is that successful businesses create ecosystems. For example, a decade's worth of financial reports and earnings calls of Big Tech corporations shows that the term 'ecosystem' is frequently used by their executives to describe themselves, especially by Alphabet/Google and Apple: they use the term to refer to their products, processes, products, and relationships. Helmond's concept of platformization gets at some of this with her argument that platformization has a dual logic, decentralizing a platform's component parts (e.g. APIs) while recentralizing 'platform ready data' (p.8). In my own work with Troy Cochrane, we made a more explicit attempt to theorize Big Tech in ecosystem terms, in order to capture the distinctiveness of this techno-economic configuration.[9] As we argue, an ecosystem represents more than a platform, instead can be theorized as "heterogenous assemblages of technical devices, platforms, users, developers, payment systems, etc. as well as legal contracts, rights, claims, standards, etc." (p.45). We emphasize that an ecosystem is a techno-economic arrangement, including both the technical infrastructure (e.g. a website or app like Facebook) as well as the rules for using that infrastructure (e.g. the rules external developers must follow when developing apps for Facebook). Binding the techno-economic sides together are the users, their personal data, and metrics used to measure, value, and govern them as valuable assets (e.g. likes, messages, comments, views, etc.).

[9] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58.

In thinking through the importance of this distinction between platform and ecosystem over the last few years, I've concluded that the best term for capturing the operations and power of Big Tech, and their emulators, is *enclave*, hence the title of this book. 'Enclave' captures the technical and socio-economic infrastructure, relations, *and* political economy of Big Tech, reflecting, amongst other things, the fact that these corporations develop digital infrastructures to attract users, collect and curate the personal data of these users, encourage other businesses to plugin to their infrastructures to access these users, establish private rules for how other businesses need to operate, and, through all of this, they establish quasi-markets that others want access to and which Big Tech controls. Such enclaves end up being distinct from normal markets where public rules prevail (i.e. those enforced by a third party, like the state). Big Tech (and their emulators) may not have started with this intent, but you can see this process play out across different Big Tech corporations in different ways as the building of data enclaves has become a dominant business model and innovation strategy today.

As data enclaves, Big Tech corporations control an ecosystem of devices, apps, platforms, and other products; the data that users and other businesses generate through participation in the ecosystem; the rules they set for users, developers, and other businesses; and the enforcement of the standards and rules in their ecosystems. Data enclaves are constituted by restricting access to those enclaves and their crucial and valuable data assets, while also locking in users and other businesses into the data enclave through technical (e.g. interoperability limits) and socio-legal mechanisms (e.g. contracts). The latter of these is evident in self-preferencing by Big Tech, directing users to new products and services Big Tech has developed; for example, the 2019–2020 US Congressional investigation into competition in digital markets concluded that "Apple leverages its control of iOS and the App Store to create and enforce barriers to competition and discriminate against and exclude rivals while preferencing its own offering".[10] Now, while there might be some product and service competition between enclaves, Big Tech aims to constrain the mobility of users and other businesses. For example, in Facebook's 2020 Fourth Quarter earnings call, its executives identified Apple as its "biggest competitor" because they thought "iMessage is a

[10] US House of Representatives (2020) *Investigation of Competition in Digital Markets*, Washington DC: House of Representatives, quote from p. 17.

key linchpin of their ecosystem"; rather than competing with their products and services, however, Facebook was concerned with attracting and controlling its user base, so that it can monetize them (through online advertising), and Apple threatened to draw away that user base with its messaging product.

Obviously, scale and scalability matter to Big Tech corporations, and most digital technology businesses, but it's only part of what makes data enclaves so dominant in today's economy. As important is *modularity*; that is, making it easy for others to plugin or slot into an ecosystem. This necessitates a capacity to integrate a range of other political-economic actors into an ecosystem, using a number of techno-economic mechanisms like the boundary assets I've already mentioned. These boundary assets are centrally implicated in the construction of the ecosystem boundary, providing the means to integrate others so they complement the ecosystem and generate positive feedback effects *and* to generate or extract data and other valuable inputs from those complementors, which can be used to improve the performativity of the ecosystem. Data enclaves need a flexible and permeable interface, rather than strict separation, so that users, businesses, technologies, and so on can cross the boundary. Controlling those boundary assets enables Big Tech to establish the private rules through the contractual arrangements (e.g. terms and conditions, privacy policies) others have to abide by, but which enable Big Tech to collect, curate, and reuse personal and other data indefinitely. The more users that these boundary assets can enrol and integrate into an enclave, the more valuable it becomes for Big Tech.

Data Enclave Case Study: AdTech and Google

At this point, it's probably a good idea to provide an example of what I mean by data enclave. For this, I'll turn to online advertising and how Alphabet/Google monetizes the personal data it collects.

Considering the ubiquity of critical takes about Alphabet/Google and online advertising, especially well-known examples like Shoshana Zuboff's book *The Age of Surveillance Capitalism*, I thought it'd be relatively easy to dissect the Alphabet/Google's data enclave. I was wrong on that! To understand Alphabet/Google as a data enclave necessitates exploring online advertising and the advertising technology sector, or 'adtech'. I'm only going to provide a brief overview of adtech, so any readers wanting more can go to the academic literature and policy materials I cite in the

rest of this section, or you could check out Lee McGuigan's new book for a history of the topic.[11]

To start, it's important to note that online advertising is split between 'search' advertising, in which online search terms or keywords are matched with advertising bids, and 'display' advertising, which represent the text, images, or videos displayed on websites, or in apps, or elsewhere online (e.g. YouTube).[12] In the early days, there was no ad targeting using personal data. Over time, online advertising has become dependent upon the adtech sector as online advertising has become more sophisticated, especially with the advent of programmatic advertising. Although I'll explain it in more detail below, it's pretty safe to define adtech, for now, as the intermediaries that sit between advertisers, who are trying to buy online ad space, and publishers, who are trying to sell online ad space; that ad space is called 'ad inventory'. Online advertising itself can be traced back at least to 1994, when advertisers and publishers usually dealt with each other directly. Advertisers purchased ad inventory on the basis of 'cost-per-mille', meaning on the basis of 1000 ('mille') impressions or

[11] McGuigan, L. (2023) *Selling the American People*, Cambridge MA: MIT Press.

[12] I drew on the following academic, business, and policy literature for much of this discussion: ClearCode.cc *Adtech Book*, available online at: https://adtechbook.clearcode.cc/; Gerardin, D. and Katsifis, D. (2019) An EU competition law analysis of online display advertising in the programmatic age, *European Competition Journal* 15(1): 55–96; Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; Bitton, D. and Lewis, S. (2020) Clearing up misconceptions about Google's ad tech business, available at: https://www.competitionpolicyinternational.com/clearing-up-misconceptions-about-googles-ad-tech-business/; Srinivasan, D. (2020) Why Google Dominates Advertising Markets: Competition Policy Should Lean on the Principles of Financial Market Regulation, *Stanford Technology Law Review* 24(1): 55–175; ACCC (2020) *Digital advertising services inquiry*, Canberra: Australian Competition and Consumer Commission, available at: https://www.accc.gov.au/publications/digital-advertising-services-inquiry-final-report; CMA (2020) *Online platforms and digital advertising market study*, UK Competition & Markets Authority, available at: https://www.gov.uk/cma-cases/online-platforms-and-digital-advertising-market-study; Scott Morton, F. and Dinielli, D. (2020) Roadmap for a Digital Advertising Monopolization Case Against Google, Omidyar Network, available at: https://publicknowledge.org/policy/roadmap-for-a-digital-advertising-monopolization-case-against-google/; Sweeney, M. (2022) Understanding the Complicated World of Advertising Technology (AdTech) & Programmatic Advertising, Clearcode, available at: https://clearcode.cc/blog/understanding-advertising-technology/; and McGuigan, L. (2023) *Selling the American People*, Cambridge MA: MIT Press; and MacKenzie, D., Caliskan, K. and Rommerskirchen, C. (2023) The longest second: Header bidding and the material politics of online advertising, *Economy & Society*.

views of the ad inventory (e.g. website). As online advertising became more complicated in the latter half of the 1990s, adtech companies, including ad servers and ad networks, began to appear as the proliferation of websites meant it became increasingly difficult for advertisers and publishers to negotiate directly with one another. One of the earliest examples of an adtech business was the ad network DoubleClick, which was founded in 1995—and acquired by Alphabet/Google in 2008—and which would aggregate 'remnant' ad inventory from publishers to auction to advertisers. Remnant ad inventory included anything that publishers could not sell directly to advertisers; DoubleClick helped to sell this inventory through an automated auction.

As publishers and advertisers sought to optimize their operations, they looked for more efficient and profitable ways to sell and buy ad inventory, respectively. This opened the way for the adtech sector to grow and expand, establishing ad servers to automate the buying and selling of online ads; ad networks to auction ad inventory; and various other platforms to make this process as efficient as possible. Most ad inventory ended up being sold in a so-called 'waterfall' system, in which publishers sold their most valuable ad inventory directly to publishers, then sold the rest through ad networks and exchanges with a declining fee cascade. As a consequence of automating the process, publishers used 'historical average performance' to determine which ad network or exchange to use in this waterfall system, which meant they wouldn't always get the best bids for their ad inventory. Into this context, DoubleClick introduced something called 'dynamic allocation' in 2007 as a way both to "fuse" direct deals and remnant auctions and replace historical average performance metrics with real-time bids in their own ad exchange (called AdX at the time).[13] Subsequently, 'real-time-bidding' (RTB) took off and replaced the 'waterfall' system from the late 2000s onwards, helped by the creation of adtech industry protocols like the OpenRTB Consortium. Since then, RTB has become the main way publishers sell their ad inventory; it enables the auctioning of a publisher's ad inventory during the milliseconds before a webpage loads. To facilitate RTB, an array of

[13] MacKenzie, D., Caliskan, K. and Rommerskirchen, C. (2023) The longest second: Header bidding and the material politics of online advertising, *Economy & Society*, p. 10.

adtech intermediaries now sits between publishers and advertisers (see Fig. 5.1).[14]

The adtech intermediaries outlined in Fig. 5.1 include:

- Traditional **advertising agencies** hired by advertisers to manage their ad campaigns.
- **Ad servers (advertisers)** that help advertisers to manage and automate their ad campaigns.
- **Demand-side platforms** (DSP) that buy ad inventory for advertisers (via advertiser ad servers and ad agencies) from ad exchanges or ad networks.
- **Data management platforms** that collect, store, and analyse personal data collected from various sources; they help advertisers (and publishers) to segment audiences and analyse the success of ad campaigns.
- **Data brokers** that collect and trade personal data.

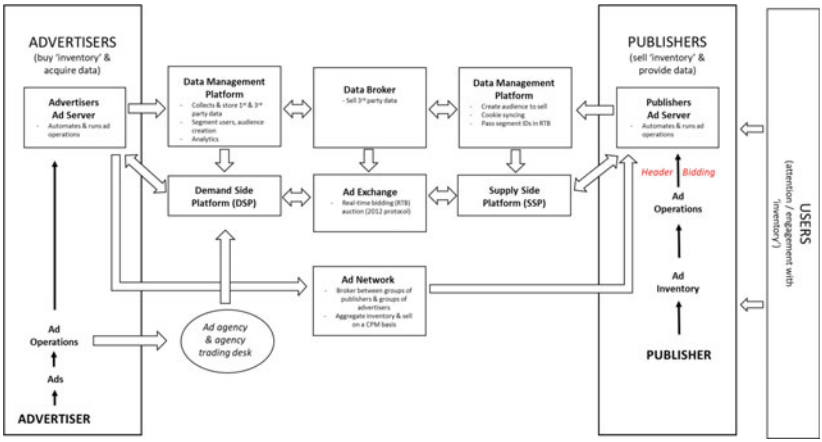

Fig. 5.1 The Adtech Sector (*Source* adapted from ClearCode.cc *Adtech Book*, available online at https://adtechbook.clearcode.cc/)

[14] Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; also see ClearCode.cc *Adtech Book*, available online at https://adtechbook.clearcode.cc/

- **Ad exchanges** that operate real-time auctions for ad inventory, usually through personal targeting of individual viewers/users.
- **Ad networks** that aggregate ad inventory and broker deals between groups of advertisers and groups of publishers.
- **Supply-side platforms** that sell ad inventory for publishers (via publisher ad servers).
- **Ad servers (publishers)** that help publishers manage selling their ad inventory.[15]

There is some overlap between adtech intermediaries, so the visual representation in Fig. 5.1 is somewhat stylized. Moreover, it shows the current state of the adtech industry resulting from the rise of 'programmatic advertising' from the late 2000s onwards, so doesn't represent a visualization of the evolution of online advertising. Today, online advertising is dominated by programmatic advertising, which took off from the mid-2010s and is "fuelled by various categories of user data" that "are used to sell and purchase ad inventory within fragments of a second", according to Damien Gerardin and Dimitrios Katsifis.[16] With programmatic advertising, personal data are used to identify the online ads to show website visitors, or app users, or YouTube watchers; this contrasts with context-based advertising which depends upon, for example, a website's content. Programmatic advertising works as follows:

- A user visits an online site.
- Before the site loads, the user's visit automatically triggers the website publisher's ad server to identify the user (via personal data) and to send its ad inventory to one or more ad exchanges.
- The ad exchanges send requests for bids for that ad inventory to ad buying platforms (which operate on behalf of advertisers looking for users to advertise to).
- The ad buying platforms submit their automated bids to this auction.
- The ad exchanges then pick their winners and return that bid information to the ad server, which picks the highest bid.

[15] See ClearCode.cc *Adtech Book*, available online at https://adtechbook.clearcode.cc/.

[16] Gerardin, D. and Katsifis, D. (2019) An EU competition law analysis of online display advertising in the programmatic age, *European Competition Journal* 15(1): 55–96; quote at p. 61.

- The site finishes loading, showing the ad from the winning bidder (i.e. advertiser).[17]

Programmatic advertising quickly dominated online advertising, and now represents around 85 percent of online ad revenues.[18] It's dependent upon the collection, transfer, and analysis of personal data, whether user-generated, social, behavioural, locational, demographic, or personally identifying (see Chapter 1). It's also been driven by the expansion of the collection of these personal data and the spread of the digital architecture used to collect these data.[19]

And this brings me to Alphabet/Google. Over time, Alphabet/Google has created a dense, interlocking, and concentrated ecosystem of adtech products and services, thoroughly dominating programmatic advertising worldwide to the extent that it has a 39 percent market share of all *global* digital advertising; the other dominant business is Facebook with around 21 percent market share.[20] In its 2020 investigation of digital advertising, the UK's Competition and Markets Authority (CMA) states that Alphabet/Google collects personal data from over 50 "consumer-facing services", giving it a significant data advantage in online advertising.[21] On their website, Alphabet/Google's explain how they monetize personal data by stating that:

> We use data to show ads that are useful to you, whether they are on Google or they are on websites and mobile apps that partner with us. We do not sell your personal information to anyone.[22]

[17] Srinivasan, D. (2020) Why google dominates advertising markets: Competition policy should lean on the principles of financial market regulation, *Stanford Technology Law Review* 24(1): 55–175, at p. 76.

[18] See the Statista report on "digital advertising" by Chanthadumrongrat (2022) *Digital Advertising—Market Data Analysis & Forecast*, Hamburg: Statista.

[19] Esteve, A. (2017) The business of personal data: Google, Facebook, and privacy issues in the EU and the USA, *International Data Privacy Law* 7(1): 36–47,

[20] Chanthadumrongrat (2022) *Digital Advertising—Market Data Analysis & Forecast*, Hamburg: Statista.

[21] CMA (2020) *Online platforms and digital advertising market study*, UK Competition & Markets Authority, available at: https://www.gov.uk/cma-cases/online-platforms-and-digital-advertising-market-study, p. 155 and p. 280.

[22] https://howwemakemoney.withgoogle.com/

Notice that they stress that "we do not sell your personal information to anyone". As early as 2004, the year they became a publicly listed corporation, their 2004 10K Annual Report noted that:

> Concerns about our collection, use or sharing of personal information or other privacy-related matters, even if unfounded, could damage our reputation and operating results. Recently, several groups have raised privacy concerns in connection with our Gmail free email service which we announced in April 2004 and these concerns have attracted a significant amount of public commentary and attention. The concerns relate principally to *the fact that Gmail uses computers to match advertisements to the content of a user's email message when email messages are viewed using the Gmail service*. Privacy concerns have also arisen with our products that provide improved access to personal information that is already publicly available, but that we have made more readily accessible by the public (p. 54, emphasis added).

Alphabet/Google puts a lot of emphasis on its role as an adtech business. For example, in their 2013 Q4 earnings call, Alphabet/Google executives state that: "Overall, our monetization solutions like our Ad Exchange, AdSense and AdMob are helping major publishers maximize their revenues from digital advertisers. So that's our core ad business". Monetization through advertising is their stated and long-term business model.[23]

As a result of its market dominance, however, Alphabet/Google has been described as a "walled garden" or "enclave" of adtech platforms, products, and services, all operating within a single and often closed ecosystem. I've tried to illustrate this in Figs. 5.2 and 5.3, differentiating between Alphabet/Google's adtech ecosystem "before 2018" and "after 2018" respectively. I chose this date as that is when the corporation significantly restructured their adtech ecosystem, merging their ad exchange, SSP, and ad server (for publishers) into Google Ad Manager; they also changed from a consecutive second-price auction to a "unified first-price" auction the following year, in 2019, which I'll come back to below. Alphabet/Google is described as a walled garden because while

[23] According to their annual financial reports, in 2015 around 90 percent of Alphabet/Google's revenues came from advertising, which has slowly declined over time, to around 80 percent in 2022. Advertising revenue in 2022 was split between: "Google Search & other" (US$162b), "YouTube ads" (US$29b), and "Google Network" (US$33b).

their adtech competitors can buy and sell ad inventory to Alphabet/Google adtech properties, and vice versa, access to certain Alphabet/Google properties is only available through their own adtech products and services (e.g. YouTube ad inventory through DV360).[24]

Alphabet/Google monetizes the 'traffic' (i.e. users) on its suite of products and services, described as "properties" by the corporation itself, as well as broader 'Google Network' of other companies that sign up to its advertising products (e.g. AdMob, AdSense). User monetization entails selling advertising, and personal data are central to this monetization as they are supposed to enable the individual targeting of users with online ads. Alphabet/Google executives described the role of personal data in online advertising in a 2006 Q4 earnings call:

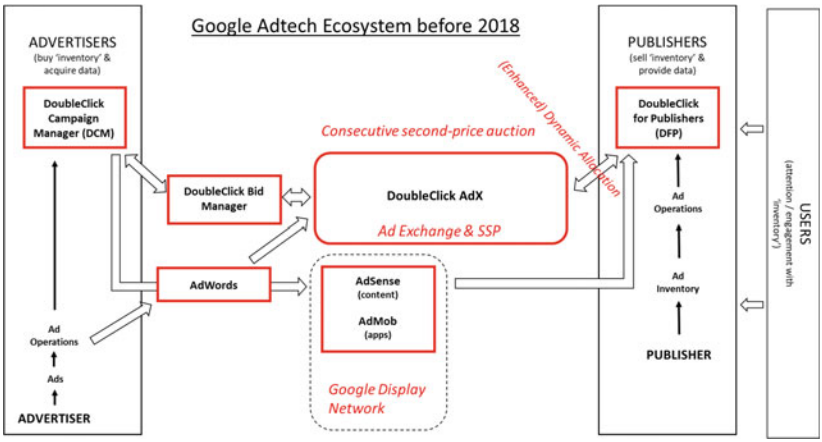

Fig. 5.2 Alphabet/Google's Adtech Ecosystem (before 2018) (*Source* red bold represents Alphabet/Google properties; various sources, including ClearCode.cc *Adtech Book*, available online at https://adtechbook.clearcode.cc/, and Geradin and Katsifis (2019, 2020); Bitton and Lewis (2020); Srinivasan (2020), see note 12)

[24] Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54.

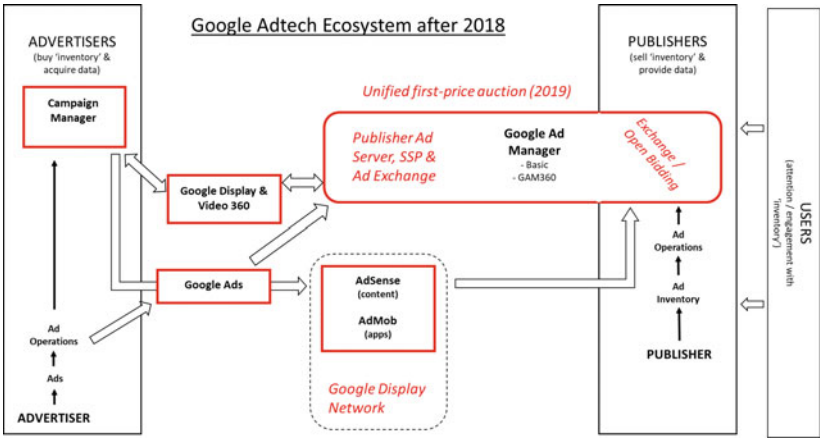

Fig. 5.3 Alphabet/Google's Adtech Ecosystem (after 2018) (*Source* red bold represents Alphabet/Google properties; various sources, including ClearCode.cc *Adtech Book*, available online at https://adtechbook.clearcode.cc/, and Geradin and Katsifis (2019, 2020); Bitton and Lewis (2020); Srinivasan (2020), see note 12)

> The strength of our ability to target this personal information continues to improve based on user feedback, better technology, et cetera, and this results in our advertisers being able to spend in the most effective way to contribute to everyone's bottom line positively. The net effect, of course, is strong revenue growth, good advertiser satisfaction and a real value to the end user.

By 2011, Alphabet/Google executives, in a Q1 earnings call, stressed the importance of the "signals" (i.e. personal data) coming from users to the development of their products and services:

> We do see social as very important. Google uses well over 200 signals in terms of how we think about [Search] ranking today. And when we think about identity and relationships, those are our key signals that can and should be integrated in the experience. So it is important, but it's one of the many that we use. *In terms of assets that apply to that, we do have a very, very large number of users coming to our door every day*. A considerable percentage of them are logged-in users that are using multiple of our products. So there is a large variety of signals that we'll be able to

use with user support and users seeing value from it to make the overall experience better (emphasis added).

Note the definition of users as "assets" in this quote. Alphabet/Google has been able to generate personal data at a massive scale, especially through the creation of detailed user profiles. In 2016, the corporation changed its privacy policies to explain that it was going to start combining data collected from across its properties and other sources, enabling it to create single user IDs to better target users with advertising as well as improve its own products and services. In collecting data from a range of companies via its boundary assets (e.g. APIs, SDKs, plugins), Alphabet/Google could thereby combine data it generated with data generated by others in its ecosystem.[25]

Parasitic Innovation?

By outlining Alphabet/Google's adtech ecosystem, I want to lay the groundwork for dissecting its innovation strategies in the development of advertising technologies, especially as these relate to the generation and exploitation of personal data in data enclaves. In what follows, I'm going to outline innovation strategies underpinned by a logic of growing and entrenching Alphabet/Google's market position by creating an enclave ecosystem that others have to plug into and buttress for their own economic survival. I'm going to use the term 'parasitic innovation' deliberately here to describe this ecosystem, mainly to emphasize that innovation is often driven by a strategic attempt to dominate markets, avoid competition, and undermine competitors—and this requires binding others to your own ecosystem by reducing their options to go elsewhere. To be clear, this tendency is most evident with data enclaves because of the particular techno-economic context in which they emerged and which is defined by a low interest rate regime (making capital cheap), weak antitrust regime (making acquisitions easy), and weak regulatory regime (eroding privacy costs).

To start, it's important to stress that generating personal data from users is a major cost for Alphabet/Google: they define it as 'traffic acquisition cost', or TAC, in their financial reports. As a 'cost of revenues',

[25] Stucke, M. (2022) *Breaking Away: How to Regain Control Over Our Data, Privacy, and Autonomy*, Oxford: Oxford University Press.

TAC amounted to US$49 billion in 2022, representing the direct costs of selling their products or services to their customers, where their customers are advertisers, not users. Reducing TAC relative to advertising revenues has been a key performance metric for Alphabet/Google since the early 2000s, as evidenced in their financial reports and earnings calls since then. In 2006, TAC represented 32 percent of advertising revenues but by 2022 this was down to 22 percent. In simple terms, TAC is the cost of attracting users to their ecosystem: it includes the cost of increasing the number of "access points" to their ecosystem, such as setting defaults on new devices through a range of contractual agreements,[26] and "revenue sharing" with device manufacturers, especially smartphone manufacturers. For example, in a 2012 Q3 earnings call, executives state:

> On the – sorry, just want to talk about the TAC line and just to say that, yes, I mean obviously all the TAC that we paid to Apple, it is just another partner for distribution, so it all is tied to our – the TAC line that we have for Google.com. On your question of the default [product on devices], I mean although there is – nothing is changed, right, I mean, when you use Google, we are a great partner with Apple, we're a great partner with many of them, and in doing so, when you do search, I mean we have a great, we have the – the fact that they've changed from Google to search is still kind of run by our engines.

As this quote illustrates, Apple is an important manufacturer that Alphabet/Google pays to ensure its products and services are set as defaults. According to the US Congressional investigation into digital markets and competition, "Apple also reportedly made $9 billion in 2018 and $12 billion in 2019 to set Google as the default search engine on the Safari browser".[27]

[26] According to the UK's Competition and Markets Authority (CMA), "Google Search has default agreements covering much more of the mobile device sector (at least 94%) than the desktop PC sector (29%). In turn, Google has a relatively higher share of supply in mobile search (97%) than it does in desktop search (84%)"; see CMA (2020) *Online platforms and digital advertising market study*, UK Competition & Markets Authority, available at: https://www.gov.uk/cma-cases/online-platforms-and-digital-advertising-market-study, p. 102.

[27] US House of Representatives (2020) *Investigation of Competition in Digital Markets*, Washington DC: House of Representatives, quote at p. 345.

The reason for mentioning TAC first is that it helps explain Alphabet/Google's innovation strategies in their adtech business, which, it's important to remember, is their main business and currently represents nearly 80 percent of their revenues. They actively innovate to attract more users and to generate more personal data from their adtech properties, introducing new technologies, processes, mechanisms, and rules to do so. I'll go over several examples here that I think reflect a good illustration of parasitic innovation (i.e. that are designed to dominate markets, avoid competition, and undermine competitors), drawing primarily upon the work of legal scholars like Damien Geradin and Dimitrios Katsifis.[28]

One of the earliest examples was Alphabet/Google's introduction of 'dynamic allocation' in late 2009. Within its own dominant adtech ecosystem, dynamic allocation enabled Alphabet/Google's ad exchange (AdX at the time) to bid in the waterfall process for remnant line items (i.e. non-direct advertising deals) on the basis of real-time performance rather than historical performance, giving it an advantage over other ad exchanges which couldn't do this. Alphabet/Google ended up with a 'right of first refusal' as dynamic allocation allegedly enabled AdX to see average historical bids and then bid just above those historical bids.[29] In 2014, the corporation introduced 'enhanced dynamic allocation' which enabled AdX to "jump even ahead of direct deals in the waterfall".[30] Consequently, publishers using DoubleClick for Publishers (DFP) could not benefit from real-time demand signals except by using Alphabet/Google's own ad exchange. According to a lawsuit brought by the State

[28] Gerardin, D. and Katsifis, D. (2019) An EU competition law analysis of online display advertising in the programmatic age, *European Competition Journal* 15(1): 55–96; Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": Analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; and Geradin, D. and Katsifis, D. (2020) Competition in Ad Tech: A Response to Google, *TILEC Discussion Paper No. DP2020-038*, available at SSRN: https://ssrn.com/abstract=3617839 or http://dx.doi.org/10.2139/ssrn.3617839.

[29] *In Re: Google Digital Advertising Antitrust Litigation*, 14 Jan 2022, available at: https://www.documentcloud.org/documents/21179902-3rd-complaint-for-texas-google-antitrust-case.

[30] Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": Analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; quote at p. 18.

of Texas and several other US states in early 2022,[31] dynamic allocation "foreclosed competition in the market for exchanges, the market for buying tools for small advertisers, and the market for buying tools for large advertisers" (p. 94). In response to dynamic allocation, publishers ended up introducing their own technological mechanism in 2015, something called 'header bidding'. Header bidding entailed browsers forcing ad exchanges to compete with one another in real-time auctions before contacting an ad server; this was designed to increase the fees publishers would receive from their ad inventory. Alphabet/Google's response was to introduce 'exchange bidding' in 2018, which opened up dynamic allocation to selected competing exchanges: however, exchange bidding still leaves Alphabet/Google with "access to the bidding data of its rivals" and its "impossible to verify whether Google runs a fair auction or treats AdX more favourably".[32]

A second example is the use of second-price auctions in Alphabet/Google's adtech ecosystem. According to the Texas lawsuit, Alphabet/Google created a "secret program" called Project Bernanke designed to exploit auction bidding in AdX. AdX used a second-price auction, meaning that the winning bidder (i.e. advertiser) would pay the second-highest bid rather than their own highest bid; this is meant to encourage bidders to bid according to their actual preferences rather than try to game the system.[33] According to the Texas lawsuit, though:

> Google's secret Bernanke program surreptitiously switched AdX from a second-price auction to a third-price auction on billions of impressions per month. Bernanke dropped the second-highest bid from the AdX auction when the two highest bids were above the floor and from Google Ads advertisers. The price to be paid, then, was the lower third-place bid. With Bernanke, AdX ran third-price auctions rather than second-price auctions. (p. 106).

[31] *In Re: Google Digital Advertising Antitrust Litigation*, 14 Jan 2022, available at: https://www.documentcloud.org/documents/21179902-3rd-complaint-for-texas-google-antitrust-case.

[32] Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; quotes at p. 19.

[33] Birch, K. (2023) There are no markets anymore: From neoliberalism to Big Tech, *State of Power Report 2023*, The Transnational Institute, available at: https://www.tni.org/en/article/there-are-no-markets-anymore.

This might sound complicated, but it's not really. Basically, the lawsuit is alleging that Alphabet/Google designed their ad exchange to operate as a second-price auction for advertisers but operate as a third-price auction for publishers. Alphabet/Google then kept the difference between the two auctions, income which should have gone to publishers (see Fig. 5.4). The reason for introducing Project Bernanke was revealed in internal documents, according to the Texas lawsuit: "According to Google, prior to Bernanke, advertisers bidding through non-Google buying tools were winning too often over advertisers bidding through Google Ads" (p.108).

A third example is Alphabet/Google's shift to a unified first-price auction in 2019, following the merger of their ad exchange, SSP, and ad server for publishers in 2018 to create Google Ad Manager. The shift to a first-price auction—meaning bidders pay what they bid—is not my focus here. Rather, alongside this change, Alphabet/Google restricted access to bidding/auction data through limits on "data transfer files". Publishers use these data transfer files to look at bids on their ad inventory and to look at impression-level data (e.g. price of an ad impression) so they can optimize yields on their ad inventory. However, in 2019 Alphabet/Google introduced a new system which restricted the ability to link these data, claiming that:

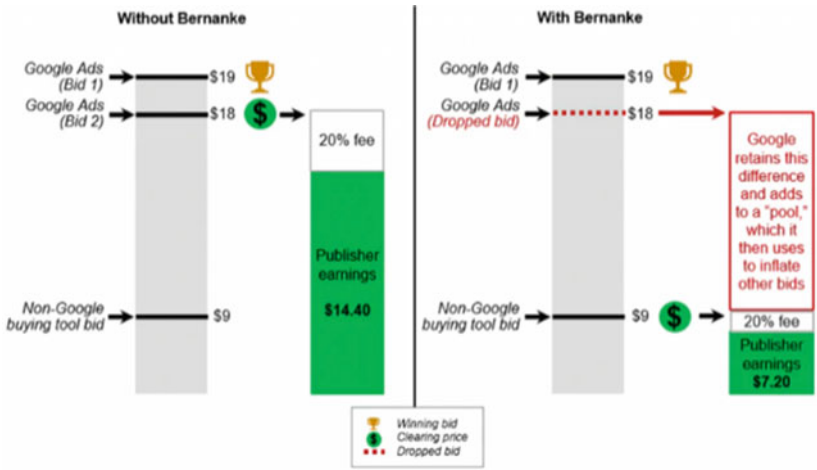

Fig. 5.4 Project Bernanke (*Source In Re: Google Digital Advertising Antitrust Litigation*, 14 Jan 2022)

> In order to prevent bid data from being tied to individual users, you will not be able to join the Bid Data Transfer file with other Ad Manager Data Transfer files.[34]

This followed a 2018 decision to limit access to user IDs by ad buyers, with Alphabet/Google referencing privacy concerns for this decision as well. As a result of both these policy changes, publishers and advertisers have become more reliant upon using Alphabet/Google's adtech properties to access user data and user profiles.[35] As Geradin and Katsifis note, Alphabet/Google have restricted access to data within their adtech ecosystem and "forced marketers to increase their dependence on Google" through the use of Alphabet/Google's data analytics and 'Ads Data Hub' services.[36]

Parasitic innovation is not defined by what Alphabet/Google does, or necessarily emblematic of their innovation and business strategies. Instead, I see it as a characteristic of data *enclaving*, which is driven by the need to generate user and customer dependence on an ecosystem by creating and dominating a pseudo-market, reducing substitutes or alternatives, and tying competitors to the success of that ecosystem rather than building their own or building a different setup. Unfortunately, we've all ended up locked into these data enclaves through our increasing dependence upon the techno-economic infrastructures they provide for our digital lives and upon the data assets these infrastructures generate through our actions and behaviours. Big Tech now sets the rules of the game in these data enclaves, limiting access to the data assets that could (and should) stimulate socially beneficial innovation across our economies and societies, whether that is in healthcare, public transit, climate adaptation, or simply making friends and meeting people. But, unlike a market, these data enclaves are designed to stymie competition, to limit innovation except when it comes to a self-reinforcing propping up of the ecosystem, and

[34] Alphabet/Google cited in Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": Analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54.

[35] CMA (2020) *Online platforms and digital advertising market study*, UK Competition & Markets Authority, available at: https://www.gov.uk/cma-cases/online-platforms-and-digital-advertising-market-study.

[36] Geradin, D. and Katsifis, D. (2020) "Trust me, I'm fair": Analysing Google's latest practices in ad tech from the perspective of EU competition law, *European Competition Journal* 16(1): 11–54; quote at p. 43.

to generate the data resources that undermine our economies and societies. Unfortunately, all of this entails a number of damaging paradoxes that have led to an unravelling of the usefulness of these infrastructures but have also left us to deal with the fallout from an array of increasingly dysfunctional products and services.

CHAPTER 6

Data Paradoxes

Abstract The growth of data enclaves has led to the emergence of a series of data paradoxes, relating to the way that digital personal data are constructed, configured as assets, and then valued. These paradoxes include, first, the fact that the most socially beneficial use of data entails freely or openly sharing and combining it with other data, while its economic value is defined by its enclaving. Without new regulations to force them to open up their enclaves, Big Tech will simply exploit policy initiatives towards open data. Second, the fact that personal data are reflexive, meaning that people and organizations start to work out how to game the system and exploit unforeseen outcomes. And third, the fact that the enclaving of our personal data means that Big Tech effectively controls the information which underpins markets, including our preferences, choices, decisions, actions, and so on. As a result of these paradoxes, markets are not working anymore, or maybe they never did.

Keywords Personal data · Data paradoxes · Big Tech · Data enclaves · Markets · Competition

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Today, many people, from media pundits through academics to politicians and policymakers, are worried about the rise of disinformation.[1] And many of those people blame social media and other digital technology for the spread of disinformation. A number of ex-tech workers have waded into these debates, whether as disillusioned ex-employees or whistleblowers, highlighting inaction within digital technology companies when it comes to the problematic effects of their platforms and ecosystems, whether that entails the spread of disinformation or bigotry. For example, in 2020 Netflix released a documentary film called *The Social Dilemma* about the effects of social media on our mental health, our political systems, and on prejudice.[2] The film itself drew heavily upon the work of Tristan Harris, founder of the US-based Center for Humane Technology (CHT). Harris is a former 'design ethicist' at Alphabet/Google who left the corporation in 2015 to set up the CHT and publicize the issues with digital design practices, especially those focused on sucking more and more of our attention into our screens.[3] In STS circles, the film caused some mirth when Harris stated, "No one got upset when bicycles showed up. Right?": foundational research in STS showed precisely that people did get very upset during the development of bicycles in the 1800s.[4] A more relevant comment Harris makes in the film, though, is the phrase, "If you're not paying for the product, you are the product". Now, it's not difficult to find others making the same comment when it comes to personal data, but as I've tried to outline in previous chapters, I think this idea misses the point and goes someway to illustrating where we might be going wrong in the ways we're currently dealing with data governance—which I'll come back to in the book's Conclusion.

We're not the 'product'; nor are we 'not paying' for the product. We're very much paying for something, whether that's an iPhone and their apps or Google search and the results, and we're paying with our time and data. Users, user data, and user engagement are very valuable for Big Tech and other digital tech companies; they are critically important assets that underpin their enclaves, locking us into this ecosystem or that. But, to see

[1] Just one example is the EU's policy on tackling disinformation: https://digital-strategy.ec.europa.eu/en/policies/online-disinformation.

[2] .https://www.thesocialdilemma.com/

[3] http://minimizedistraction.com/.

[4] Bijker, W. (1995) *Of Bicycles, Bakelites, and Bulbs*, Cambridge MA: MIT Press.

this we have to rethink our perspective because we are still epistemically tied into a worldview in which the production, sale, and consumption of goods and services are considered the driver of our economies and lives. Instead, my point in this book is that we need to conceptualize and understand personal data as an asset; that is, something that can be controlled (if not owned), something that can generate future benefits, and something that can be capitalized.

A useful example of how we need to reorient our perspective is currently going on in Canada at time of writing. The Canadian Federal Government introduced the *Online News Act* (Bill C-18) in November 2021 in an attempt to get Big Tech corporations, especially Alphabet/Google and Meta/Facebook, to pay for the news content they link to in their ecosystems.[5] Now, the *Online News Act* has been a contentious piece of legislation, with Big Tech corporations lobbying actively against it (including full page ads in national newspapers). Its premise is that corporations like Alphabet/Google and Meta/Facebook have come to dominate online advertising in Canada to such an extent that news organizations are being starved of ad revenues, forcing them to reduce coverage, cut journalist jobs, or even shut altogether.[6] A regular claim made by Meta/Facebook throughout the passage of the Act is that they provide Canadian news producers with C$230 million in 'free' value annually through the links in their ecosystems to news stories; that is, people on Facebook or Instagram seeing a link and going to the news website, which the news producer can monetize through online advertising. News journalist Greg O'Brien, however, sees things differently: he points out that Meta/Facebook never comments on the value that the Big Tech corporation gets from all the user engagement on their platforms generated by news stories (e.g. likes, shares, comments, etc.), all of which it can monetize.[7]

[5] It follows Australia's introduction of the *News Media Bargaining Code* in 2021, which was designed to get Big Tech to pay news producers for their content: https://www.bnnbloomberg.ca/what-can-canada-learn-from-australia-s-bid-to-make-big-tech-pay-for-news-1.1939992.

[6] See the reports produced by the Global Media Concentration Project for more on the concentration of advertising revenues in Canada: https://gmicp.org/; also see Tepper, J. and Hearn, D. (2019) *The Myth of Capitalism*, New Jersey: Wiley.

[7] https://www.chch.com/commentary-facebooks-fallacy/.

The point of mentioning this is to highlight that we often don't see or consider what is valuable in digital and data-driven economies, often because it's not visible or not made transparent, even in the terms and conditions we sign regularly without reading. Even now, I'm continually learning about further intricacies of data monetization. While all of this might seem like a straightforward issue of large, monopolistic corporations exploiting our behaviours, choices, decisions, preferences, and so on—which is very much the way that critics like Shoshana Zuboff frame it—I think there is something else, perhaps even more problematic, going on. In rethinking how we understand personal data, we are able to get a better handle on this something else, which I think of as a number of paradoxes underpinning the emergence of data enclaves that has led to growing dysfunctionality in our economies and societies. I'll start this chapter by discussing this increasing dysfunctionality, before then outlining how we got to this point as a result of the techno-economic configuration of data enclaves and what it means for our economies.

The 'Enshittification' of the Digital Economy

The scifi author and digital activist Cory Doctorow has emerged as one of the most vocal critics of Big Tech and digital technology in the last few years. His regular blogging about the increasingly dysfunctional nature of our digital and data-driven economies provides a wonderful and terrifying insight into where we're heading and why.[8] The same goes for his non-fiction writing, including books like *How to Destroy Surveillance Capitalism* and *Chokepoint Capitalism*. I thoroughly recommend browsing through his writings. Perhaps the most evocative concept Doctorow has come up with is the notion of 'enshittification'. According to Doctorow, enshittification can be defined as:

> First, they [platforms, ecosystems] are good to their users; then they abuse their users to make things better for their business customers; finally, they abuse those business customers to claw back all the value for themselves. Then, they die.[9]

[8] https://pluralistic.net/.

[9] https://www.wired.com/story/tiktok-platforms-cory-doctorow/.

Enshittification is a consequence of how data enclaves are configured. They are designed, sometimes deliberately but sometimes inadvertently, as two-sided or multi-sided markets. On the one side, users receive a product or service notionally for 'free' (e.g. Google search, Facebook, app, etc.), enabling the data enclave to build up its user base from which it can generate data assets; these users become valuable as they attract new users through network effects. Some of these network effects are direct, in that the number of users makes that ecosystem more useful for other users who then join the ecosystem; but other network effects are indirect, in that the number of users on one side attracts 'others' to the other side of the market.[10] On that other side are the actual customers of the data enclave, which sits in the middle acting as an intermediary and collecting fees for providing access to users and their data. Hence why people often talk about 'you being the product', as data enclaves generate their revenues from selling access to us (or some semblance of us).

Doctorow argues that all digital ecosystems come to exemplify enshittification, as this is the business model they follow, and he provides several examples of how this happens with TikTok, Amazon, Facebook, and so on. At the start of its existence, a social media platform like TikTok or Facebook is designed to be useful and fun for its users so that that they sign up and use the website or app: this is important, users have to use the app for it to become valuable so that it generates data. Often, this means that only those apps with the capital resources behind it can and will become successful data enclaves because to do so requires significant investment to attract enough users to ensure that monetizing them is worthwhile. Troy Cochrane and I have described this as a form of 'expected monopoly rent' in which investors seek to performatively shape a particular sector by flooding one startup with enough capital to eventually dominate that sector, discouraging others from competing in it.[11] You can see this happen over and over again in the digital technology sector: examples include Uber, which received billions from venture capitalists

[10] Haucap, J. (2019) Competition and competition policy in a data-driven economy, *Intereconomics* 54(4): 201–208; and Hein, A., Schreieck, M., Riasanow, T., Setzke, D., Wiesche, M., Bohm, M. and Krcmar, H. (2020) Digital platform ecosystems, *Electronic Markets* 30: 87–98.

[11] Birch, K. and Cochrane, D.T. (2022) Big Tech: Four emerging forms of digital rentiership, *Science as Culture* 31(1): 44–58.

without making any profits, or WeWork, which also received billions from venture capitalists without being profitable and eventually collapsing.

Back to Doctorow … once an enclave has a significant user base, these users become valuable and the real customers come calling: these customers are the businesses that will pay the enclave for access to the users and their data. In many cases, these customers are advertisers. The enclave starts pandering to its customers, since they're paying it, resulting in a worsening experience for its users; we've all experienced the deluge of ads on social media apps or websites like Facebook. Although it's becoming less useful, the enclave now dominates its market, meaning there is nowhere else for users to go: we're stuck basically. Then, Doctorow explains, the enclave starts to exploit its real customers, the businesses paying it money. For example, he argues that Facebook introduced payment systems for businesses to boost their position in newsfeeds; more vividly, after Twitter's acquisition by Elon Musk, it required that we for 'Twitter Blue' to ensure that our tweets are actually seen by anyone. The problem is, by the time an enclave is able to monetize its users, its attractiveness to those users is in decline while its attractiveness to its customers—like the advertisers who give them money—starts to corrode too.

Dysfunctionality sets in, making an enclave increasingly worthless. Online advertising represents a good example of this: it has ended up being increasingly gamed by large and small businesses, by government psyops agencies, or by random clickbait websites. This has been going on for some time, as Tim Hwang spells out in his 2020 book, *Subprime Attention Crisis*.[12] He argues that online advertising is all about the buying and selling of "attention assets", meaning the viewers watching a screen. As this buying and selling was automated with programmatic advertising, it led to an explosion in scams, fraud, and other attempts to game the automation of the advertising process. Examples range from click farms through the use of bots to pump up website popularity to subtler shifts in publishing practices, such as clickbait.

- **Click farms** are businesses that hire workers to pretend to be 'real' users on thousands of devices, literally stacked up on shelves; they are meant to simulate real human users, and their impressions or clicks

[12] Hwang, T. (2020) *Subprime Attention Crisis*, New York: FSG Originals x Logic.

or follows are then sold to many influencers, celebrities, companies, and others wanting to boost (the appearance of) their importance.[13]
- Non-human '**bots**' now make up around half of internet traffic, according to some commentators, meaning that a significant proportion of all the user metrics used to denote the value of a particular website or app (i.e. how many humans are viewing it) are wrong.[14]
- Big Tech corporations like Meta/Facebook have been accused of **inflating viewing metrics** on their social media properties by up to 900 percent.[15]

As I argued at the end of Chapter 1, these are examples of the massive growth in *impersonal* data; that is, fake users and user data that has been generated to mimic or parrot humans. It's unclear throughout these discussions, however, to what extent these impersonal data are entangled or enmeshed with our own personal data: how much do they shape the revenues a particular website or app can generate? How much do they impact AI training models? How much do they undermine corporate planning and strategies of Big Tech corporations? And so on. How we got to this point is an important issue, though, which I turn to next by considering three important paradoxes underpinning the emergence of data enclaves.

Reflexive Data

The first paradox concerns the argument I made earlier in the book that personal data entail reflexivity and have emergent properties, which we cannot predict or might not expect. Reflexivity is more important for what I want to argue here. Generally, in social theory reflexivity can be thought of as a recursive and self-referential process in which our knowledge claims about the world end up changing the world (and our behaviours) that we

[13] Birch, K. (2020) Automated neoliberalism? The digital organisation of markets in technoscientific capitalism, *New Formations* 100–101: 10–27.

[14] https://nymag.com/intelligencer/2018/12/how-much-of-the-internet-is-fake.html.

[15] Tepper, J. and Hearn, D. (2019) *The Myth of Capitalism*, New Jersey: Wiley; and see https://variety.com/2019/digital/news/facebook-settlement-video-advertising-lawsuit-40-million-1203361133/.

first sought to describe and explain.[16] Sometimes this ends up being self-fulfilling: a claim that everyone is 'selfish', for example, might lead people to change their behaviours or their institutions (markets) so that we end up acting *more* selfish than before, reinforcing the original claim.[17] Other times, though, it might have the opposite effect: a claim that we're all selfish, for example, might lead us to a change our behaviours and institutions to resist this assumption—by creating incentives to promote altruism—and thereby undermine the original claim.

Basically, reflexivity can be thought of as the idea that knowing the world also changes it. And I think this has important implications for our understanding of personal data as a political-economic object, especially as an asset. If we accept Sean Matin McDonald's argument that personal data have two sides (see Chapter 1),[18] then we need to understand how this plays out in our digital economies. As a refresher, McDonald argues that personal data are both 'truthful' statements of fact and 'fallible' representations, which can overlap (e.g. this belongs to Dave) but are highly contextually dependent and effects of these contexts (e.g. this belongs to Dave within this particular legal framework enforced by this particular state). As such, we have to start thinking of personal data not as a static resource but, rather, as a highly dynamic asset, which not only changes constantly but is also understood as changing constantly by individuals themselves as well as the businesses that try to make money off it (depending upon the context and institutions in that context).

Such a reflexive logic ends up impinging on the political-economic (and societal) choices, decisions, and futures we get or want to make—or even just imagine. Personal data are generated from observations of our behaviours, actions, and choices—e.g. web searches, emails, viewing habits, etc.—and the context in which those observations are constructed; that is, through the techcraft practices of digital technology businesses seeking to generate an asset that can be both measurable and legible as

[16] Giddens, A. (1984) *The Constitution of Society*, Cambridge: Polity Press.

[17] There's a wonderful article about how the assumptions in orthodox economics do exactly this: Ferraro, F., Pfeffer, J. and Sutton, R. (2005) Economics language and assumptions: How theories can become self-fulfilling, *Academy of Management Review* 30(1): 8–24.

[18] McDonald, S.M. (2021) Data Governance's New Clothes, Waterloo: Centre for International Governance Innovation, available at: https://www.cigionline.org/articles/data-governances-new-clothes/.

such (see Chapter 2). The value of that personal data, though, comes from the use of digital architectures by individuals, which can be captured by digital technology businesses. However, techcraft—the measurement, legibility, and valuation of personal data—is impacted by data's reflexivity since data generators (e.g. Big Tech) and data subjects (e.g. you and I) are both aware that it's happening—some are more aware than others—and recognize how our actions and claims can affect the world and are then able to act upon that.

Consequently, a Big Tech corporation knows that we don't want them to have our personal data to do with as they please, so they construct an array of socio-legal mechanisms (e.g. privacy policies) to buttress their business strategies and they change them as needed in response to changing public and political attitudes to their business model. Now, where this gets really interesting is when it comes to our own reflexive understandings of personal data generation: if our data are a valuable asset, we can change our user habits, disrupt their data generation architectures, and undermine their business models through gaming the system (e.g. lying about our names, preferences, choices, etc.). Ultimately, it might be possible that the compound effects of all the little 'lies' (or reflexive reworkings of ourselves) we tell and retell on a daily basis and across multiple ecosystems are undermining, or will undermine, the automation of our economies.

Making up the Rules of the Game

The second paradox concerns the argument that personal data are socially beneficial when they are shared widely, enabling their combination and aggregation in ways that generate emergent properties; yet, at the same time, personal data are most economically valuable when access to them is restricted and fees are charged for that access. Data enclaves are defined by this arrangement. Moreover, they benefit from attempts to encourage or support the free and open sharing of data, since they get access to those data and can combine or aggregate them with the data in their own enclave without having to contribute anything back. How they are able to do this is a key element in understanding the emergence of data enclaves and it relates directly to the importance of contract law rather than property rights in establishing control over data as an asset. As Fabian Muniesa and I argue, the asset form is defined as much by a contractual regime as by a regime of property rights; in particular, contractual

arrangements enable social actors to limit or delineate how a resource can be used, while property rights cannot (i.e. once transferred, property can be used by its new owners as they see fit, within the broader socio-legal setup). A contractual regime provides the mechanisms needed to generate revenues especially from intangibles assets, which often have non-rivalrous or non-excludable characteristics: for example, it's difficult to stop people who've bought knowledge from you from sharing it freely without a contractual arrangement in place to stop that. Consequently, we have witnessed an expansion of licensing and subscription fee systems in recent years, for quite a range of things, as our economies have become more asset-based.[19]

A good example of this shift in 'modes of ownership and control' is Amazon when it entered the eBook market. Although Amazon's ecommerce websites use the rhetoric of buying and selling (i.e. property rights), when it comes to eBooks and other easily shareable intangible objects, we aren't actually 'buying' anything. So, despite amazon.com stating "Buy now with 1-click", if you purchase an eBook from Amazon, you don't own it and are, instead, paying for a licence and certain use rights. This is made explicit in their Kindle terms and conditions:

> Use of Kindle Content. Upon your download or access of Kindle Content and payment of any applicable fees (including applicable taxes), the Content Provider *grants you a non-exclusive right to view, use, and display such Kindle Content* (for Subscription Content, only as long as you remain an active member of the underlying membership or subscription program), solely through Kindle Software or as otherwise permitted as part of the Service, solely on the number of Supported Devices specified in the Kindle Store, and solely for your personal, non-commercial use. *Kindle Content is licensed, not sold, to you by the Content Provider.* The Content Provider may include additional terms for use within its Kindle Content. Those terms will also apply, but this Agreement will govern in the event of a conflict. Some Kindle Content, such as interactive or highly formatted content, may not be available to you on all Kindle Software.[20]

[19] See Perzanowski, A. and Schultz, J. (2016) *The End of Ownership*, Cambridge MA: MIT Press.

[20] https://www.amazon.com/gp/help/customer/display.html?nodeId=201014950, emphasis added.

As I've highlighted in this quote, Amazon is licensing content to Kindle users and not selling them anything; moreover, it limits the rights that users have so that they can only read their eBooks on a specified device and for specific purposes. Such licensing terms and conditions constitute the contractual arrangements that define the value of an asset, representing the construction of a revenue stream through a limitation on access. Contractual arrangements also have two further implications important for understanding data enclaves: they enable the placement of access limits and they enable the construction of privately run rules of the (economic) game.

First, data enclaves are defined by a set of contractual arrangements underpinning their generation of personal data across all types of data (e.g. usage, sociodemographic, locational, etc.). The major data enclaves primarily emerged within the 'notice-and-consent' regime of the USA in which data collection and use practices were defined by the need to provide notice to users as part of a contractual arrangement for consent to use their data.[21] Here, consent is subsumed within contract law and has nothing to do with property rights. However, contractual arrangements, especially standard or boilerplate contracts like the terms and conditions agreements we click on every time we download an app or sign up to a website,[22] are frequently defined by asymmetric power relations between individual people and multinational corporations. Personal data collection and use is enabled—and very much not constrained—by our signing of terms and conditions agreements we don't read and, more importantly perhaps, cannot read within a reasonable timeframe. As a result, we often end up with less control over how our personal data are used and reused by signing such contracts, since they usually include clauses stating that the company can use our data as they see fit in perpetuity, including for unforeseen uses.[23] Of course, businesses do not frame it this way:

[21] Nissenbaum, H. (2017) Deregulating collection: Must privacy give way to use regulation? Available at SSRN: https://ssrn.com/abstract=3092282 or https://doi.org/10.2139/ssrn.3092282; and Guay, R. and Birch, K. (2022) A comparative analysis of data governance: Socio-technical imaginaries of digital personal data in the USA and EU (2008–2016), *Big Data & Society* 9(2): 1–13.

[22] Obar, J. and Oeldorf-Hirsch, A. (2020) The biggest lie on the internet: Ignoring the privacy policies and terms of service policies of social networking services, *Information, Communication & Society* 23(1): 128–147.

[23] Fourcade, M. and Kluttz, D. (2020) A Maussian bargain: Accumulation by gift in the digital economy, *Big Data & Society*, https://doi.org/10.1177/2053951719897092.

terms and conditions are usually presented as necessary for the successful delivery of a product or service (see Fig. 6.1).

Second, contract law has been defined as the creation of privately made law.[24] Contracts are defined as 'legally enforceable promises', reflecting an agreement between two (or more) contractual parties. They are private arrangements, enforceable by a third party but their contents are not defined or determined by that third party—this is important to note because this was not always the case. Contract law has evolved over time, shifting from a more 'formal' approach in the nineteenth century—centred on the notions of individualism and rational will—through a

"What Personal Information About Customers Does Amazon Collect?

We collect your personal information in order to provide and continually improve our products and services.

Here are the types of personal information we collect:

- **Information You Give Us:** We receive and store any information you provide in relation to Amazon Services. Click here to see examples of what we collect. You can choose not to provide certain information, but then you might not be able to take advantage of many of our Amazon Services.
- **Automatic Information:** We automatically collect and store certain types of information about your use of Amazon Services, including information about your interaction with products, content, and services available through Amazon Services. Like many websites, we use "cookies" and other unique identifiers, and we obtain certain types of information when your web browser or device accesses Amazon Services and other content served by or on behalf of Amazon on other websites. Our physical stores may use cameras, computer vision, sensors, and other technology to gather information about your activity in the store, such as the products and services you interact with. Click here to see examples of what we collect.
- **Information from Other Sources:** We might receive information about you from other sources, such as updated delivery and address information from our carriers, which we use to correct our records and deliver your next purchase more easily. Click here to see additional examples of the information we receive".

Fig. 6.1 Amazon.com privacy notice (*Source* https://www.amazon.com/gp/help/customer/display.html?nodeId=GX7NJQ4ZB8MHFRNJ)

[24] Much of this discussion on contract law draws on my previous work in Birch, K. (2017) *A Research Agenda for Neoliberalism*, Cheltenham: Edward Elgar, especially Chapter 8.

'relational' or 'realist' approach emerging in the early twentieth century—influenced by the idea that markets are imperfect and entail power asymmetries—and back to a 'neo-formalist' approach at the end of the twentieth century. Contract law today is characterized by an attempt to frame contractual arrangements in notions of economic efficiency and as voluntary and private arrangements between rational economic actors.[25] Whatever principles underly it, contract law enables the construction of private arrangements between different legal parties (e.g. individuals, corporations, etc.).

When it comes to data enclaves, this enables them to establish the rules of the (economic) game in their ecosystem; users, clients, developers, and others have to all abide by the rules contractually established for participation in an ecosystem in order to access that ecosystem and the benefits that it might provide (e.g. access to customers for an app developer). And these rules enable data enclaves to not only collect data (from users, clients, etc.) but also set restrictions on access to the valuable data they collect and set restrictions on the use of that data. As the report of US Congressional investigation into competition in digital markets noted, when it came to Facebook:

> Facebook's data also enables it to act as a gatekeeper because Facebook can exclude other firms from accessing its users' data. Beginning in 2010, Facebook's Open Graph provided other companies with the ability to scale through its user base by interconnecting with Facebook's platform. Some companies benefited immensely from this relationship, experiencing significant user growth from Open Graph and in-app signups through Facebook Connect, now called Facebook Login. Around that time, investors commented that Open Graph gave some companies 'monstrous growth' referring to it as 'steroids for startups'. For example, documents produced by Facebook indicate that it was the top referrer of traffic to Spotify, driving 7 million people 'to install Spotify in the month after [Facebook] launched Open Graph'.[26]

[25] It's worth emphasizing that much of the world's 'economic' law is shaped or influenced by common law principles derived from English and US common law: see Pistor, K. (2019) *The Code of Capital*, Princeton: Princeton University Press.

[26] US House of Representatives (2020) *Investigation of Competition in Digital Markets*, Washington DC: House of Representatives, quote at p. 148.

The report went on to note, however, that Meta/Facebook has the ability to give or withhold access to their social graph and the personal data underpinning it, thereby "effectively picking winners and losers online" (p. 148). On top of that, Meta/Facebook also gained insights into the strategies of other companies through their control over measurement metrics.

There Are No Markets Anymore

The third paradox I'm going to consider concerns the idea that generating more personal data will improve market outcomes, such as making personalized advertising more relevant for us users or improving our 'experience'—popular refrains by Big Tech when it comes to justifying their data collection and use. For example, Meta/Facebook's privacy policy makes these claims repeatedly.[27] Market outcomes are supposed to improve as markets generate more 'information' on which market actors can rely to make their decisions and reveal their preferences. The paradox is that data enclaves need to generate data about us to make money, but they also need to limit access to that data to make money: they need to be in control of it. This becomes contradictory in a political and policy climate defined by the idolization of 'free' markets, which has pretty much been the case for the last few decades. Often defined as 'neoliberalism', this pro-market worldview has dominated government policy, international financial decision-making, and think tank posturing to such an extent that other perspectives have become almost unfathomable. And all while the reality of markets rarely lived up to the vaunted benefits or were ignored when convenient.[28] The failure of market thinking has become even more stark with the dominance of Big Tech. These data enclaves increasingly control or dominate much of the market infrastructure our economies now rely upon, including the underlying market information on which markets are supposedly dependent. To understand this, I've got

[27] https://www.facebook.com/privacy/policy/?entry_point=data_policy_redirect&entry=0.

[28] On this, see Birch, K. (2017) *A Research Agenda for Neoliberalism*, Cheltenham: Edward Elgar.

to provide a brief introduction to the case for market dominance made by neoliberals.[29]

Neoliberalism is usually defined as a political-economic *and* moral project to redesign our societies by putting markets at the centre of government, business, and even individual decision-making; contradictorily, this redesign of our societies is often premised on a naturalistic notion of markets as inherent to human nature and behaviours. A particularly important figure in promoting this idea was the Austrian economist Friedrich Hayek: he argued that no single agent, including and especially government, can coordinate our economies or societies because that agent simply does not have the cognitive capacity to process all the information we produce or use everyday.[30] Instead, Hayek and other neoliberals argued—and still argue—that markets are the only information processors that can coordinate all our economic or social decisions and actions. According to Hayek:

> The reason for this [economic problem] is that the "data" from which the economic calculus starts are never for the whole society "given" to a single mind which could work out the implications and can never be so given.[31]

Markets make the best coordination machines because they are able to process all the information we generate in our lives, enabling us to make the *right* economic and moral decisions. Markets generate prices through supply and demand dynamics, which provides us with the information we need to decide what to produce, what to consume, what preferences we should change, and how to manage our collective resources. All of this is why Hayek (and others) thought information is a critical problem for us to solve and provided the rationale for letting markets spontaneously emerge by removing government or societal 'interference'. As time went

[29] Much of this discussion in this section draws on: Birch, K. (2023) There are no markets anymore: From neoliberalism to Big Tech, *State of Power 2023 Report*, The Transnational Institute (3 Feb): https://www.tni.org/en/article/there-are-no-markets-anymore.

[30] See Dardot, P. and Laval, C. (2014) *The New Way of the World*, London: Verso; and Mirowski, P. and Nik-Khah, E. (2017) *The Knowledge We Lost in Information*, Oxford: Oxford University Press.

[31] Hayek, F. (1945) The use of knowledge in society, available at: https://www.econlib.org/library/Essays/hykKnw.html.

by, however, this epistemic notion of markets being the best information processors led to a more deliberate approach to designing markets to achieve the societal objectives policymakers and others wanted (rather than expecting them to emerge spontaneously).[32]

Since the 1980s, policymakers and others have deliberately and actively designed markets to achieve the ends those designers want and a new economics field of 'mechanism design' has emerged around this.[33] We can see examples of mechanism design in a range of government attempts to privatize public assets, or deregulate economic sectors, or auction off entitlements like radio or cellphone spectrum. Mechanism design has a relatively short intellectual history, going back to work by people like William Vickery who examined pricing in auctions and whose name is often associated with second-price auctions (see Chapter 4's discussion of Alphabet/Google). In designing market mechanisms, economists assume that individuals are rational and self-interested, seeking to maximize their own interests, and that these individual behaviours can lead to collective, social benefits if we find the right incentive structure for a market. Markets can be designed to reveal our 'true' preferences through the 'choice architecture' that forces us to be truthful when making our decisions. The usefulness of markets depends on this; all market agents have to reveal information about themselves on which others can act, or one side ends up being able to game the market and exploit the information asymmetries that pervade markets. To do this, market designers turn the usual economics perspective on its head by creating the markets they want in order to achieve the outcomes they want; and to do this, they construct the market architecture they need to incentivize us to do what they want.

Data enclaves are defined by market design. An array of digital and algorithmic technologies has enabled digital technology businesses, especially Big Tech, not only to generate the information on which markets are supposedly dependent (e.g. personal data about our preferences, our decisions, our actions, etc.) but also to monetize that information for their own benefit. And that has meant restricting access to market information in order to create deliberate information asymmetries. Information about who wants to buy what, how much they'd be willing to pay for

[32] Amadae, S. (2016) *Prisoners of Reason*, Cambridge: Cambridge University Press.

[33] Viljoen, S., Goldenfein, J. and McGuigan, L. (2021) Design choices: Mechanism design and platform capitalism, *Big Data & Society* 8(2). https://doi.org/10.1177/20539517211034312.

what, how many people view what, and so on can all be sold to other companies who want information on their customers. While this market information is meant to be transparent and truthful to ensure the benefits of market competition are spread widely, its increasing control by data enclaves means that digital technology businesses have gone far beyond the 'surveillance' fears of their many critics. In particular, Big Tech have become *the* key intermediaries in our daily lives, including when it comes to the information we rely upon to connect with one another, to make decisions, to choose our politicians, to judge the usefulness of things, and much else besides. In becoming data enclaves, Big Tech has designed digital technologies with the specific goal of generating an increasing mass of personal data that cements their dominance across their ecosystem of devices, applications, users, platforms, and so on. However, their ecosystems are not markets; as data enclaves, these corporations rely upon the creation of information asymmetries in order to generate revenues: for example, adtech businesses make money precisely *because* others don't have access to the personal data they collect and restrict access to.

A very clear example of the implications of this increasing domination and control over the market information we need to live our lives relates to the issue of dynamic pricing or price discrimination. The legal scholar Frederik Zuiderveen Borgesius has written about the specifics of price discrimination in the digital economy, noting that online retailers in the USA are already charging people from different places different prices.[34] While price differentiation is not new—businesses have often priced things differently for different people—current and future digital technologies provide the means to turbo-charge this practice while obscuring its problematic premises and effects. The massification of personal data collection and analysis also enables other forms of dynamic pricing not available previously. A contemporary example is Uber's surge pricing which is only possible because of the data they collect.[35] While they design their pricing algorithms around an economistic notion of supply and demand, Uber also designed their pricing algorithms to adjust depending upon a range of other elements: this enables surge pricing which is meant to attract

[34] Borgesius, F.Z. (2020) Price discrimination, Algorithmic decision-making, and European non-discrimination law, 31, *European Business Law Review* 31(3): 401–422.

[35] Rosenblat, A. and Stark, L. (2016) Algorithmic labour and information asymmetries: A case study of Uber's drivers, *International Journal of Communication* 10: 3758–3784.

drivers to in-demand areas. The overall problem with dynamic or discriminatory pricing is that it enables unequal treatment of people depending upon a range of unknown and unknowable criteria controlled by a digital technology business seeking to maximize revenues. In their work, Marian Fourcade and Kieran Healy theorize the benefits that some people will get from this, especially those with what they call 'übercapital' which is sought after by digital technology businesses, while others will be disadvantaged: for example, people who live in wealthy neighbourhoods can be targeted and incentivized with beneficial pricing while those in the 'wrong' neighbourhoods can be ignored or charged over the odds for similar products and services (e.g. banking).

Overall, I think the most important point to remember from this chapter is that personal data are not simply a proxy for 'attention' and we, as users, are not becoming the 'product' which Big Tech sells to others. I've tried to show that the measurement, legibility, and valuation of personal data are necessarily implicated in our understanding of how markets operate, especially as personal data are generated by data enclaves and access to personal data is deliberately restricted as the monetization strategy. I think a more important debate needs to be started about how we understand personal data *as* market information: are they the same thing? Or do we need to understand them as distinct? If the former, then we need to move beyond the idea that personal data are an economic good and to start thinking of them as an asset which enables data enclaves to do certain things, especially to create ecosystems in which they control the market, avoid competition, and undermine competitors. The result of this is the dysfunctionality I outlined in the earlier part of the chapter, and which is becoming all too evident in our daily interactions with these data enclaves: online search is now dominated by ads, ecommerce is rife with scams and fakes, social media is gamed by bots, and even the software or digital infrastructures we rely upon are locking us into one system or another.

CHAPTER 7

Conclusion: Where Next for Data Governance?

Abstract If digital personal data are the underlying resource base of our societies and economies, then the configuration of data as an asset and its concentration in a series of data enclaves leaves societies and policymakers dealing with a number of problematic impacts and side effects from the resulting data paradoxes. We need to develop new ways to govern personal data which gets beyond the status quo: that is, leaving it to the business world to do as they please. Data markets do not work, and private property rights are unlikely to provide a long-lasting solution. Other forms of data governance might provide viable alternatives, but which one or ones becomes the critical question for publics and policymakers to decide.

Keywords Personal data · Data governance · Policymaking

I want to conclude this book by outlining the key take aways from the earlier chapters and then by considering what to do about data enclaves.

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Where Are We At?

In this book, I've highlighted how Big Tech are able to control and manage the information on which markets depend, including much of the personal data that they generate through the specific techno-economic infrastructures we all now use on a daily basis. As a result of our growing dependence upon these infrastructures, we have become locked into a series of data enclaves that no longer function as markets. Rather, these data enclaves are able to monetize the very market information necessary for markets to function as we expect and desire them to function. The consequences of this are problematic across our economies, as I sought to illustrate in Chapter 6 and as many others have similarly pointed out over the last few years.[1] However, most critical analyses focus on only one end of the process.

Concerns about surveillance and advertising often dominate discussions about problems with the digital economy. For example, a lot of people, from academics to politicians to businesspeople, talk about contemporary capitalism as an 'attention economy'.[2] It's a concept that's been around for some time and reflects the notion that the digital economy can be understood and defined by the value of our attention to digital technology businesses seeking to monetize our 'views' through advertising: in effect, we are paying for free digital products and services with our time.[3] Returning to Chapter 2, Tim Hwang, ex-global policy lead for AI and machine learning at Alphabet/Google, even describes our attention as an asset (i.e. "attention asset"), which can be measured and valued as such. Similarly, the US Congressional investigation into digital markets and competition revealed that corporate managers and executives from Big Tech corporations like Meta/Facebook think in the same way:

[1] Some key examples include: Srnicek, N. (2016) *Platform Capitalism*, Cambridge: Polity Press; Zuboff, S. (2019) *The Age of Surveillance Capitalism*, New York: Public Affairs; Lehdonvirta, V. (2022) *Cloud Empires*, Cambridge, MA: MIT Press; and Stucke, M. (2022) *Breaking Away: How to Regain Control Over Our Data, Privacy, and Autonomy*, Oxford: Oxford University Press.

[2] https://econreview.berkeley.edu/paying-attention-the-attention-economy/.

[3] Brynjolfsson, E. and Oh, J.H. (2012) The attention economy: Measuring the value of free digital services on the internet, *ICIS 2012 Proceedings* 9, available at: https://aisel.aisnet.org/icis2012/proceedings/EconomicsValue/9.

> In an interview conducted by Subcommittee staff, a former employee explained that as a product manager at Facebook 'your only job is to get an extra minute. It's immoral. They don't ask where it's coming from. They can monetize a minute of activity at a certain rate. So the only metric is getting another minute'.[4]

They want our time because they can monetize it and because we, as users, then become capitalizable, as discussed in Chapter 3. Now, I'm personally sympathetic to these analytical framings of the digital economy: I'm sure we've all found ourselves sucked into one app or another for a longer period of time than it is healthy to stare at kittens on a screen. But, in writing this book, I think I'm coming to the perspective that these framings of the attention economy miss a key aspect of what's going on with data enclaves.

If we think of attention as awareness (of information, for example), then data enclaves are simultaneously trying to erode our attention as much as attract it; specifically, to erode our access to information because information asymmetries are valuable to these data enclaves. If they have information and we don't, then this asymmetry can be monetized as discussed in Chapter 6. This is why it's helpful to think of personal data as a political-economic object (i.e. asset) with reflexive and emergent properties, which both constitutes the data enclaves themselves and buttresses their dominance through the reinforcement of the techno-economic arrangement of that system. Lyotard's argument about system-reinforcing knowledge makes a lot of sense here (see Chapter 2).[5] Personal data are market information that reinforces the techno-economic arrangements underpinning data enclaves, optimizing these enclave ecosystems whether or not the personal data are 'truthful' representations of individuals increasingly locked into those enclaves. Personal data are generated by data enclaves with a certain function, and that function is to self-reinforce the system that produces it: enclaves don't need to provide transparent or 'truthful' information, they need, rather, to undermine its very generation and proliferation. They require information asymmetries. And this is why we see so much misinformation, disinformation,

[4] US House of Representatives (2020) *Investigation of Competition in Digital Markets*, Washington, DC: House of Representatives, quote at p. 135.

[5] Lyotard, J.-F. (1984) *The Postmodern Condition*, Manchester: University of Manchester Press.

nonsense, and outright bigotry and hatred proliferate across these data enclaves: unfortunately, these all generate revenues and reinforce the data enclaves' position, meaning that removing any of it will undermine the enclaves. Consequently, things are likely to get worse unless we break up the enclaves.

How these information asymmetries are configured matters. And this matters for any discussion of personal data as a political-economic object, especially as a valuable asset discussed in Chapter 4. Personal data are not property, nor do property rights provide data enclaves with the socio-legal mechanism to secure control over personal data, since it cannot be owned. Rather, personal data are generated and configured by contractual arrangements, not property rights; contract law, then, plays a central role in the assetization of personal data.[6] A contract is a legally enforceable promise that entails a claim on future revenues; as such it can be treated as an asset. Contract law is a form of private ordering or governance in which private parties are assumed to be able to transact with one another to reach a beneficial outcome for themselves: unfortunately, these assumptions often ignore the problems of asymmetrical power in bargaining between individual persons and large multinationals, like Big Tech. Instead, contract law basically enables digital technology businesses to set the rules of the game in their ecosystems, controlling what users and customers can do and how.

What Do We Do Now?

Personal data are an important and valuable political-economic asset. Personal data are important for society generally, not just for individuals or businesses, because they can provide the means for working out how to do a lot of things differently, hopefully for the benefit of the collective good (e.g. improving health outcomes, public transit, welfare, education, democratic processes, etc.). This is an important reason for why there is considerable policy and political concern with understanding how to govern personal data. For example, policymakers and experts in international standard setting and financial institutions (e.g. OECD, UN) are trying to work out how to account for personal data, as are national-level statistical offices which are trying to find ways to frame the importance

[6] Birch, K. and Muniesa, F. (eds.) (2020) *Assetization: Turning Things into Assets in Technoscientific Capitalism*, Cambridge, MA: MIT Press.

of data to our societies and economies. As a result, many policymakers, politicians, publics, activists, and experts around the world are rethinking how to deal with personal data, especially as its political-economic value clashes with prevailing data governance regimes. For example, in 2023 the Bundeskartellamt, Germany's federal competition authority, noted that "Google's business model relies heavily on the processing of user data" and that "Due to its established access to relevant data gathered from a large number of different services, Google enjoys a strategic advantage over other companies".[7] This statement reflects earlier statements by the EU's competition authority in 2021 that the European Commission "already considers data as an asset in merger assessments"; this statement came as response to an EU Parliamentary question regarding an investigation into "the way data concerning users is gathered, processed and monetised by Google".[8]

Prevailing data governance regimes tend to start from a particular position: protect people's privacy (to whatever extent expected by national conventions) but let businesses do as they wish with our personal data. However, as I've emphasized throughout, personal data are generated rather than simply lying around and ready to be collected, which requires a different way of thinking about data governance. Instead, it helps to think about data governance through the lens of 'techcraft', which I outlined in earlier chapters and which draws on the work of James C. Scott's *Seeing Like a State*.[9] Techcraft concerns the construction of personal data through techno-economic means, including standards, measurement tools, commercial logics, etc., that generate users and their data as well as, simultaneously, the means to monetize those users. As a concept, techcraft highlights how personal data are not simply observed and recorded, but rather generated within the data enclaves to function as

[7] Bundeskartellamt (2023) *Press Release: Statement of Objections Issued Against Google's Data Processing Terms*, Bonn: Bundeskartellamt, available at: https://www.bundeskartellamt.de/SharedDocs/Meldung/EN/Pressemitteilungen/2023/11_01_2023_Google_Data_Processing_Terms.html.

[8] In 2021, European Commission Executive Vice-President Vestager specifically noted that "The Commission already considers data as an asset in merger assessments", in response to a Parliamentary Question [E-000274/2021(ASW)] about personal data: https://www.europarl.europa.eu/doceo/document/E-9-2021-000274-ASW_EN.html.

[9] Scott, J. (1998) *Seeing Like a State*, New Haven: Yale University Press.

an asset self-reinforcing that enclave. Like Scott's argument about statecraft, techcraft entails the generation of measurable, legible, and valuable data rather than 'truthful' facts about individuals: this includes defining users, use, and user engagement in particular ways and controlling access to this information. Importantly, techcraft does not entail new forms of accountability for the generation of personal data or its construction as a political-economic object: in fact, I would argue the opposite, that it actually entails an attempt to avoid accountability.

In rethinking data governance, it's clear that certain approaches do not and will not work. Despite much popular and mainstream interest in it, personal data ownership rights—giving each of us property rights over our own personal data—are not a viable governance mechanism. Proponents of this include Jaron Lanier in *Who Owns the Future* and Eric Posner and Glen Weyl in *Radical Markets*. We cannot own our data, especially user data generated within digital ecosystems, without establishing a wholly new digital infrastructure to surveil and track ourselves. I'm not sure this would be popular, or even doable, and it's likely to be even more privacy-invading than the present regime and would require a continuing reliance on private businesses to do it for us. Furthermore, it's difficult philosophically to determine who should own what personal data: as Cory Doctorow notes, who should own the information that you are the child of your parents? Should it be you, or your parents, or both of you?[10]

I have more hope for collective approaches to data governance, including the establishment of collective data trusts,[11] or data collectives.[12] These could be run by governments, public agencies, or mutual collectives, depending upon the preference of citizens, and they could generate (i.e. collect, curate, store) our personal data for use by anyone who pays a fee and abides by the licensing arrangements (with stringent privacy constraints). As a data governance approach, these would provide collective control over the use of our personal data; for example, we could limit its use for things we might disagree with (e.g. facial recognition, digital red-lining, etc.) and we could limit the reuse of personal data, meaning that digital tech businesses couldn't hold onto it in perpetuity.

[10] On this issue, see Doctorow, C. (2020) *How to Destroy Surveillance Capitalism*, Medium Editions, available at: https://onezero.medium.com/how-to-destroy-surveillance-capitalism-8135e6744d59.

[11] https://www.theodi.org/article/what-is-a-data-trust/.

[12] https://www.bennettinstitute.cam.ac.uk/blog/whose-data-commons-part-one/.

There would still be issues with this setup, including: a continuing issue with how to track personal data and what that would mean for surveiling individuals; it wouldn't necessarily stop Big Tech (or others) accessing our personal data; and it would require significant oversight to run and manage, as well as public buy-in into it.

Before we get to what we might want in the future, simply introducing a more stringent data governance regime is probably the most urgent policy goal right now. It's worth remembering, though, that there is a long history to personal data governance stretching back decades and that different jurisdictions have pursued very different policy approaches, meaning that personal data can be treated quite differently in different parts of the world.[13]

Personally, I think the European Union that has taken the most concerted and coordinated policy action on data governance to date. For example, the EU has introduced a suite of data governance regulations and measures over the last couple of years, designed specifically to rein in the power of Big Tech. These include:

- *Digital Markets Act*: coming into effect in 2023, this Act is designed to establish ex ante regulations to control the behaviour of so-called 'gatekeeper' firms, covering what the EU defines as "large, systemic online platforms". The Act forbids gatekeepers from doing certain things, like combining personal data from platforms with data collected for other services, or self-preferencing, treating their own products and services more favourably than others; while it is designed to enable users to uninstall default software or apps more easily, and to stop gatekeepers from tracking users outside their platform.[14]
- *Digital Services Act*: coming into effect in 2024, the Act is designed to increase transparency in online advertising while reducing illegal and harmful content and misinformation.[15]

[13] Pasquale, F. (2015) *The Black Box Society*, Cambridge, MA: Harvard University Press.

[14] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/digital-markets-act-ensuring-fair-and-open-digital-markets_en.

[15] https://ec.europa.eu/commission/presscorner/detail/en/ip_22_2545.

- *Data Governance Act*: came into effect in 2022, the Act was designed to open up and standardize data sharing between organizations and countries in the EU, while limiting the ability of businesses to hoard data (and potentially stopping them create data enclaves).[16]

These recent data governance changes follow the EU's 2018 *General Data Protection Regulation* (GDPR), which has almost become a de facto international standard, having a significant impact on the ways digital technology businesses think about privacy and data protection around the world. Some have argued that it's an unworkable regulation, or even benefits Big Tech since large multinationals are more able to adopt GDPR regulations, but there are signs that it is going to have an even more important impact in the future. This is the result of an abuse of dominance case brought against Meta/Facebook by the German competition agency, Bundeskartellamt, for collecting personal data beyond its specific products and services (e.g. Facebook, Instagram). In 2022, this case came before the Court of Justice of the EU (CJEU), where Advocate General Rantos handed down the following opinion:

> In order to collect and process user data, Meta Platforms relies on the contract for the use of the services entered into with its users when they click on the 'Sign up' button, thereby accepting Facebook's terms of service. Acceptance of those terms of service is an essential requirement for using the Facebook social network. The central element of this case is the practice of *collecting* data from other group services, as well as from third-party websites and apps via integrated interfaces or via cookies placed on the user's computer or mobile device, *linking* those data with the user's Facebook account and then *using* them ('the practice at issue').[17]

In his opinion, Rantos' goes on to argue Meta/Facebook could not claim 'necessity' within the GDPR framework for collecting personal and user data outside its ecosystem (e.g. through cookies), or beyond the original terms of service. This case has enormous implications for data enclaves, especially those that rely upon advertising and the monetization

[16] https://digital-strategy.ec.europa.eu/en/policies/data-governance-act.

[17] https://curia.europa.eu/juris/document/document.jsf?docid=265901&doclang=EN.

of personal data. In 2023, the CJEU delivered its verdict, which activist organization NOYB argues:

> …has largely closed the doors for Meta to use personal data beyond what is strictly necessary to provide the core products (such as messaging or sharing content) - all other processing (like advertisement and sharing personal data) requires freely given and fair consent by users.[18]

It looks like the CJEU ruling will mean that terms and conditions agreements, the contractual basis on which much of the current data governance regime rests, can't be used to hoover up as much personal data as possible, across and beyond an ecosystem. It'll be interesting to see how this combination of data protection and competition law plays out in the coming years, and whether it'll influence other countries in their deliberations about how to deal with Big Tech and their control over our personal data.

Whichever way countries go on data governance, the key policy issue is to find ways to make businesses more accountable for the personal data they generate, not only in terms of how they collect and use personal data but also how they and others understand it as a political-economic object (i.e. asset). Properly accounting for personal data will change the way it's dealt with and working towards an international standard on this front is a vital policy shift we need to see today, especially considering its possible repercussions on privacy regulations, competition policies, and taxation treatment. There are signs this is happening, so there is hope that we can find a way to make Big Tech and others more accountable for personal data, but we need to keep pushing in this direction to ensure it happens.

[18] https://noyb.eu/en/cjeu-declares-metafacebooks-gdpr-approach-largely-illegal.

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